

BitZ: Proofs in arbitrary domains with commitments in binary fields

BitZ: proofs and commitments over arbitrary finitely generated rings through binary fields

BitZ: proofs and commitments in arbitrary finitely generated rings via low entropy grand products in binary fields

BitZ: proofs and commitments in arbitrary rings through binary fields

immediate

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Abstract

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1 Introduction

Albert: Interesting stuff to mention: 1) cost of BitZ do not depend on the ring, only on bit-size. 2) BitZ PCS can be used as drop-in replacement of any PCS? 3) Why more than

A Succinct Non-interactive Argument of Knowledge (SNARK) for a relation $\mathcal{R}$ is a non-interactive protocol that enables a prover to convince a verifier that, for a given public input $\mathbf{x}$, they know a witness $\mathbf{w}$ such that $(\mathbf{x}, \mathbf{w}) \in \mathcal{R}$, and where communication and verifier runtime are sublinear (i.e. the scheme is succinct).

In the vast majority of scenarios, $\mathcal{R}$ is defined over one (or more) algebraic object R , usually a finite field, and $(\mathbf{x}, \mathbf{w}) \in \mathcal{R}$ if and only if $(\mathbf{x}, \mathbf{w})$ consists of elements that satisfy some polynomial

equations. The choice of R has a crucial impact on the design and end-to-end performance of the proof system, for two reasons:

1. Usually the computations one wants to prove are not native to R , so one has to perform a more or less complex emulation of the computation into polynomial equations over R . This can affect the size of the input-witness pair $\mathbf{x}, \mathbf{w}$ significantly [?]. Additionally, the emulation step can be technically involved and prone to security errors [?].
2. The prover and verifier algorithms efficiency and design is affected by the choice of R : for example, modern CPUs are more performant for operations in certain small fields (e.g. M31 [?, HLP24], Babybear [?], Goldilocks [?]) than in arbitrary fields, especially if these are larger. Importantly, another class of fields where CPUs operate exceptionally fast are *binary fields*, i.e. fields of order a power of two.

Recently, two independent lines of research have received increased attention:

- Designing proof systems for relations expressed over $R = \mathbb{Z}$ the ring of integers, and over other rings that are not fields [CHA24, GWV⁺25, AGK⁺26, SSPP26, CXNB26, ?]. The motivation behind this is to address Item 1 above: in many practical scenarios, the computation to be proved is most succinctly and easily expressed over such rings, see Remark 3.4 Albert: more refs?.
- Designing proof systems for $R = \mathbb{F}_{2^\nu}$ a binary field, for example [BSBHR18b, ?, DP24, ?, BRW26]. The reason for this is twofold: first, some computations like classic hashes (SHA-256, Blake, etc.) are easier to express over $\mathbb{F}_{2^\nu}$ than over other structures such as prime fields (however, as we discuss in ??, these hash functions remain relatively un-native to $\mathbb{F}_{2^\nu}$ due to the hardness of emulating operations modulo 2^{32} or 2^{64}). Second, binary fields are highly performant on modern CPU's, and further enable using a boon of protocol techniques that further benefit performance, most importantly *packed* polynomial commitment schemes [DP23, DP24].

1.1 Our contributions

Albert: It should be very clear throughout the paper: The PIOP machinery is for the most part Zinc+'s (which builds on Zinc and Zaratan). The dual field PCS primitive is novel; its instantiation is pretty much novel (though ideas can be traced back to [Bin26, Sou26]), and its enabling optimizations are novel. The PCS in binary fields is completely known.

In this work we bring these two lines of research together. We present BitZ, a hash-based proof system that:

1. BitZ can prove polynomial constraints over essentially arbitrary rings R , including $\mathbb{Z}$, prime fields, lattices, binary fields, arbitrary field extensions, polynomial rings, etc., addressing Item 1 above (BitZ can handle constraints over different rings at once).
2. In its initial commitment step, BitZ's decomposes the witness $\mathbf{w}$ into a vector of bits $\text{bits}(\mathbf{w})$, and commits to $\text{bits}(\mathbf{w})$ over a binary field $\mathbb{F}_{2^\nu}$ with an error correcting code (leveraging the packing technique from [DP23, DP24]).
3. The initial polynomial constraints over the ring(s) R are reduced to an MLE evaluation claim (or more generally, an inner product claim) over a finite field $\mathbb{F}$, of the form $\langle \mathbf{v}, \pi(\mathbf{w}) \rangle = \mu \in \mathbb{F}$, where π is a projection from R to $\mathbb{F}$ and $\mathbf{v}$ is a vector of elements from $\mathbb{F}$. The field $\mathbb{F}$ can be a fixed field extension, a random prime field (following [CHA24]), or a fixed prime field, depending on what the ring(s) R is (are).

4. BitZ’s Polynomial Commitment Scheme (PCS) enables proving such linear claim efficiently. It operates, for the most part, over $\mathbb{F}_{2^\nu}$. Thus, BitZ benefits of many of the advantages of running proof systems over binary fields (Item 2 discussed earlier one). See below for a brief description of BitZ’s PCS.

BitZ provides, among others, a compiler that takes any PCS over $\mathbb{F}_{2^\nu}$ and transforms it into a PCS over any of the above rings R , following the “compiler” frameworks of [SSPP26, CXNB26].

We implement and open-source BitZ [?], and use it to prove a number of computations of practical interest. In ?? we provide a summary of the results. In all computations but one, BitZ outperforms the current state-of-the-art approaches, sometimes by orders of magnitudes (for example: ***). BitZ is even able to prove Babybear multiplications more efficiently than Plonky3, a mature proof system that operates natively on the Babybear field, and which is used in industry within a variety of projects [?]. **Albert: Mention what is the security parameter**

We manage to instantiate our protocol using $\mathbb{F}_{2^\nu} = \mathbb{F}_{2^{128}}$ in GHASH representation, attain 128 bits of security, and have no limitation on the size of the ring R we start with. See ?? and Section 5.7 for more comments and details on this. **Albert: For BabyBear bench, give separate numbers for PIOP part and the rest**

Computation	BitZ	Other schemes
SHA-256 (7KB) + ECDSA		Albert: Zinc+,Vega,Longfewllow
2^{17} 32 bit integer mults		Albert: Binius,Limber,Zinc+
2^{20} Babybear mults		Albert: Plonk3
RSA accumulator + Poseidon has (MultiSwap [?], ??) [CXNB26]		Albert: Limber
Falcon		
SHA-256 in 8KB message (2^7 compressions)		Albert: flock/binius
SHA-256 in 2MB message (2^{15} compressions)		Albert: flock/binius

Table 1: **Albert: Include memory.** **Albert: Revisit: should we include the experiemnts on hashing big messages here?** Performance of BitZ on a suite of various computations of interest. For each such computation, we also benchmark the performance of prior state of the art protocols. All experiments were run on a MacBook Air M4 16GB, multi-threaded. **Albert: Mention what is the security parameter** **Albert: Maybe report numbers for both the AI and human implementation**

BitZ’s PCS in a nutshell. BitZ proves the aforementioned claim $\langle \mathbf{v}, \pi(\mathbf{w}) \rangle = \mu$ over a finite field $\mathbb{F}$ as follows. We describe first an initial, mostly insufficient, attempt.

1. **Bitification.** First, it transforms the claim into a claim of the form $\langle \mathbf{u}, \pi(\text{bits}(\mathbf{w})) \rangle = \mu$ over $\mathbb{F}$, for some $\mathbf{u}$.
2. **Lift to integers.** Assuming $\mathbb{F}$ is a prime field of size q and $\pi_q : \mathbb{Z} \rightarrow \mathbb{F}$ is reduction modulo $|\mathbb{F}|$, the prover sends $\mu' \in \mathbb{Z}$ such that $\langle \pi_q^{-1}(\mathbf{u}), \text{bits}(\mathbf{w}) \rangle = \mu'$ in $\mathbb{Z}$.
3. **Proof in the exponent of $\mathbb{F}_{2^\nu}$.** Now take a generator g of the multiplicative subgroup $\mathbb{F}_{2^\nu}^*$ of $\mathbb{F}_{2^\nu}$, and prove that $g^{\langle \pi_q^{-1}(\mathbf{u}), \text{bits}(\mathbf{w}) \rangle} = g^{\mu'}$ in $\mathbb{F}_{2^\nu}$. Concretely, we have

$$g^{\langle \pi_q^{-1}(\mathbf{u}), \text{bits}(\mathbf{w}) \rangle} = \prod_i \left((g^{\pi_q^{-1}(\mathbf{u}_i)} - 1) \cdot \text{bits}(\mathbf{w})_i + 1 \right) = g^{\mu'}, \quad (1)$$

which is a polynomial expression over $\mathbb{F}_{2^\nu}$ on the vector $\text{bits}(\mathbf{w})$ that we committed over $\mathbb{F}_{2^\nu}$. This is a standard situation where there is no “wrong-field” arithmetic, hence in principle one can use any proof system over $\mathbb{F}_{2^\nu}$ to prove (1).

There are several issues with this attempt:

- First, $\mathbb{F}$ may not be a prime field, in which case there is no natural lift to integers in Step 2; rather, there is a lift to integer polynomials $\mathbb{Z}[X]$.
- Second, even if $\mathbb{F}$ is a prime field, $q = |\mathbb{F}|$ may be large and the linear expression $\langle \pi_q^{-1}(\mathbf{u}), \text{bits}(\mathbf{w}) \rangle$ may overflow the order of $g \in \mathbb{F}_{2^\nu}^*$, which is $2^\nu - 1$. In this case, proving $\langle \pi_q^{-1}(\mathbf{u}), \text{bits}(\mathbf{w}) \rangle = \mu'$ “in the exponent of g ” is unsound.
- Third, even if neither of these problems hold, (1) is, in principle, an expensive constraint to prove: indeed, it is a length $|\text{bits}(\mathbf{w})|$ grand product of, essentially, arbitrary factors from $\mathbb{F}_{2^\nu}$ (typically ≈ 128 bit elements). Grand products can be expensive to prove, and furthermore the fact that the factors are large prevents us from using known sumcheck optimizations such as univariate skips [?] and [Albert: what else?](#).
- Four, when proving (1), typically one ends up with an MLE evaluation claim on the vector $(g^{\pi_q^{-1}(\mathbf{u})_i})_i$ (and another on $\text{bits}(\mathbf{w})$, which is proved with any PCS over $\mathbb{F}_{2^\nu}$). If the verifier checks the claim itself, then it pays a linear cost. If the prover proves this claim, then it is essentially back to square one.

We solve these issues as follows: 1) if $\mathbb{F}$ is a proper extension, at Step 2 we lift to a claim $\langle \pi_{\text{canon}}^{-1}(\mathbf{u}), \text{bits}(\mathbf{w}) \rangle = \mu' \in \mathbb{Z}[X]$ over $\mathbb{Z}[X]$, where $\pi_{\text{canon}} : \mathbb{Z}[X] \rightarrow \mathbb{F}$ is a canonical projection of $\mathbb{Z}[X]$ onto $\mathbb{F}$. Then the verifier sends a random prime q' of $\approx |\mathbb{F}_{2^\nu}^*|/|\text{bits}(\mathbf{w})|$ size (small enough to avoid the issue of overflowing in the exponent), and a random element $\alpha \in \mathbb{F}_{q'}$; then the previous claim is replaced by $\langle \pi_{q',\alpha} \circ \pi_{\text{canon}}^{-1}(\mathbf{u}), \pi_{q',\alpha}(\text{bits}(\mathbf{w})) \rangle = \pi_{q',\alpha}(\mu') \in \mathbb{F}_{q'}$ where $\pi_{q',\alpha}(f(X)) = \pi_{q'}(f)(\alpha) \in \mathbb{F}_{q'}$. Now we re-enter Step 2 with this new claim.

2) If $\mathbb{F}$ is too large of a prime field, we follow a similar approach as above: once we have the integer claim $\langle \pi_q^{-1}(\mathbf{u}), \text{bits}(\mathbf{w}) \rangle = \mu'$, the verifier samples a random “small” prime q' , reduce the claim modulo q' , and then re-enter Step 2 with this new claim.

[Albert: remove?](#) We emphasize again that we manage to instantiate BitZ with 128 bits of security, using $\mathbb{F}_{2^\nu} = \mathbb{F}_{2^{128}}$, while having no limitation on the size of the ring R we start with.

3) The main idea is to split (1) into $\approx \sqrt{|\text{bits}(\mathbf{w})|}$ grand products of the same form as (1), so that, even though in total there are $|\text{bits}(\mathbf{w})|$ factors among all of them, these factors can only take $\sqrt{|\text{bits}(\mathbf{w})|}$ values. More concretely, we assume $\mathbf{u} = \mathbf{u}^{(1)} \otimes \mathbf{u}^{(2)}$ tensor-decomposes into $\approx \sqrt{|\text{bits}(\mathbf{w})|}$ sized factors (this is typically the case since $\mathbf{v}$ is usually an “eq” vector), and then, assuming the notation of Step 3 above, we aim to prove $\langle \pi_q^{-1}(\mathbf{u}^{(1)}), \text{bits}(\mathbf{w})_{*j} \rangle = \mu'_j$ in $\mathbb{Z}$ for all $j \in [|\mathbf{u}^{(2)}|]$, such that $\sum_j \pi_q(\mu'_j) \cdot \mathbf{u}_j^{(2)} = \mu$, where we index $\text{bits}(\mathbf{w})$ with indices $i \in [|\mathbf{u}^{(1)}|], j \in [|\mathbf{u}^{(2)}|]$. We then set up a grand product for each of these claims, obtaining

$$g^{\langle \pi_q^{-1}(\mathbf{u}^{(1)}), \text{bits}(\mathbf{w})_{*j} \rangle} = \prod_i (g^{\pi_q^{-1}(\mathbf{u}_i^{(1)})} - 1) \cdot \text{bits}(\mathbf{w})_{ij} + 1 = g^{\mu'_j} \quad \text{for all } j. \quad (2)$$

There are $|\text{bits}(\mathbf{w})|$ factors in total, but only $\approx \sqrt{|\text{bits}(\mathbf{w})|}$ different values these factors can take.

We say (2) is a *low entropy batched grand product*. To prove (2) we use a batched GKR protocol with the following optimization, among others: because of the low-entropy phenomenon, we build several small lookup tables of size $O(\sqrt{|\text{bits}(\mathbf{w})|})$ (of a few MB at most) which are used during the most expensive sumcheck rounds of the GKR protocol. Concretely, the last sumcheck (the most expensive one), uses 3 lookup tables to compute the first three messages of the prover; the second last sumcheck does the same for the first two prover messages (with two tables); and the third last sumcheck does the same for the first prover message (with one table).

The above solution also takes care of Issue 4: the batched GKR protocol outputs MLE evaluation claims on $(g^{\pi_q^{-1}(\mathbf{u}^{(1)})_i})_i$ (and $\text{bits}(\mathbf{w})$), which are dealt with the PCS over $\mathbb{F}_{2^\nu}$. This is an $O(\sqrt{|\text{bits}(\mathbf{w})|})$ vector, and the verifier can check this claim by itself cheaply.

Virtual $\mathbb{F}_2$ -linear transforms, NP-complete dually linear relations, and arithmetization of SHA-256 Observe that, once we have a standard linear claim on $\text{bits}(\mathbf{w})$ in $\mathbb{F}_{2^\nu}$, we could replace $\text{bits}(\mathbf{w})$ by a linear expression on some other committed vector $\mathbf{f}$ over $\mathbb{F}_{2^\nu}$, so that $\text{bits}(\mathbf{w}) = M \cdot \mathbf{f}$ for some matrix M . The final linear claim can then be replaced by a linear claim for $\mathbf{f}$. Hence, we can, instead of committing to $\text{bits}(\mathbf{w})$, commit to $\mathbf{f}$ and treat $\mathbf{w}$ and $\text{bits}(\mathbf{w})$ as virtual vectors. See ?? for details.

This allows BitZ to prove polynomial constraints over R on vectors $\mathbf{w}$ that are a virtual $\mathbb{F}_2$ -linear combination of a committed vector $\mathbf{f} \in \mathbb{F}_{2^\nu}^{\ell'}$. For example, BitZ can efficiently prove the following R1CS-like relation:

$$\mathcal{R}_{\text{R1CS-bits-virtual}}(\mathbb{Z}) := \left\{ \begin{array}{l} \mathbf{x} = (\mathbf{x}, A, B, C, M) \in \mathbb{F}_2^k \times (\mathbb{Z}^{m \times \ell})^3 \times \mathbb{F}_2^{\ell \times (k + \ell')} \\ \mathbf{w} = \mathbf{f} \in \{0, 1\}^{\ell'} \subseteq \mathbb{F}_2^{\ell'} \end{array} \mid \begin{array}{l} (A \cdot \mathbf{w}) \circ (B \cdot \mathbf{w}) = C \cdot \mathbf{w} \text{ in } \mathbb{Z} \\ \mathbf{w} = \pi_2^{-1}(M \cdot (\mathbf{z}, \mathbf{f})) \end{array} \right\},$$

where π_2 denotes reduction modulo 2. A particularly interesting subclass of this relation is that obtained by setting $A = B = \mathbf{0}$, in which case the resulting relation, which we call the *CM relation*, degenerates to linear constraints over $\mathbb{Z}$ and over $\mathbb{F}_2$, i.e. a “dually linear” constraint.

In Remark 3.3 we show that the CM relation is NP-complete [Albert: Or NP-hard?](#). In ?? we arithmetize SHA-256 as an instance of the CM relation, with *** smaller witnesses than Binius64 [Bin26] and Flock [BRW26] [Albert: add numbers. Mention that we don't prove hashes faster though \(but we have better proof size and memory so...\)](#)

1.2 Related work

Proof systems over integers and other rings. In [CHA24], Campanelli and Hall-Andersen propose proving polynomial constraints over integers, by projecting them onto a random prime field (sampled by the verifier), a technique which the authors call *fingerprinting*. Subsequent works have adopted this idea [GWV⁺25, SSPP26, AGK⁺26, CXNB26]. [CHA24] also proposes a hidden-order group based PCS suitable for their PIOP, similar to the ones in [BFS19, BHR⁺21]. The PCS is not implemented and so we do not compare against it.

The Zinc [GWV⁺25] work adopts *fingerprinting* and designs a hash-based PCS over the rationals $\mathbb{Q}$, suitable for the fingerprinting approach, providing the first relatively practical proof system over the integers. [GWV⁺25] addresses subtle technical issues that arise when aiming to commit to *integer* vectors with error correcting codes (namely, malicious provers can potentially use rational numbers that are very close to being integers of the target magnitude).

The Zinc+ [AGK⁺26] work extends [GWV⁺25] in two directions: 1) It proposes a constraint system that includes polynomial constraints simultaneously over many types of rings¹ and fields,

¹In the terminology of [?], the constraints are over *polynomial rings with ideal membership constraints*, for certain types of ideals (generated by a monic and non-constant polynomial, or zero). However this is equivalent to saying the constraints are over different types of rings (within a certain class)

and provides a method for constructing PIOP’s for those. This method extends the fingerprinting approach by not only projecting $\mathbb{Z}$ onto a random prime field, but projecting all sorts of rings and fields onto a random prime field. 2) It improves the soundness analysis of the PCS in [GWV⁺25], plus it proposes a new linear code over $\mathbb{Z}$ or $\mathbb{Q}$, called Integer Pseudo Reed-Solomon (IPRS) code, obtained by “lifting” (in a suitable way) an FFT encoding algorithm over a small finite field, to the integers. **Albert: compare BitZ to Zinc+?**

In [SSPP26] the authors propose Qaren, a compiler that takes any PCS over a finite field of sufficiently large characteristic and creates a PCS for integer polynomials (or, more precisely, rational polynomials close to the target integer polynomials, see the overview of [GWV⁺25] above). The authors instantiate their compiler to obtain the first transparent and fully succinct PCS over integers/rationals. The prior work [GWV⁺25, AGK⁺26] is not succinct on the bit-size of the integers being committed. No implementation or performance reports is provided in [SSPP26].

The concurrent work Limber [CXNB26] proposes a new compiler that transforms any PCS over fields of sufficiently large characteristic q (in this case, q is $\Omega(\lambda^2 \log(N))$, N being the witness size, while in [SSPP26] it is $\Omega(2^\lambda)$). Additionally Limber compiles into strictly integer PCS, and does not present the issue that a malicious prover could use rationals that are “almost integers”. Limber is instantiated in practice with the Hyrax PCS (in contrast to [GWV⁺25, AGK⁺26] which use a hash based PCS). **Albert: Wait for the implementation to come out so we can discuss its performance against prior work better.** In [?] we compare BitZ against Limber, obtaining *** **Albert: fill in.**

Proof systems over binary fields. Hash-based SNARKs over binary fields were pioneered by the line of work of Ben-Sasson et al. [BSBHR18a, BSBHR18b, ?, ?], but did not see widespread adoption until the introduction of *packed* polynomial commitment schemes by Diamond and Posen [DP23, DP24]; see also [Bin26]. Such schemes allow one to commit to a bit-valued witness over an extension field without incurring embedding overhead, while still providing evaluation proofs for its original “uncompressed” representation. The *ring-switching* technique of Diamond and Posen [DP24] is generic, enabling the use of codes over larger alphabets without introducing representational redundancy.² A particularly attractive choice is Reed–Solomon codes over $GF(2^k)$ [Bin26, ?], which offer an excellent trade-off between distance and encoding cost, although several works [BCF⁺24, ?, ?] explore this design space using more efficient codes.

At the time of writing, two further works push the performance envelope of binary proof systems. Bünz, Rothblum, and Wang [BRW26] combine [Bin26] with the batched R1CS strategy inspired by [?], achieving a striking proving throughput of several hundred thousand hashes per second on an M4 Max CPU. BinarySpartan [?], in turn, incorporates the optimizations of [Bin26]-Flock into Spartan, resulting in a proof system with again high hash-proving throughput.

Commitments in binary fields with openings in rings R of characteristic different than 2.

To our knowledge this primitive is studied only in Soukhanov’s note [Sou26] and in the present work. The note [Sou26] proposes a scheme for committing over the binary field $\mathbb{F}_{2^{127}}$ and then executing a PCS opening over $\mathbb{F}_{2^{127}-1}$ (a prime field), by running the PCS in the exponent of a generator of $\mathbb{F}_{2^{127}}^*$. The approach, substantially different to ours, is limited to $R = \mathbb{F}_{2^{127}-1}$ (or analogous configurations where $2^n - 1$ is prime) and does not seem to be able to leverage our

²Alternative approaches based on Galois conjugates have been proposed thereafter [?, ?]. Remarkably, the Galois approach is present in prior work [BSBHR18b, Appendix F], although not explicitly framed in full generality.

aforementioned “low-entropy” based optimizations. The note does not provide an implementation nor performance reports. We discuss this further in [Section 3.3.1](#).

Qaren and Limber [SSPP26, CXNB26] provide a related primitive when $R = \mathbb{Z}$, but can only support commitments over fields of relatively large characteristic (compared to 2).

Albert: more? Swirl, Frepack?

1.3 Acknowledgements

We thank Lev Soukhanov and Giacomo Fenzi for insightful and helpful discussions. Albert: More if not authors: Marcin

Albert: World Foundation

Usage and non-usage of AI This document was essentially written without the assistance of AI. It was only used to identify typos, make refactorings, and for the cost accounting in [Section 6](#).

We used AI in our implementation and experiments, as explained in ??.

Albert: Ideas for the “Our contributions”section: 1) PCS for $\mathbf{f} \in S^{\ell_0}$ and constraints $\langle \psi(\mathbf{f}), \mathbf{v} \rangle = \mu$ in R . We commit to the bits of $\mathbf{f}$ over a binary fields (in packed form), unlocking the boons of binary fields. 2) Technique for making structured (low entropy) grand products efficient. 3)

2 New technical overview

1. Start with PCS with $\mathbf{f} \in \mathbb{Z}^{\ell_0}$ and R is a prime field.
2. Then low entropy grand product.
3. Generalizations of PCS. $\mathbf{f} \in S^{\ell}$ and R an extension field (e.g. Babybear proving) or a f.g. ring (for the sake of lattices).
4. General proof system.
5. Virtual $\mathbb{F}_2$ -linear transforms in $\mathbb{Bit}\mathbb{Z}$ and NP-complete dually linear relations.

3 Technical overview

Albert: Need to discuss somewhere that range checks are free

Albert: discuss somewhere (technical overview) the succinctness/non-succinctness of verifier

3.1 A simple version of $\mathbb{Bit}\mathbb{Z}$

We explain how $\mathbb{Bit}\mathbb{Z}$ works. Throughout the exposition we make several simplifying assumptions, for the sake of clarity. Then we address the general cases in [Section 3.4](#).

Let R be a ring, for example R may be the integers $\mathbb{Z}$, a prime field, a lattice ring, a finite field extension, etc. For simplicity, for now let R be either $\mathbb{Z}$ or a finite field $\mathbb{K}$ (possibly an extension field). In [Section 3.4](#) we treat the general case. If $R = \mathbb{Z}$, then we assume a public parameter B specifying the maximum bit-size of a witness element.

In $\text{Bit}\mathbb{Z}$, given a public vector $\mathbf{x} = \mathbf{x} \in R^k$, the prover wants to prove it knows $\mathbf{w} = \mathbf{w} \in R^n$ such that

$$\mathbf{p}_i(\mathbf{x}, \mathbf{w}) = 0 \quad \text{in } R \quad \text{for all } i \in [m], \quad (3)$$

where $\mathbf{p} = (\mathbf{p}_i)_{i \in [m]}$ is a collection of polynomials over R . For example, in an *R1CS constraint*,

$$\mathbf{p}_i(\mathbf{x}, \mathbf{w}) = (A \cdot (\mathbf{x}, \mathbf{w}))_i \cdot (B \cdot (\mathbf{x}, \mathbf{w}))_i - (C \cdot (\mathbf{x}, \mathbf{w}))_i$$

where A, B, C are $m \times (k + n)$ matrices over R .

Step 1: Commitment in $\mathbb{F}_{2^\nu}$. The prover creates a vector $\text{bits}(\mathbf{w}) \in \{0, 1\}^\ell \subseteq \mathbb{Z}^\ell$ of integer bits, according to some public specification of how to represent elements $r \in R$ as a linear combination of bits, i.e. $r = L(\text{bits}(r))$ for $L : \{0, 1\}^{\ell/n} \rightarrow R$ a linear map. For example, if $R = \mathbb{Z}$, then $L(b_1, \dots, b_B) = \sum_{i \in [B]} 2^{i-1} \cdot b_i$.

Then the prover looks at $\text{bits}(\mathbf{w})$ as a vector from $\mathbb{F}_2^\ell$, encodes it with a Reed-Solomon code (or any other code) over $\mathbb{F}_{2^\nu}$, and sends a Merkle root of the resulting codeword to the verifier.

When encoding $\text{bits}(\mathbf{w})$, every ν bits $b_1, \dots, b_\nu$ are packed as a single field element $\sum_{i \in [\mathbb{F}_{2^\nu}]} X^{i-1} \cdot b_i$ from $\mathbb{F}_{2^\nu}$, following [DP23] (one can use a different $\mathbb{F}_2$ -base than the monomial base $(X^i)_i$).

Albert: We may want to mention that we do an OOD sample if we are working beyond the UDR

Step 2: Projection to a finite field $\mathbb{F}$. **Albert:** If necessary, the verifier samples a random map π from R to a finite field $\mathbb{F}$ of $\approx \lambda$ bit size (not necessarily a prime field). Then the claim (3) is replaced by

$$\pi(\mathbf{p}_i)(\pi(\mathbf{x}), \pi(\mathbf{w})) = 0 \quad \text{in } \mathbb{F} \quad \text{for all } i. \quad (4)$$

The form of the samplable maps π depends on the type of ring R we started with. For example, if $R = \mathbb{Z}$, then π is reduction modulo a random prime q , following [CHA24]. If $R = \mathbb{K}$ is a finite field (not necessarily a prime field), then π is always the identity. In Section 3.4 we discuss other rings and other projections.

This step can be skipped if one already has an efficient PIOP for polynomial constraints over R ; for example, if R is a finite field of reasonable size.

If R is a finite field of very large size, one may prefer to lift the initial polynomial constraints to $\mathbb{Z}$ first (or $\mathbb{Z}[X]$ if R is field extension), and then project onto a random prime field $\mathbb{F}$ of bit-size $\approx \lambda$ during this step. This can compress the bit-size of the witness in the next steps by a very large factor.

Step 3: PIOP over $\mathbb{F}$. As in [CHA24, AGK⁺26, CXNB26], a PIOP is run over $\mathbb{F}$, reducing (4) to a linear claim over $\mathbb{F}$ or a field extension $\mathbb{F}_{\text{ext}(e)}$ of $\mathbb{F}$. The claim is of the form

$$\langle \mathbf{v}, \pi(\mathbf{w}) \rangle = \mu \quad \text{in } \mathbb{F}_{\text{ext}(e)}, \quad (5)$$

for some $\mu \in \mathbb{F}_{\text{ext}(e)}$, $\mathbf{v} \in (\mathbb{F}_{\text{ext}(e)})^\ell$.

In general, if $\mathbb{F}$ is a prime field of $\approx \lambda$ bits (e.g. if we started with $R = \mathbb{Z}$), then $\mathbb{F}_{\text{ext}} = \mathbb{F}$. The field $\mathbb{F}_{\text{ext}}$ may be a proper extension if, for example, we started with $\mathbb{R} = \mathbb{K}$ a small finite field (in which case $\pi = \text{Id}$ and $\mathbb{K} = \mathbb{F}$), and the PIOP had to use challenges in an extension of $\mathbb{F}$ to guarantee soundness.

We assume $\mathbf{v}$ admits a tensor decomposition $\mathbf{v} = \mathbf{v}^{(1)} \otimes \mathbf{v}^{(2)} \in \mathbb{F}_q^{n_1 \cdot n_2}$. This is typically the case since $\mathbf{v}$ is usually an “eq” vector. This will be leveraged crucially, for efficiency purposes, at the later Step 5.2.

For simplicity, we assume $\mathbb{F}_{\text{ext}} = \mathbb{F}$ is a prime field. We discuss the general case in [Section 3.4](#).

Step 4: Bitification of the linear claim. The claim (5) is replaced by

$$\langle \mathbf{u}, \pi(\text{bits}(\mathbf{w})) \rangle = \mu \quad \text{in } \mathbb{F}, \quad (6)$$

for a new $\mathbf{u} \in \mathbb{F}^\ell$. This transformation is done using the linear bit representation L mentioned in Step 1. Concretely, representing L as matrix multiplication and indexing vectors appropriately, $\langle \mathbf{v}, \pi(\mathbf{w}) \rangle = \langle \mathbf{v}, \pi(L \cdot \text{bits}(\mathbf{w})) \rangle = \langle \pi(L)^\top \cdot \mathbf{v}, \pi(\text{bits}(\mathbf{w})) \rangle$.

The new vector $\mathbf{u}$ still admits tensor decompositions $\mathbf{u} = \mathbf{u}^{(1)} \otimes \mathbf{u}^{(2)}$ with, say $\mathbf{u}^{(1)} \in \mathbb{F}^{\ell_1}$ and $\mathbf{u}^{(2)} \in \mathbb{F}^{\ell_2}$. Indeed, one can see that $L = \text{Id}_{n \times n} \otimes L'$ for some vector $L' \in \mathbb{F}^{\ell/n}$ and Id the identity matrix of dimension n . Then for example $\mathbf{u}^{(1)} = \mathbf{v}^{(1)}$ and $\mathbf{u}^{(2)} = \mathbf{v}^{(2)} \otimes \pi(L')$, see [Remark 5.14](#) for details [Albert: Review. Maybe I could prove it here](#). Other choices of decompositions may lead to better efficiency later.

We assume that $|\mathbb{F}| < (|\mathbb{F}_{2^\nu}| - 1)/\ell_1$, a condition needed for soundness later in Step 5.2. Below in [Remark 3.1](#) we explain how to handle larger $\mathbb{F}$. We note that this is a very constraining assumption because, when for example aiming to use $\mathbb{F}_{2^\nu} = \mathbb{F}_{2^{128}}$ (a highly performant binary field) and 128 bits of security, one can easily have $\mathbb{F}$ with $|\mathbb{F}| \geq (|\mathbb{F}_{2^\nu}| - 1)/\ell_1$.

Step 5: BitZ Polynomial Commitment Scheme (PCS). Our BitZ PCS enables efficiently proving (6) with $\text{bits}(\mathbf{w})$ being committed over $\mathbb{F}_{2^\nu}$ (in packed form). This is done as follows:

Step 5.0: reducing the prime modulus if necessary. This step triggers if the condition $|\mathbb{F}| < (|\mathbb{F}_{2^\nu}| - 1)/\ell_1$ is not met. We explain this step below in [Remark 3.1](#) and assume for now that the condition holds.

When $\mathbb{F}_{\text{ext}}$ is a proper field extension, this step always triggers.

Step 5.1: Transformation of (6) into a batch of low-magnitude integer claim. The prover organizes $\text{bits}(\mathbf{w})$ into a $\ell_1 \times \ell_2$ matrix with columns $\text{bits}(\mathbf{w})_{*j} \in \{0, 1\}^{\ell_1}$ ($j \in [\ell_2]$). Then it publishes $\eta_1, \dots, \eta_{\ell_2} \in \mathbb{Z}$ defined as $\eta_j = \langle \pi^{-1}(\mathbf{u}^{(1)}), \text{bits}(\mathbf{w})_{*j} \rangle$ in $\mathbb{Z}$, for all $j \in [\ell_2]$. Then the verifier checks that [Albert: Add a remark about the possibility of proving the below with a sumcheck](#)

$$\sum_{j \in [\ell_2]} \pi(\eta_j) \cdot \mathbf{u}_j^{(2)} = \mu \quad \text{in } \mathbb{F},$$

and (6) is replaced by the integer claim

$$\langle \pi^{-1}(\mathbf{u}^{(1)}), \text{bits}(\mathbf{w})_{*j} \rangle = \eta_j \quad \text{in } \mathbb{Z} \quad \text{for all } j \in [\ell_2], \quad (7)$$

where π^{-1} is the canonical representation of elements from $\mathbb{F}$ as integers in the range $[0..|\mathbb{F}| - 1]$.

Step 5.2: Exponentiation in $\mathbb{F}_{2^\nu}$ and grand product in $\mathbb{F}_{2^\nu}$. Since we assumed $|\mathbb{F}| < (|\mathbb{F}_{2^\nu}| - 1)/\ell_1$, (7) is true if and only if

$$\begin{aligned} g^{\eta_j} &= g^{\langle \pi^{-1}(\mathbf{u}^{(1)}), \text{bits}(\mathbf{w})_{*j} \rangle} \\ &= g^{\sum_{i \in [\ell_1]} \pi^{-1}(\mathbf{u}_i^{(1)}) \cdot \text{bits}(\mathbf{w})_{ij}} = \prod_{i \in [\ell_1]} g^{\pi^{-1}(\mathbf{u}_i^{(1)}) \cdot \text{bits}(\mathbf{w})_{ij}} \\ &= \prod_{i \in [\ell_1]} \left(\left(g^{\pi^{-1}(\mathbf{u}_i^{(1)})} - 1 \right) \cdot \text{bits}(\mathbf{w})_{ij} + 1 \right) \quad \text{in } \mathbb{F}_{2^\nu} \quad \text{for all } j \in [\ell_2], \end{aligned} \tag{8}$$

Xander: superscript missing $\mathbf{u} = \mathbf{u}^{(1)}$ **Albert:** Yes, thanks! where g is a generator of the multiplicative subgroup of $\mathbb{F}_{2^\nu}$, and in the middle expression, $\text{bits}(\mathbf{w})_{ij}$ denotes the i, j -th entry (a bit) in $\text{bits}(\mathbf{w})$ seen as an element of $\mathbb{F}_{2^\nu}$.

The prover and verifier run a batched grand product PIOP over the binary field $\mathbb{F}_{2^\nu}$ for the claim (8), obtaining as a result a new claim of the form

$$\langle \mathbf{z}, \text{bits}(\mathbf{w})_{*j} \rangle = c_j \quad \text{in } \mathbb{F}_{2^\nu} \quad \text{for all } j \in [\ell_2], \tag{9}$$

for some $\mathbf{z} \in \mathbb{F}_{2^\nu}^{\ell_1}$ and $c_j \in \mathbb{F}_{2^\nu}$. In general, these claims can be batched to

$$\langle \mathbf{z}', \text{bits}(\mathbf{w}) \rangle = c \quad \text{in } \mathbb{F}_{2^\nu} \tag{10}$$

for $\mathbf{z}' \in \mathbb{F}_{2^\nu}^{\ell_1}$, $c \in \mathbb{F}_{2^\nu}$ with, for example, $\mathbf{z}' = (\alpha_1, \dots, \alpha_{\ell_2}) \otimes \mathbf{z}$ for $\alpha_1, \dots, \alpha_{\ell_2}$ verifier's randomness, and $c = \sum_{j \in [\ell_2]} \alpha_j \cdot c_j$. However, in most instantiations this step may not be necessary, especially if using an interleaved code to commit to $\text{bits}(\mathbf{w})$ **Albert:** In the GKR grand product section of the main body I may need to mention batching?.

This exponentiation trick is inspired by [Bin26, Sou26], where the same idea is used in different contexts, see Section 3.3.1.

This is by far the most expensive step of $\mathbb{B}it\mathbb{Z}$. Our grand product PIOP is a simple GKR protocol for grand products, for which we design novel **Albert:** to our knowledge prover optimizations, see Section 3.3 below and Section 6. These optimizations exploit the “low entropy” of the factors $(g^{\pi^{-1}} - 1) \cdot \text{bits}(\mathbf{w})_{ij} + 1$ across $(i, j) \in [\ell_1] \times [\ell_2]$, in the sense that across $\ell = \ell_1 \cdot \ell_2$ factors, there are only $\ell_1 + 1 = O(\sqrt{\ell})$ possible values taken by these factors. This is why it is important to have the vectors $\mathbf{v}$ and $\mathbf{u}$ admit a tensor split into $\approx$ square-root sized factors.

Step 5.3: Proving linear claims over $\mathbb{F}_{2^\nu}$. At this point, since $\text{bits}(\mathbf{w})$ was committed over $\mathbb{F}_{2^\nu}$ (in packed form), one can prove (9) (or the batched version of it) via standard methods such as in [DP23, DP24, BRW26]. Concretely, we use the so-called ring switch from [DP23, DP24] to turn (9) into linear claims about the packed binary field elements we encoded, and then use WHIR [ACFY25] over $\mathbb{F}_{2^\nu}$ to prove these claims. See Section B and ?? for details on ring-switching.

In Section 5.7 we discuss how to efficiently instantiate $\mathbb{B}it\mathbb{Z}$ with $\mathbb{F}_{2^\nu} = \mathbb{F}_{2^{128}}$, and attain 128 bits of security.

Remark 3.1 (Description of Step 5.0). **Albert:** In the actual protocol this step is done after Step 5.1. I think they are equivalent but should unify Assume that $\mathbb{F}_{\text{ext}} = \mathbb{F}$ is a prime field (the general

case is described in [Section 3.4](#)), and that in Step 5.0 the condition $|\mathbb{F}| < (|\mathbb{F}_{2^\nu}| - 1)/\ell_1$ does not hold. Then the prover sends an integer $\mu' \in \mathbb{Z}$ such that

$$\langle \pi^{-1}(\mathbf{u}^{(1)}) \otimes \pi^{-1}(\mathbf{u}^{(2)}), \text{bits}(\mathbf{w}) \rangle = \mu' \quad \text{in } \mathbb{Z},$$

The verifier checks that $\pi(\mu') = \mu$, and responds with a random prime q such that $q < (|\mathbb{F}_{2^\nu}| - 1)/\ell_1$. Then prover and verifier proceed to Step 5.1 after updating $\mathbb{F} \leftarrow \mathbb{F}_{q'}$ the prime field with q' elements, $\pi \leftarrow \pi_{q'}$, $\mu \leftarrow \pi_q(\mu')$, and $\mathbf{u} \leftarrow \pi_q \circ \pi^{-1}(\mathbf{u}^{(1)}) \otimes \pi_q \circ \pi^{-1}(\mathbf{u}^{(2)})$.

Remark 3.2 (Other usages of the “prove-in-the-exponent” approach). There are other projects (e.g. Binius64 [\[Bin26\]](#) and Shoukhanov’s note [\[Sou26\]](#)) that use exponentiation to prove integer (or mod $|\mathbb{F}_{2^\nu}| - 1$) claims in binary fields, none of these (to our knowledge) is used or is usable in a context that results in the low entropy grand products of ours, where our optimizations apply. We discuss this further in [Section 3.3.1](#).

3.1.1 Intuition of soundness proofs

We prove that the above protocol (and the more general version in [Section 3.4](#)) is round-by-round knowledge sound. We next provide the main ideas of why this is the case.

[Albert: todo](#)

3.2 Virtual $\mathbb{F}_2$ -linear transforms in $\mathbb{Bit}\mathbb{Z}$ and NP-complete dually linear relations

Observe that, at the end of the day, Step 5.2 outputs a linear claim of the form³

$$\langle \mathbf{z}, \text{bits}(\mathbf{w}) \rangle = c \quad \text{in } \mathbb{F}_{2^\nu}, \tag{11}$$

for some $\mathbf{z} \in \mathbb{F}_{2^\nu}^\ell$ and $c \in \mathbb{F}_{2^\nu}$. This is a standard linear claim over $\mathbb{F}_{2^\nu}$ for the vector $\text{bits}(\mathbf{w})$ which we committed to previously.

Now, say that we know that $\text{bits}(\mathbf{w})$ is a linear transformation of a vector $\mathbf{f} \in \mathbb{F}_2^{\ell'} \subseteq \mathbb{F}_{2^\nu}^{\ell'}$, so that

$$\text{bits}(\mathbf{w}) = M \cdot \mathbf{f} \quad \text{in } \mathbb{F}_{2^\nu},$$

for some matrix $M \in \mathbb{F}_2^{\ell \times \ell'}$. Then

$$\langle \mathbf{z}, \text{bits}(\mathbf{w}) \rangle = \langle M^\top \cdot \mathbf{z}, \mathbf{f} \rangle = c \quad \text{in } \mathbb{F}_{2^\nu}. \tag{12}$$

Hence, in this scenario, one can commit to $\mathbf{f}$ instead of $\text{bits}(\mathbf{w})$ (again, in packed formed if necessary), replace the claim (12), and prove (12) in Step 5.3 of $\mathbb{Bit}\mathbb{Z}$.

Thus, $\mathbb{Bit}\mathbb{Z}$ can be expanded so as to prove claims where we are given a public vector $\mathbf{x} = \mathbf{x} \in R^k$, and the prover claims it knows $\mathbf{w} = \mathbf{f} \in \mathbb{F}_{2^\nu}^{\ell'}$ such that

$$\begin{cases} \mathbf{p}_i(\mathbf{x}, \mathbf{w}) = 0 & \text{in } R \quad \text{for all } i \in [m], \\ \text{bits}(\mathbf{w}) = \pi_2^{-1}(M \cdot \mathbf{f}) \end{cases} \tag{13}$$

where $\mathbf{p} = (\mathbf{p}_i)_{i \in [m]}$ is a collection of polynomials over R ; and π_2 is reduction modulo 2 [Albert: I may have been loose about when \$\text{bits}\(\mathbf{w}\)\$ is in \$\mathbb{F}_2\$ and when in integers during the technical overview, review.](#)

³We start with the batched version of the claim, but the ideas in this section can be adapted to (9) as well.

A particularly interesting instantiation of this relation is the following R1CS-like relation over the integers, where $\mathbf{w}$ is a vector of bits (hence $\mathbf{w} = \text{bits}(\mathbf{w})$) and $\mathbf{w} = \pi_2^{-1}(M \cdot \mathbf{f})$:

$$\mathcal{R}_{\text{R1CS-bits-virtual}}(\mathbb{Z}) := \left\{ \begin{array}{l} \mathbf{x} = (\mathbf{x}, A, B, C, M) \in \mathbb{F}_2^k \times (\mathbb{Z}^{m \times \ell})^3 \times \mathbb{F}_2^{\ell \times (k + \ell')} \\ \mathbf{w} = \mathbf{f} \in \{0, 1\}^{\ell'} \subseteq \mathbb{F}_2^{\ell'} \end{array} \left| \begin{array}{l} (A \cdot \mathbf{w}) \circ (B \cdot \mathbf{w}) = C \cdot \mathbf{w} \text{ in } \mathbb{Z} \\ \mathbf{w} = \pi_2^{-1}(M \cdot (\mathbf{z}, \mathbf{f})) \end{array} \right. \right\}$$

An interesting subset of $\mathcal{R}_{\text{R1CS-bits-virtual}}(\mathbb{Z})$ is that where A and B are zero matrices. We call the resulting relation the CM relation because of the matrices C and M :

$$\mathcal{R}_{\text{CM}}(\mathbb{Z}) := \left\{ \begin{array}{l} \mathbf{x} = (\mathbf{x}, C, M) \in \mathbb{F}_2^k \times \mathbb{Z}^{m \times \ell} \times \mathbb{F}_2^{\ell \times (k + \ell')} \\ \mathbf{w} = \mathbf{f} \in \{0, 1\}^{\ell'} \subseteq \mathbb{F}_2^{\ell'} \end{array} \left| \begin{array}{l} C \cdot \mathbf{w} = 0 \text{ in } \mathbb{Z} \\ \mathbf{w} = \pi_2^{-1}(M \cdot (\mathbf{x}, \mathbf{f})) \end{array} \right. \right\}$$

In a sense, the CM relation places only linear constraints over its witness (with some constraints happening in $\mathbb{F}_{2^\nu}$ and other in $\mathbb{Z}$). Yet, the CM relation is NP-complete, as shown below in [Remark 3.3](#).

Indeed, in ?? we **arithmetize SHA-256 as a CM instance**. Our arithmetization leads to witnesses of size (in bits) **** **Albert: report comparison with Flock/Binius in witness size. Mention as well the performance comparison.**

Remark 3.3 (The $\mathcal{R}_{\text{CM}}$ relation is NP-complete). Any Boolean circuit can be arithmetized as an instance of $\mathcal{R}_{\text{CM}}$. Indeed, since any boolean circuit can be written as a system of equations using the operations XOR and AND, it suffices to prove that, given $x, y, z \in \{0, 1\}$, we can express both of the following constraints as an instance of $\mathcal{R}_{\text{CM}}$:

$$x \text{ XOR } y = z, \quad x \text{ AND } y = z.$$

Indeed, $x \text{ XOR } y = z$ if and only if $\pi_2^{-1}(\pi_2(x) + \pi_2(y)) = z$, where here $+$ means addition modulo 2. Further, $x \text{ AND } y = z$ if and only if

$$2 \cdot z = x + y - \pi_2^{-1}(\pi_2(x) + \pi_2(y)).$$

One can verify this last equivalence for example by directly inspecting that it holds for the four possible combinations of $x, y \in \{0, 1\}$.

Albert: Todo: Expressiveness result of Remco

3.3 GKR for low-entropy batched grand products

Recall that, in Step 5.2 of $\mathbb{B}\text{it}\mathbb{Z}$ ([Section 3.1](#)) the prover must prove that

$$g^{\langle \pi^{-1}(\mathbf{u}), \text{bits}(\mathbf{w})_{*j} \rangle} = \prod_{i \in [\ell_1]} \left(g^{\pi^{-1}(\mathbf{u}_i)} - 1 \right) \cdot \text{bits}(\mathbf{w})_{ij} + 1 = g^{\eta_j} \quad \text{in } \mathbb{F}_{2^\nu} \quad \text{for all } j \in [\ell_2]. \quad (14)$$

For efficiency purposes, we assume both ℓ_1 and ℓ_2 are $O(\sqrt{\ell})$ (see [Section 5.7](#) for exact parameter instantiations). At a protocol level, we simply prove (14) with a standard GKR protocol for

grand products [?]. Concretely, let $\ell_1 = 2^{r_1}$ and $\ell_2 = 2^{r_2}$. We set up multilinear polynomials $L_0, L_1, \dots, L_{r_1}$ on $r_2 + i$ variables for $i \in [0..r_1]$, respectively, such that

$$L_{r_1}(\mathbf{b}, \mathbf{c}) = \left(g^{\pi^{-1}(\mathbf{u}_i)} - 1 \right) \cdot \text{bits}(\mathbf{w})_{ij} + 1 \quad \text{for all } (\mathbf{b}, \mathbf{c}) \in \{0, 1\}^{r_1} \times \{0, 1\}^{r_2},$$

and, for $i = 0, \dots, r_1 - 1$

$$L_i(\mathbf{b}, \mathbf{c}) = L_{i+1}(\mathbf{b}, 0, \mathbf{c}) \cdot L_{i+1}(\mathbf{b}, 1, \mathbf{c}) \quad \text{for all } \mathbf{b} \in \{0, 1\}^i \times \{0, 1\}^{r_2}. \quad (15)$$

We then use a series of sumchecks to certify Eq. (15) for all $i = 0, \dots, r_1 - 1$, and $L_0(\mathbf{c}) = g^{\eta_{\text{int}}(\mathbf{c})}$ for all $\mathbf{c} \in \{0, 1\}^{r_2}$.

We emphasize again that, at a protocol level, this GKR scheme is completely standard. See Section 6 for all the details.

At the end of the GKR grand product, the prover and verifier are left with MLE evaluation claims of the form:

$$\widetilde{g^{\pi^{-1}(\mathbf{u})}}(\mathbf{r}) = a, \quad \widetilde{\text{bits}(\mathbf{w})_{*j}}(\mathbf{r}) = a_j \quad \text{for all } j \in [\ell_2] \quad (16)$$

for some $\mathbf{r} \in \mathbb{F}_{2^\nu}^{r_1}$, where $\widetilde{}$ denotes MLE extension, and $g^{\pi^{-1}(\mathbf{u})}$ is the vector with entries $g^{\pi^{-1}(\mathbf{u}_i)}$.

Since $\mathbf{u}$ is public, the verifiers checks the first claim of (16) on its own in time $O(\sqrt{\ell})$. This is one, among several, reasons why the tensor split of $\mathbf{u} = \mathbf{u}^{(1)} \otimes \mathbf{u}^{(2)}$ and the splitting the original claim into a batch of ℓ_2 claims (see Step 5.1 of Section 3.1) pays off.

We design novel (to our knowledge [Albert: check](#)) prover optimizations for the GKR grand product (without altering the protocol itself), see [Construction 6.1](#) for details. Namely, we exploit the following property, which is enabled, again, by the tensor split of $\mathbf{u} = \mathbf{u}^{(1)} \otimes \mathbf{u}^{(2)}$ and the splitting the original claim into a batch of ℓ_2 claims:

Low entropy grand products: There are at most $\ell_1 = O(\sqrt{\ell})$ values that $L_{r_1}(\mathbf{b}, \mathbf{c})$ can take on the hypercube of size $\ell_1 \cdot \ell_2 = \ell$.

This property trickles down to different stages of the GKR protocol. For example, the above property implies that $L_{r_1-1}(\mathbf{b}', \mathbf{c})$ as well can take only an $O(\sqrt{\ell})$ amount of values (with worse constants now) across all $(\mathbf{b}', \mathbf{c}) \in \{0, 1\}^{r_1-1} \times \{0, 1\}^{r_2}$. Similarly, the coefficients of the first messages sent by the prover at the last stages of the GKR protocol (which involve the most expensive sumchecks), also benefit from a low entropy property.

This allows the prover to compute small lookup tables that are kept on cache and reused across all values of $\mathbf{c} \in [\ell_2]$. We build different lookup tables used at different stages of the GKR protocol. For example, we use 4 lookup tables in the first round of the last sumcheck of GKR (the most expensive sumcheck, being on $r_1 + r_2$ variables). These tables are used to compute the first prover message of such sumcheck. We then construct 3 more tables to compute the second's prover message, and finally we use one last table to compute the third prover message. Similarly, we build lookup tables for the first rounds of the second and third last sumchecks of the GKR protocol.

3.3.1 Comparison to other “prove-in-the-exponent” usages

[Albert: Missing refs?](#) [Longfellow?](#)

Binius64 For example, [Bin26] multiplies pairs of 64-bit integer witness elements in the exponent of g . This is substantially different from our setting where, when binary fields come into play, we are only interested in proving an integer claim about a linear combination of elements of the form “(public values) · (witness elements)”, i.e. $\langle \mathbf{v}, \mathbf{w} \rangle = \mu$ in $\mathbb{Z}$, with the public values $\mathbf{v}$ admitting a tensor split $\mathbf{v} = \mathbf{v}^{(1)} \otimes \mathbf{v}^{(2)}$ (also BitZ can prove multiplication of all sorts of ring elements, not just 64-bit integers). This simpler setting is what allows us to use the low entropy optimizations designed in this work. Crucially, **the reason why we only need to address this setting is because we start off with constraints and a PIOP over integers/fields/rings (Steps 2 to 4 inclusive), and only ever move to work on binary fields when we need to prove an MLE evaluation of the witness (a linear claim) over a “small” prime field (Step 5).**

[Bin26] on the other hand multiplies 64-bit integer witness in the exponent directly. More precisely, [Bin26] proves the identity $g^{\mathbf{w}_{1i} \cdot \mathbf{w}_{2i}} = g^{2^{64} \cdot \mathbf{w}_{3i}} \cdot g^{\mathbf{w}_{4i}}$ for $\mathbf{w}_1, \dots, \mathbf{w}_4$ vectors of 64-bit integers (together with a small additional constraint to address possible overflow). This is done through a combination of LogUp* [?], a GKR grand product (for relatively small products), and the Frobenius automorphism trick. It does not seem possible to follow a similar optimization strategy to ours.

We emphasize again that BitZ supports proving operations of all kinds of ring elements (small and large integers, field elements, lattice ring elements, extension field elements), as opposed to ?? where multiplication is limited to 64-bit integers.

In ?? we compare the performance of BitZ and Binius64 when multiplying integers. BitZ outperforms Binius64 by [Albert: fill in](#).

The note [Sou26]. In the note [Sou26], Soukhanov proposes a scheme for committing to a vector of bits $\mathbf{w}$ over the binary field $\mathbb{F}_{2^{127}}$, and proving MLE evaluation claims over the prime field $\mathbb{F}_{2^{127}-1}$, for a vector of elements from $\mathbb{F}_{2^{127}-1}$ obtained by packing bits from $\mathbf{w}$ together. To do so, at a high level, the MLE evaluation claims are proved “in the exponent” in the binary field $\mathbb{F}_{2^{127}}$, leveraging that the multiplicative subgroup of $\mathbb{F}_{2^{127}}$ has prime order $|\mathbb{F}_{2^{127}}^*| = 2^{127} - 1$. The proposed approach is substantially different from ours: [Sou26] essentially takes a polylogarithmic PCS over the prime field $\mathbb{F}_{2^{127}-1}$, such as Basefold [ZCF23], WHIR [ACFY25], etc., and executes it “in the exponent of $\mathbb{F}_{2^{127}}$ ”. To do so, it relies on the lookup argument LogUp* [?], grand product arguments (for small products), the Frobenius trick, and additional commitments, for every of the $\log \ell$ rounds.

In its execution, [Sou26] and Binius64 [Bin26] seem to us to be more similar than any of these two to BitZ. Indeed, they both use, mostly, the same collection of subprotocols, in a similar manner. Because of this we expect both approaches to have comparable performance profiles, though we do not validate this hypothesis due to [Sou26] not reporting performance nor providing an implementation.

Conceptually, [Sou26] shares with BitZ the aim of committing to bits in a binary field, while proving MLE claims of those bits in some other structure R . Note that [Sou26] is limited to $R = \mathbb{F}_{2^{127}-1}$, while BitZ can handle essentially all rings and fields R of practical interest, achieving 128 bits of security and using $\mathbb{F}_{2^{128}}$ as binary field. Additionally it is not clear whether [Sou26] can leverage any of our low-entropy oriented optimizations without substantially modifying the approach, incorporating some of our ideas. **Albert: Ask Lev if he has implemented his thing so we can compare. Also ask him if Binius64 is essentially following his method (Logup*, grand product, Frobenius)**

Albert: The comparison of Lev and BitZ right now has several unbacked claims (which I believe are true, but can’t really prove them, especially without an implementation). So probably it’s better

to not push it, remove the unbacked claims, and leave it as it is

3.4 BitZ in full generality

In general, BitZ can prove polynomial constraints over multiple rings R_0 (in the same instance) of the form $R[X]/(h(X))$, where R is either the ring of integers $\mathbb{Z}$ or a finite field (not necessarily prime), and $h(X) \in R[X]$ is either the zero polynomial, or a monic non-constant polynomial. Examples of such rings R_0 include: the integers $\mathbb{Z}$, arbitrary finite fields, and lattice rings of the form $\mathbb{F}_q[X]/(\Phi(X))$ for $\Phi(X)$ a cyclotomic polynomial and q a prime power. More generally, R_0 can be any two-generated ring of zero or prime characteristic.

Concretely, we define a target relation, which we call *Universal Polynomial Satisfiability (UPESAT)*, as follows. Let ℓ be a witness length and k a public input length. Let R be either $\mathbb{Z}$ or a finite field. Let $\mathbf{R} = (R_i)_{i \in [\ell]}$ be a collection of rings of the form $R_i = R[X]/(h_i(X))$ for $h_i(X) \in R[X]$ either zero or a monic non-constant polynomial. For each $i \in [\ell]$, let $\mathbf{p}^{(i)}$ be a vector of n polynomials

$$\mathbf{p}^{(i)} = \left(\mathbf{p}_j^{(i)} \mid j \in [n] \right) \in ((R[X])[Y_1, \dots, Y_{k+\ell}])^n$$

with coefficients in $R[X]$ and on $k+\ell$ variables $Y_1, \dots, Y_{k+\ell}$. Given $a \in R[X]$ and $i \in [\ell]$, we say that $a = 0$ holds in R_i if the image of a under the canonical projection $R[X] \rightarrow R_i$ is zero. Finally, given a degree bound d and a bit-size bound, we let $R_{\leq B}^{\leq d}[X]$ be the subset of $R[X]$ formed by polynomials of degree $< d$ and coefficients in R of bit-size $\leq B$ (see Step 1 of Section 3.1 for a formalization of how we bit-decompose elements from R).

We define our target relation as

$$\mathcal{R}_{\text{UPESAT}}(R_{\leq B}^{\leq d}[X], \mathbf{p}, \mathbf{R}) = \left\{ \begin{array}{l} \mathbf{x} = \mathbf{x} \in \left(R_{\leq B}^{\leq d}[X] \right)^k \\ \mathbf{w} = \mathbf{w} \in \left(R_{\leq B}^{\leq d}[X] \right)^\ell \end{array} \mid \mathbf{p}_j^{(i)}(\mathbf{x}, \mathbf{w}) = 0 \text{ in } R_i \text{ for all } j \in [n], i \in [\ell] \right\}$$

Remark 3.4 (Non prime characteristics or arithmetic modulo arbitrary integers). UPESAT does not allow to work with non-prime and non-zero characteristics m . I.e. it does not support, for example, $\mathbb{Z}/2^n\mathbb{Z}$ (i.e. arithmetic modulo $m = 2^n$). Following [CHA24, GWV⁺25], we arithmetize such characteristics in $\mathbb{Z}$ or $\mathbb{Z}[X]$ via the standard trick of adding a term of the form $+m \cdot w$ to the constraints, for w new witness elements. The correctness and soundness of this step follows from the fact that $a + b = c$ in $\mathbb{Z}/m\mathbb{Z}$ if and only if $a + b = c + m \cdot w$ in $\mathbb{Z}$ for some $w \in \mathbb{Z}$. Similar identities hold for rings of non-prime and non-zero characteristic other than $\mathbb{Z}/m\mathbb{Z}$. See the introduction and technical overviews of [GWV⁺25, CXNB26] for more details.

We note that UPESAT is a subset of the more general UCS relation from [AGK⁺26]. See ?? for a comparison. [Albert: Mention we could allow for different characteristics in the same relation?](#)

Similarly as in Section 3.2, BitZ can also handle a variation of $\mathcal{R}_{\text{UPESAT}}$ where $\text{bits}(\mathbf{w})$ is a $\mathbb{F}_2$ -linear transformation of a vector of bits $\mathbf{f} \in \mathbb{F}_2^{\ell'}$.

Example 3.5. [Albert: I could briefly explain here how to arithmetize the computations in our experiments.](#)

We now explain how to modify the basic version of BitZ in Section 3.1 for the relation $\mathcal{R}_{\text{UPESAT}}$. There are two aspects that need modification:

1. In Step 2 we need to design a method for transforming constraints in R_i to constraints in a finite field $\mathbb{F}$.
2. In Step 3, $\mathbb{F}_{\text{ext}}$ could be a proper extension of a prime field $\mathbb{F}_p$. This occurs if for example $R_i = \mathbb{F}_p$ for all $i \in [\iota]$ (i.e. all our constraints are over the prime field $\mathbb{F}_p$), and p is small (say, 32-64 bits), and so the PIOP in Step 3 needs to work on a field extension of $\mathbb{F}_p$.

Generalized Step 2. Regarding Step 2, we discuss first the case that $\iota = 1$ and $h_1 = (0)$, so that all constraints are over $R[X]$, for R the integers or a finite field, with witnesses belonging to $R_{\leq B}^d[X]$. In this case the verifier samples a map π of one of the following forms: If $R = \mathbb{Z}$ and $d > 0$,

$$\pi : \mathbb{Z}[X] \rightarrow \mathbb{F}_q, \quad g(X) \mapsto \pi_q(g)(\alpha)$$

for a random prime q and a random $\alpha \in \mathbb{F}_q$, where π_q denotes reduction modulo q .

If $R = \mathbb{F}$ is a finite field of characteristic q , then

$$\pi : \mathbb{F}[X] \rightarrow \mathbb{F}', \quad g(X) \mapsto g(\alpha)$$

for $\mathbb{F}'$ possibly a field extension of $\mathbb{F}$ (so that $|\mathbb{F}'| = O(2^\lambda)$) and $\alpha \in \mathbb{F}'$. [Albert: Say a few words of why this is sound \(once I've written the soundness intuition for the basic protocol\)](#)

The general case where not all ideals $(h_i(X))$ are zero is dealt in the same way as in Zinc+ [\[AGK⁺26\]](#). We omit the details here for the sake of brevity. We refer to ?? (Items 3 and 4 of Round 1, and Round 2, of ??) or to the technical overview of [\[AGK⁺26\]](#) [Albert: Should I describe this here?](#).

The case when $\mathbb{F}_{\text{ext}}$ is a proper extension Next we discuss the scenario where $\mathbb{F}_{\text{ext}}$ is a proper field extension at the end of Step 3 of [Section 3.1](#). In this case, Step 4 proceeds in the same way as in the normal case. At the end of Step 4 prover and verifier are left with the claim

$$\langle \mathbf{u}, \pi(\text{bits}(\mathbf{w})) \rangle = \mu \quad \text{in } \mathbb{F}_{\text{ext}},$$

for some $\mathbf{u} \in \mathbb{F}_{\text{ext}}^\ell$ and $\mu \in \mathbb{F}_{\text{ext}}$.

After Step 4, Step 5.0 triggers [Albert: Again, in the main body this step triggers after step 5.1 and is slightly different.](#) Since $\mathbb{F}_{\text{ext}}$ is a finite field, it is of the form $\mathbb{F}_{\text{ext}} = \mathbb{Z}[X]/(q, f(X))$ for an irreducible polynomial $f(X)$ over $\mathbb{F}_q[X]$. Let $\pi_{\text{canon}} : \mathbb{Z}[X] \rightarrow \mathbb{F}_{\text{ext}}$ be the canonical homomorphism sending $g(X) \in \mathbb{Z}[X]$ to its reduction modulo q and modulo $f(X)$.

The prover sends an integer polynomial $\mu' \in \mathbb{Z}[X]$ such that

$$\langle \pi_{\text{canon}}^{-1}(\mathbf{u}^{(1)}) \otimes \pi_{\text{canon}}^{-1}(\mathbf{u}^{(2)}), \text{bits}(\mathbf{u}) \rangle = \mu' \quad \text{in } \mathbb{Z}[X],$$

The verifier checks that $\pi_{\text{canon}}(\mu') = \mu$, and responds with a random prime q' such that $q' < (|\mathbb{F}_{2^\nu}| - 1)/\ell_1$ and with a random prime field element $\alpha \in \mathbb{F}_{q'}$. Then prover and verifier proceed to Step 5.1 after updating $\mathbb{F}_{\text{ext}} \leftarrow \mathbb{F}_{q'}$, $\pi \leftarrow \pi_{q', \alpha}$, $\mu \leftarrow \pi_{q', \alpha}(\mu') \in \mathbb{F}_{q'}$, and

$$\mathbf{u} \leftarrow \pi_{q', \alpha} \circ \pi_{\text{canon}}^{-1}(\mathbf{u}^{(1)}) \otimes \pi_{q', \alpha} \circ \pi_{\text{canon}}^{-1}(\mathbf{u}^{(2)}),$$

where $\pi_{q', \alpha} : \mathbb{Z}[X] \rightarrow \mathbb{F}_{q'}$ is defined as

$$\pi_{q', \alpha}(g(X)) = \pi_{q'}(g)(\alpha) \in \mathbb{F}_{q'}.$$

Albert: Again, say a few words of why this is sound (once I've written the soundness intuition for the basic protocol)

One can, instead of sampling morphisms of the form $\pi_{q',\alpha}$, sample other types of $\mathbb{Z}$ -module homomorphisms $\pi' : \mathbb{Z}^{\deg(\mathbb{F}_{\text{ext}})}[X]$ to a prime field. See ???. This can modestly improve the soundness errors at the expense of increasing the communication complexity. Albert: It may be a good idea to present the protocols as parametrized by a family of module homomorphisms where we sample from. Right now the protocols have the fixed family above. Let's see if there's time

The rest of BitZ proceeds exactly as in Section 3.1.

Albert: To discuss somewhere: this technique allows us to do things such as: 1) start with a constraint in Babybear, do the PIOP in an extension, and then open with BitZ, 2) run PIOP's with primes of size, say ≥ 128 bits, and then change the prime to a 108 bit prime to fit in the exponent of $\mathbb{F}_{2^\nu}^*$ without overflowing

Albert: Thoughts: We can handle arbitrary linear weights in our claims, of arbitrary bit-size and in arbitrary extension degrees, and still use a smallish binary field like GHASH during our protocol (not obvious at all because "proving in the exponent in principle only allows for using $\lesssim 90$ -ish integer bits. Albert: This is a good story for the technical overview

Scratch from another section. Delete? Albert: These two examples would be illustrative in the intro and TO To fix intuition, $\deg(\mathbb{K})$ and q are determined by the field or ring a PIOP runs on. For example, in our full proving system in ??, constraints can be (among others) over $\mathbb{Z}$ or $\mathbb{Z}[X]$, then they are projected to a random prime field $\mathbb{F}_q$, a PIOP is ran on them, and an MLE claim over $\mathbb{F}_p$ is produced. Then Construction 5.13 is used to prove those claims. The size of q is set so that the PIOP achieves λ bits of security. A very conservative choice in this setting is $q \leq 2^{256}$, but also $q \leq 2^{128}$ is feasible if $\lambda = 100$ or one is willing to use grinding during the PIOP.

Another setting is to have constraints over a finite field $\mathbb{K}$ such as a degree 4 or 5 extension of the Babybear field, or a degree 2 or 3 extension of the Goldilocks field

4 Preliminaries

General notation Given an integer $n \geq 1$, we write $[n]$ to denote the set $\{1, \dots, n\}$. Given integers $m \leq n$, we let $[m..n]$ denote the set of integers $\{m, m+1, \dots, n\}$.

We use $\circ$ to denote composition of maps, so that for maps $f : S_1 \rightarrow S_2$ and $g : S_2 \rightarrow S_3$, we have $f \circ g : S_1 \rightarrow S_3$ defined as $f \circ g(x) = f(g(x))$.

Any map $f : S_1 \rightarrow S_2$ applies entry-wise to vectors from S_1^n , for any n , and similarly for matrices. A map f also applies coefficient-wise to polynomials with coefficients in S_1 . A multiplication of a vector or a matrix by a scalar means that we multiply each entry of the vector or matrix by that scalar.

We denote vectors with boldface letters, e.g. $\mathbf{u}, \mathbf{v}$. By $\mathbf{u}_i$ we denote the i -th position of $\mathbf{u}$. Sometimes we index the entries of a vector $\mathbf{u}$ with elements of a boolean hypercube $\{0, 1\}^t$.

We look at vectors both as row vectors and column vectors. When performing matrix-vector multiplications, it will always be clear which one we intend. Given a ring R , a matrix $M \in R^{n \times m}$ and vectors $\mathbf{u} \in R^m$, $\mathbf{v} \in R^n$, one has

$$\langle M \cdot \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{u}, M^\top \cdot \mathbf{v} \rangle, \quad (17)$$

where $M^\top$ denotes transposition and $\langle \cdot, \cdot \rangle$ denotes inner product.

4.1 Algebra

Concrete fields and rings We write $\mathbb{F}_q$ for the finite field with q elements (where q is a prime or a prime power), $\mathbb{Q}$ for the field of rational numbers, and $\mathbb{Z}$ for the ring of integers. For an integer $n \geq 1$, $\mathbb{Z}_n$ denotes the ring of integers modulo n , with the convention $\mathbb{Z}_0 := \mathbb{Z}$. When n is prime, $\mathbb{Z}_n = \mathbb{F}_n$. We let

$$\pi_n : \mathbb{Z} \rightarrow \mathbb{Z}_n$$

denote reduction modulo n . We fix the inverse $\pi_n^{-1} : \mathbb{Z}_n \rightarrow [0..n-1] \subseteq \mathbb{Z}$ mapping each element a in $\mathbb{Z}_n$ to the integer in the range $[0..n-1]$ representing a .

Rings in general All rings in this work are associative, commutative, have a multiplicative identity. Given a ring R , we write $R[X_1, \dots, X_n]$ for the ring of polynomials with coefficients in R and variables $X_1, \dots, X_n$ (not necessarily multilinear).

The *characteristic* of a ring R , denoted $\text{char}(R)$, is the smallest $c \geq 1$, if it exists, such that $1 + \dots + 1 = 0$ in R . Otherwise, if $1 + \dots + 1 \neq 0$ for all c , the characteristic is zero. By $[0..n]_R$ we denote all elements in R of the form $1 + \dots + 1$ for $i = 0, \dots, n$. For every ring R we have a canonical embedding

$$\mathbb{Z}_{\text{char}(R)} \subseteq R$$

of the ring of integers modulo $\text{char}(R)$ into R . Namely, the set $\{n \cdot 1 \mid n \in \mathbb{Z}\} \subseteq R$ for 1 the identity element of S , forms the ring $\mathbb{Z}_{\text{char}(R)}$. Further, $\pi_{\text{char}(R)}$ induces a natural homomorphism (not necessarily surjective)

$$\pi_{\text{char}(R)} : \mathbb{Z} \rightarrow R \tag{18}$$

sending $n \in \mathbb{Z}$ to $n \cdot 1$ for 1 the identity of R .

Given a set or vector of integers $\mathbf{w} \in \mathbb{Z}^m$, we define

$$\mathbf{w}_R = \pi_{\text{char}(R)}(\mathbf{w}). \tag{19}$$

Finitely generated rings Further, all rings in this work are *finitely generated* as rings. Equivalently, every ring R in this work is isomorphic to a quotient

$$R = \mathbb{Z}[X_1, \dots, X_k]/I$$

of a polynomial ring over $\mathbb{Z}$ by an ideal I . This condition is satisfied by, for example:

1. All finite fields and the rings $\mathbb{Z}_n$;
2. $\mathbb{Z}$, and the polynomial rings $\mathbb{Z}[X]$ and $\mathbb{F}_q[X]$
3. Arbitrary quotients of the previous rings, such as $\mathbb{Z}_n[X]/(h(X))$ for an arbitrary polynomial $h(X)$ and an arbitrary integer $n \geq 0$, which includes lattice rings.

It is not satisfied by, for example, $\mathbb{Q}$ and $\mathbb{Q}[X]$, which are not finitely generated as rings. Some constructions in this work impose further requirements on the rings they use. We state these when needed.

Given a finitely generated ring R with fixed generator set $\gamma_1, \dots, \gamma_k$, we denote $R_{\leq B}$ the set of elements of R of the form $h = \sum_{i \in [d]} h_i \cdot \gamma_i$ for

$$h_i \in [0..2^B - 1]_S \subseteq R$$

, i.e. $R_{\leq B}$ is the set of ring elements that can be expressed as a linear combination of the generators $\gamma_1, \dots, \gamma_d$ where the coefficients can be obtained by adding 1 together (in R) less than 2^B times.

Albert: Not used? Given a ring R , n variables $X_1, \dots, X_n$, and a bit-size bound $B \geq 1$, we define

$$R_{\leq B}[X_1, \dots, X_n]$$

as the subset of $R[X_1, \dots, X_n]$ formed by polynomials all whose coefficients are in the set $[0..2^B - 1]_R$.

Albert: Needed? A subset $S \subseteq R$ of a ring R is called *exceptional* if for all two distinct $a, b \in R$, the element $a - b$ has a multiplicative inverse in R .

Lemma 4.1 (Generalized Alon–Füredi Theorem [?]). *Let R be a ring and let $S \subseteq R$ be a finite exceptional subset of R . Let $f(\mathbf{Y}) \in R[\mathbf{Y}]$ be a nonzero multilinear polynomial in μ variables. Then $\Pr_{\mathbf{r} \leftarrow S^\mu} [f(\mathbf{r}) = 0] \leq \mu/|S|$.*

4.2 Linear codes and relations

Linear codes Let $\mathcal{C} \subseteq \mathbb{F}^m$ be a linear code over $\mathbb{F}$ of dimension ℓ and blocklength m . For $h \geq 1$, the h -interleaving of $\mathcal{C}$ is the code $\mathcal{C}^{\equiv h} \subseteq (\mathbb{F}^h)^m$ of dimension $\ell \cdot h$ and blocklength $m \cdot h$, whose codewords have the form

$$((c_{i1}, \dots, c_{ih}))_{i \in [m]} \quad \text{for } (c_{1j}, \dots, c_{mj}) \in \mathcal{C} \quad \text{for all } j \in [h].$$

Given two words w_1, w_2 of length n , the relative Hamming distance between w_1 and w_2 is denoted

$$\Delta(w_1, w_2).$$

The *relative minimum distance* of $\mathcal{C}$, denoted $\Delta(\mathcal{C})$, is the minimum of $\Delta(c_1, c_2)$ over distinct codewords $c_1, c_2 \in \mathcal{C}$. Given $\delta \in (0, 1)$ and an integer $L \geq 1$, we say $\mathcal{C}$ is (δ, L) -list-decodable if for every word $w \in \mathbb{F}^m$ there are at most L codewords $c \in \mathcal{C}$ with $\Delta(w, c) < \delta$. Every code is $(\delta, 1)$ -list-decodable for $\delta \leq \Delta(\mathcal{C})/2$. The value $\Delta(\mathcal{C})/2$ is called the *unique decoding radius* of $\mathcal{C}$.

Relations A *relation* $\mathcal{R}$ consists of triples $(\mathbf{x}, \mathbf{y}; \mathbf{w})$, where $\mathbf{x}$ is the explicit instance, $\mathbf{y}$ the implicit instance or oracle, and $\mathbf{w}$ the witness. The corresponding language is $\text{LANG}(\mathcal{R}) := \{(\mathbf{x}, \mathbf{y}) \mid \exists \mathbf{w} \text{ such that } (\mathbf{x}, \mathbf{y}, \mathbf{w}) \in \mathcal{R}\}$.

Polynomial Equation Satisfiability (PESAT) over finite fields We recall a relation from [?] **Albert: Linear time accumulation paper, or earlier?** over finite fields, which captures popular constraints such as R1CS, CCS, AIR (cf. [?]), and lookup and permutation constraints (cf. [GWV⁺25]).

Fix a finite field $\mathbb{F}$, a witness length ℓ , an input length k , a number of rows n , a collection of n polynomials $\mathbf{p} = (\mathbf{p}_1, \dots, \mathbf{p}_n)$ on $k + \ell$ variables, for each $j \in [n]$.

Definition 4.2. The PESAT relation with respect to these parameters is defined as

$$\mathcal{R}_{\text{PESAT}}(\mathbb{F}, \mathbf{p}) = \left\{ \begin{array}{l} \mathbf{x} = \mathbf{x} \in \mathbb{F}^k, \\ \mathbf{y} = \perp, \\ \mathbf{w} = \mathbf{f} \in \mathbb{F}^\ell \end{array} \middle| \mathbf{p}_i(\mathbf{x}, \mathbf{f}) = 0 \text{ for all } i \in [n] \right\}$$

Let $\mathcal{C} \subseteq \mathbb{F}^m$ be a linear code of dimension ℓ' and blocklength m over $\mathbb{F}$. Let $W \subseteq \mathbb{F}^\ell$ be a subset and let $T : W \rightarrow \mathbb{F}^{\ell'}$ be an arbitrary map (looking ahead, for us T will be a packing map that packs elements from $\mathbb{F}_2$ into elements of a larger binary field, so that $W = \mathbb{F}_2^\ell \subseteq \mathbb{F}^\ell$). We define the *Encoded PESAT relation* as:

$$\mathcal{R}_{\text{PESAT}}(\mathbb{F}, \mathbf{p}, \mathcal{C}, T) := \left\{ \left(\begin{array}{l} \mathbf{x} = \mathbf{x} \in \mathbb{F}^k, \\ \mathbf{y} = f \in \mathbb{F}^m, \\ \mathbf{w} = \mathbf{f} \in W \end{array} \right) \mid \begin{array}{l} \text{Enc}_{\mathcal{C}}(T(\mathbf{f})) = f \\ \wedge \mathbf{p}_i(\mathbf{x}, \mathbf{f}) = 0 \text{ for all } i \in [n] \end{array} \right\}$$

We will drop the reference to T whenever it is understood from the context. When $n = 1$ and $\mathbf{x}$ are such that $\mathbf{x} = (\mathbf{v}, \mu) \in \mathbb{F}^\ell \times \mathbb{F}$; $\mathbf{p}_1(\mathbf{x}, \mathbf{f}) = \langle \mathbf{f}, \mathbf{v} \rangle - \mu$ (and $\ell = k$), the relation $\mathcal{R}_{\text{PESAT}}(\mathbb{F}, \mathbf{p}, \mathcal{C}, T)$ is called a *constrained code* [?] [Albert: WHIR?](#). We denote it

$$\mathcal{R}_{\text{Lin}}(\mathbb{F}, \mathcal{C}, T).$$

Definition 4.3 (Relaxation of $\mathcal{R}_{\text{PESAT}}$ and $\mathcal{R}_{\text{Lin}}$). Given $\delta \in (0, 1)$, the *relaxation* of $\mathcal{R}_{\text{PESAT}}$ and $\mathcal{R}_{\text{Lin}}$, denoted

$$\mathcal{R}_{\text{PESAT}}^\delta, \quad \mathcal{R}_{\text{Lin}}^\delta$$

are the relations defined exactly as $\mathcal{R}_{\text{PESAT}}$ and $\mathcal{R}_{\text{Lin}}$, respectively, except that the constraint $\text{Enc}_{\mathcal{C}}(T(\mathbf{f})) = f$ is replaced by

$$\Delta(\text{Enc}_{\mathcal{C}}(T(\mathbf{f})), f) < \delta.$$

4.3 Interactive Oracle Reductions (IOR) of proximity

A public-coin *Interactive Oracle Reduction (IOR) of proximity* $\Pi = (\mathbf{P}, \mathbf{V})$ from a relation $\mathcal{R}$ to a relation $\mathcal{R}'$ is specified as follows. The prover $\mathbf{P}$ receives $(\mathbf{x}, \mathbf{y}, \mathbf{w}) \in \mathcal{R}$. The verifier $\mathbf{V}$ receives $\mathbf{x}$ and query access to $\mathbf{y}$. They interact for k rounds. In round $j \in [k]$, the prover sends an oracle π_j and the verifier sends a uniformly random message $\rho_j \in \{0, 1\}^{r_j}$. After the k rounds, the verifier makes queries to $\mathbf{y}$ and to $\pi_1, \dots, \pi_k$. Then:

- $\mathbf{V}$ either rejects or outputs a new pair $(\mathbf{x}', \mathbf{y}')$, where $\mathbf{y}'$ is specified implicitly in terms of the oracles $\mathbf{y}$ and $\pi_1, \dots, \pi_k$.
- $\mathbf{P}$ outputs a witness $\mathbf{w}'$ such that $(\mathbf{x}', \mathbf{y}', \mathbf{w}') \in \mathcal{R}'$.

We write $\langle \mathbf{P}, \mathbf{V} \rangle(\mathbf{x}, \mathbf{y}, \mathbf{w})$ for the interaction of prover and verifier during the IOR.

Definition 4.4 (Completeness). Π has (*perfect*) *completeness* if for every $(\mathbf{x}, \mathbf{y}, \mathbf{w}) \in \mathcal{R}$,

$$\Pr_{\rho_1, \dots, \rho_k} \left[(\mathbf{x}', \mathbf{y}'; \mathbf{w}') \in \mathcal{R}' \mid (\mathbf{x}', \mathbf{y}'; \mathbf{w}') \leftarrow \langle \mathbf{P}, \mathbf{V} \rangle(\mathbf{x}, \mathbf{y}, \mathbf{w}) \right] = 1.$$

Remark 4.5. An IOP for a relation $\mathcal{R}$ is an IOR from $\mathcal{R}$ to the trivial relation $\mathcal{R}_{\text{triv}} = \{(\mathbf{x} = \perp, \mathbf{y} = \perp, \mathbf{w} = \perp)\}$.

We proceed to define round-by-round knowledge soundness of an IOR, as presented in [?].

Definition 4.6 (Knowledge state function with relaxation). Let $\Pi = (\mathbf{P}, \mathbf{V})$ be an IOR from a relation $\mathcal{R}$ to $\mathcal{R}'$. A *knowledge state function with relaxation* $(\widetilde{\mathcal{R}}, \widetilde{\mathcal{R}'})$ for Π is a (possibly inefficient) function KState that, on input $(\mathbf{x}, \mathbf{y})$, interaction transcript tr , and knowledge state witness $\mathbf{w}$, outputs a bit and has the following properties.

- **Empty transcript.** If $\text{tr} = \emptyset$ is the empty transcript, then $\text{KState}(\mathbf{x}, \mathbf{y}, \text{tr}, \mathbf{w}) = 1$ if and only if $(\mathbf{x}, \mathbf{y}; \mathbf{w}) \in \widetilde{\mathcal{R}}$.
- **Prover moves.** If tr is a transcript where the prover is about to move and $\text{KState}(\mathbf{x}, \mathbf{y}, \text{tr}, \mathbf{w}) = 0$, then $\text{KState}(\mathbf{x}, \mathbf{y}, \text{tr} \parallel \pi, \mathbf{w}) = 0$ for every prover message π .
- **Full transcript.** If $\text{tr} = (\pi_1, \rho_1, \dots, \pi_k, \rho_k)$ is a full transcript, then $\text{KState}(\mathbf{x}, \mathbf{y}, \text{tr}, \mathbf{w}) = 1$ if and only if $\forall$ on tr outputs $(\mathbf{x}', \mathbf{y}')$ such that $(\mathbf{x}', \mathbf{y}'; \mathbf{w}) \in \widetilde{\mathcal{R}'}$.

Definition 4.7 (Round-by-round knowledge soundness). A k -round IOR Π has *round-by-round knowledge soundness with relaxation* with errors $(\kappa_1, \dots, \kappa_k)$ if there exists a knowledge state function KState as in Definition 4.6 and a deterministic extractor $\mathcal{E}_{\text{rbr}}$ such that for every $(\mathbf{x}, \mathbf{y})$, every $j \in [k]$, and every transcript tr where the verifier is about to send ρ_j ,

$$\Pr_{\rho_j} \left[\exists \mathbf{w} \text{ such that } \begin{array}{l} \wedge \text{KState}(\mathbf{x}, \mathbf{y}, \text{tr}, \mathcal{E}_{\text{rbr}}(\mathbf{x}, \mathbf{y}, \text{tr} \parallel \rho_j, \mathbf{w})) = 0 \\ \wedge \text{KState}(\mathbf{x}, \mathbf{y}, \text{tr} \parallel \rho_j, \mathbf{w}) = 1 \end{array} \right] \leq \kappa_j(\mathbf{x}, \mathbf{y}).$$

If $(\widetilde{\mathcal{R}}, \widetilde{\mathcal{R}'}) = (\mathcal{R}, \mathcal{R}')$ then we say the round-by-round knowledge soundness of Π is *without relaxation*.

Remark 4.8 (Composition of IORs). *Albert: Turn into a theorem/lemma. What can I cite for this? TensorSwitch, Arc, or ZOOK?* IORs can be composed together and inherit completeness and round-by-round knowledge soundness errors. More precisely, let Π_1, Π_2 be perfectly complete IORs from $\mathcal{R}_1$ to $\mathcal{R}_2$ and $\mathcal{R}_3$, respectively, with round-by-round knowledge errors with relaxation $(\widetilde{\mathcal{R}}_1, \widetilde{\mathcal{R}}_2)$ and $(\widetilde{\mathcal{R}}_2, \widetilde{\mathcal{R}}_3)$, respectively; and with errors $\kappa_{11}, \dots, \kappa_{1k_1}$ and $\kappa_{21}, \dots, \kappa_{2k_2}$, respectively. Consider the IOR Π' from $\mathcal{R}_1$ to $\mathcal{R}_3$ consisting of applying IOR Π_1 first, and then Π_2 taking as input for Π_2 the output of Π_1 . Then Π' is a perfectly complete $(k_1 + k_2)$ -round IOR from $\mathcal{R}_1$ to $\mathcal{R}_3$ with round-by-round knowledge soundness with relaxation $(\widetilde{\mathcal{R}}_1, \widetilde{\mathcal{R}}_3)$ and errors $\kappa_{11}, \dots, \kappa_{1k_1}, \kappa_{21}, \dots, \kappa_{2k_2}$.

Grinding *Albert: Review* Grinding [?] is a technique where, given an IOP Π for a relation $\mathcal{R}$ with relaxation $\widetilde{\mathcal{R}}$ with round-by-round knowledge errors $\kappa_1, \dots, \kappa_k$, produces a new IOP Π' for $\mathcal{R}$ with round-by-round knowledge with relaxation $\widetilde{\mathcal{R}}$ with errors

$$2^{-\alpha_1} \cdot \kappa_1, \dots, 2^{-\alpha_k} \cdot \kappa_k.$$

The new IOP Π' works exactly as Π , except that, at each round $i \in [t]$, the verifier samples a random string of α_i bits, called *grinding string*, which sends alongside its message, and rejects unless all bits are 0.

Clearly Π' is not necessarily complete, even if Π is. The goal is to use the BCS compilation of IOP's, and not the IOP's themselves (i.e. the compilation where oracles are replaced by Merkle roots and where interactivensness is removed through the Fiat-Shamir transform) To this end, one chooses $\alpha_1, \dots, \alpha_k$ small enough that, the prover of the BCS transform of Π' can *grind* transcripts until all of the grinding strings in the transcript are the zero string. This results in the prover having to compute per round $i \in [t]$, an average of 2^{α_i} hashes of its i -th message followed by a small string of bits (in practice, one has the prover send a small string of bits next to its message, so that during grinding, it does not need to produce fresh Merkle roots at each grinding iteration). In Table 2, we provide example grinding times for different values of α_i .

The round-by-round knowledge errors of Π' are obtained by defining a state function which is exactly as the state function of Π , except that the function outputs zero if some of the aforementioned

strings of bits are not the zero string. The extractor for Π' is defined as the extractor for Π , ignoring the additional strings of bits in the transcript.

Table 2: Cost of finding a 64-bit nonce such that $\text{BLAKE3}(\text{msg}, \text{nonce})$ has at least α leading zero bits, with msg of 256 bits. Single and multi-threaded on a MacBook Air M4 16GB. [Albert: As implemented in Flock?](#)

α	18	19	20	21	22	23	24	25
Mean, 1 thread (ms)	12.3	19.5	42.8	82.2	203.8	464.4	808.5	1291.4
Median, 1 thread (ms)	9.2	16.3	27.8	55.2	140.6	358.1	608.5	852.0
Mean, 10 threads (ms)	2.3	3.5	8.9	14.3	35.5	84.6	148.6	240.8
Median, 10 threads (ms)	1.8	2.5	6.8	9.6	26.9	68.0	111.3	161.8

5 BitZ IOP for LinBitsRings

5.1 The core LinBitsRings relation

In this section we define the main linear relation we work with. We start by fixing notation:

- Let $\mathbb{F}_{2^\nu}$ be a field with 2^ν elements. Let $\beta_1, \dots, \beta_\nu \in \mathbb{F}_{2^\nu}$ be an $\mathbb{F}_2$ -basis of $\mathbb{F}_{2^\nu}$ seen as an $\mathbb{F}_2$ -vector space of dimension ν . Given a vector $\mathbf{u} \in \mathbb{F}_2^\ell$, with ν dividing ℓ , we define

$$\text{Pack}_\nu(\mathbf{u}) \in \mathbb{F}_{2^\nu}^{\ell/\nu}$$

as the vector whose j -th entry is

$$\text{Pack}_\nu(\mathbf{u})_j = \mathbf{u}_{(j-1)\cdot\nu+1} \cdot \beta_1 + \dots + \mathbf{u}_{(j-1)\cdot\nu+\nu-1} \cdot \beta_{\nu-1} + \mathbf{u}_{j\cdot\nu} \cdot \beta_\nu \in \mathbb{F}_{2^\nu}.$$

- Let $\ell \geq 1$ be a dimension parameter divisible by ν , $\mathcal{C}$ a code over $\mathbb{F}_{2^\nu}$ of dimension ℓ/ν and blocklength m .
- Let R be a finitely generated ring (typically R is $\mathbb{Z}$, a finite field, not necessarily a prime field, or a quotient of a polynomial ring $\mathbb{Z}[X]$ or $\mathbb{F}_q[X]$) and let $\pi : \mathbb{Z} \rightarrow R$ be the homomorphism mapping an integer a to $a \cdot 1 \in R$ (i.e. reduction modulo the characteristic of R).

We begin by defining a particular subclass of our target relation.

Definition 5.1. The *Core LinBitsRings* relation is defined as

$$\mathcal{R}_{\text{Lin-BitsDual}}(\{0, 1\}, \pi, R, \mathcal{C}) := \left\{ \left(\begin{array}{l} \mathbf{x} = (\mu, \mathbf{v}) \in R \times R^\ell, \\ \mathbf{y} = f \in \mathbb{F}_{2^\nu}^m, \\ \mathbf{w} = \mathbf{f} \in \{0, 1\}^\ell \subseteq \mathbb{Z}^\ell \end{array} \right) \middle| \begin{array}{l} f = \text{Enc}_{\mathcal{C}} \circ \text{Pack}_\nu \circ \pi_2(\mathbf{f}) \\ \langle \pi(\mathbf{f}), \mathbf{v} \rangle = \mu \quad \text{in } R \end{array} \right\}$$

For $\delta \in (0, 1)$, we let $\mathcal{R}_{\text{Lin-BitsDual}}^\delta$ be the relation $\mathcal{R}_{\text{Lin-BitsDual}}$ after replacing the constraint $f = \text{Enc}_{\mathcal{C}} \circ \text{Pack}_\nu \circ \pi_2(\mathbf{f})$ by

$$\Delta(f, \text{Enc}_{\mathcal{C}} \circ \text{Pack}_\nu \circ \pi_2(\mathbf{f})) < \delta.$$

5.2 Tensor decomposition and products of constrained codes

Albert: Motivate

We start with some groundwork for our IOP for the Core LinBitsRings relation. Let $(\mathbf{x}, \mathbf{y}, \mathbf{w}) = ((\mu, \mathbf{v}), f, \mathbf{f}) \in \mathcal{R}_{\text{Lin-BitsDual}}(\{0, 1\}, \pi, R, \mathcal{C})$. We fix a tensor decomposition of the vector $\mathbf{v} \in R^\ell$ as

$$\mathbf{v} = \mathbf{v}^{(1)} \otimes \mathbf{v}^{(2)} = \left(\mathbf{v}_i^{(1)} \cdot \mathbf{v}_j^{(2)} \right)_{i \in [\ell_1], j \in [\ell_2]}$$

for some $\mathbf{v}^{(1)} \in R^{\ell_1}, \mathbf{v}^{(2)} \in R^{\ell_2}$, with $\ell_1 \cdot \ell_2 = \ell$. Note that the trivial tensor decomposition is admitted, i.e. we may effectively not decompose $\mathbf{v}$ by choosing $\ell_1 = \ell, \ell_2 = 1, \mathbf{v}^{(1)} = \mathbf{v}$, and $\mathbf{v}^{(2)} = (1)$.

We index the entries of $\mathbf{f}$ as $\mathbf{f} = (\mathbf{f}_{ij})_{i \in [\ell_1], j \in [\ell_2]}$. Note that

$$\langle \pi(\mathbf{f}), \mathbf{v} \rangle = \sum_{j \in [\ell_2]} \left(\sum_{i \in [\ell_1]} \pi(\mathbf{f}_{ij}) \cdot \mathbf{v}_i^{(1)} \right) \cdot \mathbf{v}_j^{(2)} = \sum_{j \in [\ell_2]} \langle \pi(\mathbf{f}_{*j}), \mathbf{v}^{(1)} \rangle \cdot \mathbf{v}_j^{(2)},$$

where $\mathbf{f}_{*j} = (\mathbf{f}_{ij})_{i \in [\ell_1]}$. Consider the relation (cf. Definition 4.2) $\mathcal{R}_{\text{PESAT}}(\mathbb{F}_{2^\nu}, Q_{\text{Bit}\mathbb{Z}}, \mathcal{C}, \text{Pack}_\nu)$ for ℓ_2 constraint polynomials $Q_{\text{Bit}\mathbb{Z}}$ so that

$$\mathcal{R}_{\text{PESAT}}(\mathbb{F}_{2^\nu}, Q_{\text{Bit}\mathbb{Z}}, \mathcal{C}, \text{Pack}_\nu) = \left\{ \left(\begin{array}{l} \mathbf{x} = (\mathbf{y}, \boldsymbol{\mu}) \in \mathbb{F}_{2^\nu}^{\ell_1} \times \mathbb{F}_{2^\nu}^{\ell_2}, \\ \mathbf{y} = f \in \mathbb{F}_{2^\nu}^m, \\ \mathbf{w} = \mathbf{f} = (\mathbf{f}_{*j})_{j \in [\ell_2]} \in \mathbb{F}_2^{\ell_1 \times \ell_2} \end{array} \right) \left| \begin{array}{l} f = \text{Enc}_{\mathcal{C}} \circ \text{Pack}_\nu(\mathbf{f}) \\ \bigwedge \prod_{i=1}^{\ell_1} ((\mathbf{y}_i - 1) \cdot \mathbf{f}_{ij} + 1) = \boldsymbol{\mu}_j \text{ for all } j \in [\ell_2] \end{array} \right. \right\}.$$

We omit an explicit description of the polynomials $Q_{\text{Bit}\mathbb{Z}}$, since the fact that the above relation is a particular instance of the PESAT relation is not relevant in this paper.

We assume prover and verifier have access to an IOR Π_{red} that reduces $\mathcal{R}_{\text{PESAT}}(\mathbb{F}_{2^\nu}, Q_{\text{Bit}\mathbb{Z}}, \mathcal{C}, \text{Pack}_\nu)$ to $\mathcal{R}_{\text{Lin}}(\mathbb{F}_{2^\nu}, \mathcal{C}, \text{Pack}_\nu)$, where, recall, this latter relation has predicates of the form $\langle \mathbf{g}, \mathbf{u} \rangle = \mu'$ with $\mathbf{g} \in \mathbb{F}_2^\ell$, $\mathbf{u}$ a vector over $\mathbb{F}_{2^\nu}$, and $g = \text{Enc}_{\mathcal{C}} \circ \text{Pack}_\nu(\mathbf{g})$. In our instantiation Π_{red} is an optimized GKR protocol for grand products (cf. Section 6), which reduces the initial grand product claims to MLE evaluation claims of $\mathbf{y}$ and $\mathbf{f}$ over $\mathbb{F}_{2^\nu}$.

We assume Π_{red} works so that the verifier never queries $\mathbf{y}$; and that, given an input

$$(\mathbf{x} = (\mathbf{y}, \boldsymbol{\mu}), \mathbf{y} = f, \mathbf{w} = \pi_2(\mathbf{f})) \in \mathcal{R}_{\text{PESAT}}(\mathbb{F}_{2^\nu}, Q_{\text{Bit}\mathbb{Z}}, \mathcal{C}, \text{Pack}_\nu),$$

Π_{red} outputs

$$((\mathbf{v}, \mu'), f, \pi_2(\mathbf{f})) \in \mathcal{R}_{\text{Lin}}(\mathbb{F}_{2^\nu}, \mathcal{C}, \text{Pack}_\nu)$$

for some vector $\mathbf{v}$ and field element μ' . In particular, Π_{red} outputs the same oracle f and witness $\pi_2(\mathbf{f})$ as the oracle and witness it is given as inputs.

This is indeed the case with GKR: GKR reduces the initial circuit evaluation claim with a witness input, into an MLE evaluation claim of the same witness, without querying any oracle in-between.

5.3 The BitZ IOP for the core LinBitsRings relation

Algebraic set-up Recall (Section 4.1) that all rings in this work are finitely generated as rings. We fix generators $\gamma_1, \dots, \gamma_k \in R$ and we let

$$\pi_{\text{canon}}^R : \mathbb{Z}[X_1, \dots, X_k] \rightarrow R$$

be the surjective ring homomorphism mapping each integer a to $a \cdot 1 \in R$, and X_i to γ_i for $i \in [k]$. The choice of $\gamma_1, \dots, \gamma_k$ is far from unique, see Example 5.2 below. **Albert:** Aren't we paying one generator too much? 1 can be a generator can't it?

We also fix an inverse map $\pi_{\text{canon}}^{R^{-1}} : R \rightarrow \mathbb{Z}[X_1, \dots, X_k]$ for π_{canon}^R , so that $\pi_{\text{canon}}^R \circ \pi_{\text{canon}}^{R^{-1}} = \text{id}$. We refer to $\pi_{\text{canon}}^{R^{-1}}(a)$ as the *lift* of $a \in R$. Note that, given π_{canon}^R , there may be many possible choices for $\pi_{\text{canon}}^{R^{-1}}$. In Example 5.2 we provide example inverse maps.

Recall (see Section 4.1) that π_n denotes reduction modulo n and that π_n^{-1} maps each element from $\mathbb{Z}_n$ to its integer representative in $[0..n-1]$.

Example 5.2. We provide some examples of rings R and choices for projections π_{canon}^R and their inverses:

- **Prime fields.** If $R = \mathbb{F}_q$ is a prime field with q elements, then we can take $k = 0$, so that $\pi_{\text{canon}}^{\mathbb{F}_q} = \pi_q : \mathbb{Z} \rightarrow \mathbb{F}_q$ is simply reduction modulo q , with its inverse being π_q^{-1} (see Section 4.1).
- **Integers.** Similarly, for $R = \mathbb{Z}$ we can take $k = 0$, and then $\pi_{\text{canon}}^{\mathbb{Z}}$ and its lifts are the identity map on $\mathbb{Z}$.
- **Field extensions.** If R is a field extension of a prime field $\mathbb{F}_q$ of degree $d > 1$, then we can take $k = 1$ and $\gamma_1 = x$, where x is a root of an irreducible polynomial $h(X_1) \in \mathbb{F}_q[X_1]$ of degree d (so that $R = \mathbb{F}_q(x)$). Then $\pi_{\text{canon}}^R : \mathbb{Z}[X_1] \rightarrow R$ consists in reducing coefficients modulo q , and then sending X_1 to x . Writing $a \in R$ as $a = \sum_{i=0}^{d-1} a_i \cdot x^i$ with $a_i \in \mathbb{F}_q$, one can then define a natural lift of a as $\sum_{i=0}^{d-1} \pi_q^{-1}(a_i) \cdot X_1^i$.
- **Univariate integer polynomials.** If $R = \mathbb{Z}[X]$, then we can take $k = 1$ and $\gamma_1 = X$, and then $\pi_{\text{canon}}^{\mathbb{Z}[X]} : \mathbb{Z}[X_1] \rightarrow \mathbb{Z}[X]$ is an isomorphism, with lifts given by its inverse.
- **Lattices, Galois rings, etc.** More generally, if R is finitely generated as a $\mathbb{Z}_{\text{char}(R)}$ -module and it is free as a $\mathbb{Z}_{\text{char}(R)}$ -module, with $\mathbb{Z}_{\text{char}(R)}$ -module-basis $\beta_1 = 1, \beta_2, \dots, \beta_m$ (a condition satisfied by, e.g., all finite fields, $\mathbb{Z}$, and $\mathbb{Z}_n[X]/(h(X))$ for any monic $h(X)$ and any integer $n \geq 0$), then we may take $k = m - 1$ and $\gamma_i = \beta_{i+1}$ for $i \in [k]$. Then given $a \in R$ and writing $a = \sum_{i \in [m]} a_i \cdot \beta_i$ with $a_i \in \mathbb{Z}_{\text{char}(R)}$, we can define the lift of a as $\pi_{\text{char}}^{-1}(a_1) + \sum_{i=2}^m \pi_{\text{char}}^{-1}(a_i) \cdot X_{i-1}$.

We assume without further comment that it is efficient to compute values $\pi_{\text{canon}}^R(h)$ for $h \in \mathbb{Z}[X_1, \dots, X_k]$, and to check whether $\pi_{\text{canon}}^R(h) = 0$. This is the case in all examples above.

The BitZ IOP. We provide an IOP for $\mathcal{R}_{\text{Lin-BitsDual}}$ next. We assume the notation from the previous two Sections 5.1 and 5.2, and from the preliminary Section 4.1, together with:

- Let Π_{iopp} be an IOP for $\mathcal{R}_{\text{Lin}}(\mathbb{F}_{2^\nu}, \mathcal{C}, \text{Pack}_\nu)$. In our instantiation, Π_{iopp} is the WHIR/Ligerito protocol [ACFY25, ?] over binary fields as implemented in Flock [BRW26], with ring-switching [DP23] and a preliminary sumcheck over $\mathbb{F}_{2^\nu}$ that transforms arbitrary inner products to MLE evaluation claims (since the the ring switch of [BRW26] does not support arbitrary inner product claims). We set the proximity parameter set to the Johnson bound when targeting 100 bits of security, and to the unique decoding radius for 128 bits of security, see Section 5.7 for more details.

- Let $\ell = \ell_1 \cdot \ell_2$ be dimension parameters, and R a ring with k generators $\gamma_1, \dots, \gamma_k$ **Albert: change generator symbol since γ is used heavily in the general LinBitsRings relation?**, as in the previous sections. We let $\mathcal{P}$ be a set of primes with $\max \mathcal{P} < (|\mathbb{F}_{2^\nu}| - 1)/\ell_1$. Given the tensor decomposition $\mathbf{v} = \mathbf{v}^{(1)} \otimes \mathbf{v}^{(2)}$ from [Section 5.2](#), we define **Albert: Why a subscript R in N_R and not in the others?**

$$\mathbf{u} := (\mathbf{v}^{(1)}, \mathbf{v}^{(2)}) \in R^{\ell_1 + \ell_2}$$

$$D := \max_{i \in [\mathbf{u}]} \deg(\pi_{\text{canon}}^R{}^{-1}(\mathbf{u}_i)), \quad N_R := \max_{i \in [\mathbf{u}]} \|\pi_{\text{canon}}^R{}^{-1}(\mathbf{u}_i)\|_\infty,$$

where $\deg$ denotes total degree, and $\|\cdot\|_\infty$ denotes the maximum absolute value of the coefficients. We also define a quantity **Albert: Check**

$$T \leq (D + 1)^k,$$

where k is a minimum number of generators of R , as follows. T is the maximum, over monomials m in variables $X_1, \dots, X_k$, of the number of pairs (m_1, m_2) of monomials of total degree at most D each such that $m_1 \cdot m_2 = m$.

For example, $T \leq D + 1$ if $k = 1$, and $T \leq 2$ if $D \leq 1$. In practice, for us, the bound $T \leq (D + 1)^k$ is good enough.

- For every prime $q \in \mathcal{P}$, let $\mathcal{M}_q$ be a set of ring homomorphisms

$$\pi_{\text{rand}} : \mathbb{Z}[X_1, \dots, X_k] \rightarrow \mathbb{F}_q.$$

The verifier will sample primes and morphisms from $\mathcal{P}$ and $\mathcal{M}_q$. We let $\theta(R, \ell)$ be the maximum probability that $\pi_{\text{rand}}(h) = 0$, over random $q \leftarrow \mathcal{P}$, $\pi_{\text{rand}} \leftarrow \mathcal{M}_q$, and where the maximum is taken over nonzero polynomials $h \in \mathbb{Z}[X_1, \dots, X_k]$ of total degree at most $2 \cdot D$, with $\|h\|_\infty \leq 2 \cdot T \cdot \ell \cdot N_R^2$, and such that $\pi_{\text{canon}}^R(h) \neq 0$. Precisely:

$$\theta(R, \ell) := \max_h \left\{ \Pr_{q \leftarrow \mathcal{P}, \pi_{\text{rand}} \leftarrow \mathcal{M}_q} [\pi_{\text{rand}}(h) = 0] \right\},$$

where h ranges over the polynomials just described. See [Lemma 5.3](#) for an example.

Lemma 5.3 (Example definition of $\mathcal{M}_q$ and the corresponding $\theta(R, \ell)$). *Let $\mathcal{P}$ be a set of primes. For each $q \in \mathcal{P}$ let $\mathcal{M}_q$ be the set of all maps $\pi : \mathbb{Z}[X_1, \dots, X_k] \rightarrow \mathbb{F}_q$ of the form*

$$\pi(h(X_1, \dots, X_k)) = \pi_q(h)(\alpha_1, \dots, \alpha_k), \quad \text{for } h \in \mathbb{Z}[X_1, \dots, X_k],$$

for some $\alpha_1, \dots, \alpha_k \in \mathbb{F}_q$, and where $\pi_q(h)$ is h after reducing its coefficients modulo q . Then

$$\theta(R, \ell) \leq \frac{2 \cdot D}{q_{\min}} + \frac{\log(4 \cdot T \cdot \ell \cdot N_R^2)}{|\mathcal{P}| \cdot \lfloor \log(q_{\min}) \rfloor}.$$

where $q_{\min}$ is the smallest prime in $\mathcal{P}$.

Proof. We defer the proof to later in [Section 5.4](#). □

Albert: Comment somewhere that assuming tensore decomposition is important, but also natural

Construction 5.4 (IOP for the core $\mathcal{R}_{\text{Lin-BitsDual}}$ relation). **Parameters.** $(\ell, \ell_1, \ell_2, \Pi_{\text{iopp}}, \Pi_{\text{red}}, \mathcal{P}, (\mathcal{M}_q)_{q \in \mathcal{P}})$ comprised of: a witness vector length ℓ ; tensor decomposition dimensions ℓ_1, ℓ_2 satisfying $\ell_1 \cdot \ell_2 = \ell$; the IOP Π_{iopp} described above, the IOR Π_{red} from ??; and the sets $\mathcal{P}, \mathcal{M}_q$ from above. **Albert: Parameters changed. Change when calling this construction. I put as parameters everything that is used by the protocol that is not present in the definition of the relation already**

Inputs. The prover and verifier receive $\mathbf{x} = (\mu, \mathbf{v}) \in R \times R^\ell$ and $\mathbf{y} = f \in \mathbb{F}_{2^\nu}^m$. The prover receives $\mathbf{w} = \mathbf{f} \in \{0, 1\}^\ell \subseteq \mathbb{Z}^\ell$ such that $(\mathbf{x}, \mathbf{y}, \mathbf{w}) \in \mathcal{R}_{\text{Lin-BitsDual}}(\{0, 1\}, \pi, R, \mathcal{C})$. We assume the tensor split from the previous [Section 5.2](#), and adopt its notation.

Let $g \in \mathbb{F}_{2^\nu}$ be a fixed generator of the multiplicative group $\mathbb{F}_{2^\nu}^*$. Denote π_{canon}^R simply as π_{canon} .

1. **Round 0 (optional): Out Of Domain (OOD) sample.** **Albert: I think this is not implemented?** This step can be skipped if the proximity parameter δ is set within the unique decoding radius of $\mathcal{C}$ (cf. [Section 4.2](#), and [Theorem 5.7](#)). Otherwise, the verifier samples and sends a random $\zeta \in \mathbb{F}_{2^\nu}$, and the prover responds with

$$y = \widetilde{\text{Pack}_\nu \circ \pi_2(\mathbf{f})}(\zeta), \quad \text{where } \zeta := (\zeta^{2^0}, \zeta^{2^1}, \dots, \zeta^{2^{\log(\ell/\nu)-1}}).$$

2. **Round 1 (optional): replacing R by a “small” prime field $\mathbb{F}_q$.** This step can be skipped (i.e. prover and verifier move to Round 2 directly) if R is a prime field $\mathbb{F}_q$ and $q < (|\mathbb{F}_{2^\nu}| - 1)/\ell_1$. The prover sends $\mu' \in \mathbb{Z}[X_1, \dots, X_k]$ such that

$$\langle \pi_{\text{canon}}^{-1}(\mathbf{v}^{(1)}) \otimes \pi_{\text{canon}}^{-1}(\mathbf{v}^{(2)}), \mathbf{f} \rangle = \mu' \quad \text{in } \mathbb{Z}[X_1, \dots, X_k].$$

The verifier checks that μ' has total degree at most $2 \cdot D$ and coefficients in $[-T \cdot \ell \cdot N_R^2 .. T \cdot \ell \cdot N_R^2]$ **Albert: state with the infinity norm**, checks that $\pi_{\text{canon}}(\mu') = \mu$, and replies with a random prime $q \in \mathcal{P}$ and a random morphism $\pi_{\text{rand}} : \mathbb{Z}[X_1, \dots, X_k] \rightarrow \mathbb{F}_q$ from $\mathcal{M}_q$. The prover and verifier update:

$$\begin{aligned} \mathbf{v}^{(1)} &\leftarrow \pi_{\text{rand}} \circ \pi_{\text{canon}}^{-1}(\mathbf{v}^{(1)}) \in \mathbb{F}_q^{\ell_1}, \\ \mathbf{v}^{(2)} &\leftarrow \pi_{\text{rand}} \circ \pi_{\text{canon}}^{-1}(\mathbf{v}^{(2)}) \in \mathbb{F}_q^{\ell_2}, \\ \mu &\leftarrow \pi_{\text{rand}}(\mu') \in \mathbb{F}_q, \end{aligned}$$

The prover and verifier now set to prove the updated claim

$$\langle \mathbf{v}^{(1)} \otimes \mathbf{v}^{(2)}, \pi_q(\mathbf{f}) \rangle = \mu \quad \text{over } \mathbb{F}_q. \tag{20}$$

Remark 5.5 (Skipping Round 1 when $R = \mathbb{Z}$ and $\|\mathbf{v}\|_\infty$ is “small”). **Albert: Make sure our completeness and soundness theorem proves this case as well** This step can also be skipped when $R = \mathbb{Z}$ and all entries of $\mathbf{v}^{(1)}$ are nonnegative and less than $(|\mathbb{F}_{2^\nu}| - 1)/\ell_1$. In that case the claim to prove at the start of Round 2 is $\langle \mathbf{v}^{(1)} \otimes \mathbf{v}^{(2)}, \mathbf{f} \rangle = \mu$ over $\mathbb{Z}$. In this case the map π_q in Round 2 is simply replaced by the identity map, and $\mathbb{F}_q$ is replaced by $\mathbb{Z}$. In Phase 3, γ is $\mathbf{v}^{(1)}$.

3. **Round 2: transforming (20) to ℓ_2 integer constraints with bounded integers.**

- The prover sends integers $\mu_1, \dots, \mu_{\ell_2} \in [0 \dots (q-1) \cdot \ell_1]$ such that, over $\mathbb{Z}$,

$$\langle \mathbf{f}_{*j}, \pi_q^{-1}(\mathbf{v}^{(1)}) \rangle = \mu_j \quad \text{for all } j \in [\ell_2]. \quad (21)$$

- The verifier checks that $\mu_j \in [0 \dots (q-1) \cdot \ell_1]$ for all $j \in [\ell_2]$, and that

$$\sum_{j \in [\ell_2]} \pi_q(\mu_j) \cdot \mathbf{v}_j^{(2)} = \mu \quad \text{over } \mathbb{F}_q. \quad (22)$$

Next, prover and verifier will focus on certifying that (21) holds for all $j \in [\ell_2]$.

In [Remark 5.6](#) we describe a variation in which, instead of sending the elements $\mu_1, \dots, \mu_{\ell_2}$, the prover commits to their bit-decomposition over $\mathbb{F}_{2^\nu}$, and proves (22) with [Construction 5.13](#). This saves on proof size and verifier time when ℓ is large.

4. Phase 3: Transforming the claim into a PESAT claim over $\mathbb{F}_{2^\nu}$, and invoking the IOR Π_{red} .

- Prover and verifier define

$$\gamma = \pi_q^{-1}(\mathbf{v}^{(1)}) \in [0 \dots q-1]^{\ell_1} \subseteq \mathbb{Z}^{\ell_1},$$

and set to prove the following claim:

$$\prod_{i \in [\ell_1]} g^{\gamma_i \cdot \mathbf{f}_{ij}} = g^{\mu_j} \in \mathbb{F}_{2^\nu}, \quad \text{for all } j \in [\ell_2].$$

This claim is equivalent to

$$\prod_{i \in [\ell_1]} ((g^{\gamma_i} - 1) \cdot \pi_2(\mathbf{f}_{ij}) + 1) = g^{\mu_j} \in \mathbb{F}_{2^\nu}, \quad \text{for all } j \in [\ell_2].$$

Notice that

$$(\mathbf{x} = ((g^{\gamma_i})_{i \in [\ell_1]}, (g^{\mu_j})_{j \in [\ell_2]}), \mathbf{y} = f, \mathbf{w} = \pi_2(\mathbf{f})) \in \mathcal{R}_{\text{PESAT}}(\mathbb{F}_{2^\nu}, Q_{\text{BitZ}}, \mathcal{C}, \text{Pack}_\nu). \quad (23)$$

- To prove (23), prover and verifier engage in the IOR Π_{red} ⁴ with input

$$(\mathbf{x} = ((g^{\gamma_i})_{i \in [\ell_1]}, (g^{\mu_j})_{j \in [\ell_2]}), \mathbf{y} = f, \mathbf{w} = \pi_2(\mathbf{f})).$$

At the end of this IOR, they are left with the following claim to prove:

$$((\mathbf{v}, \mu'), f, \pi_2(\mathbf{f})) \in \mathcal{R}_{\text{Lin}}(\mathbb{F}_{2^\nu}, \mathcal{C}, \text{Pack}_\nu) = \left\{ \left(\begin{array}{l} \mathbf{x} = (\mathbf{u}, \mu) \in \mathbb{F}_{2^\nu}^{\ell_1} \times \mathbb{F}_{2^\nu}, \\ \mathbf{y} = g \in \mathbb{F}_{2^\nu}^m, \\ \mathbf{w} = \mathbf{g} \in \mathbb{F}_2^{\ell_2} \end{array} \right) \middle| \begin{array}{l} g = \text{Enc}_{\mathcal{C}} \circ \text{Pack}_\nu(\mathbf{g}) \\ \langle \mathbf{u}, \mathbf{g} \rangle = \mu \end{array} \right\} \quad (24)$$

for some vector $\mathbf{v}$ and field element μ' output during the IOR Π_{red} .

⁴Recall that, in our instantiation, Π_{red} is the GKR protocol applied to multiple grand products, cf. [Section 6](#).

5. **Phase 4: Using an IOP over $\mathbb{F}_{2^\nu}$.** Prover and verifier engage in the IOP Π_{iopp} to certify the claim (24). If Round 0 (Out Of Domain sampling) triggered, they also use Π_{iopp} to certify the evaluation claim from Round 0.

Remark 5.6 (Reducing proof size by committing to the elements $\mu_1, \dots, \mu_{\ell_2}$). When $\ell = \ell_1 \cdot \ell_2$ is such that ℓ_2 is large, the elements $\mu_1, \dots, \mu_{\ell_2} \in \mathbb{Z}$ sent at Round 2 of [Construction 5.4](#) may start to dominate the proof size. For example, for $\ell_2 = 2^{13}$ and μ_i of 128 bits, the elements $\mu_1, \dots, \mu_{\ell_2}$ amount to 131 KB. To mitigate this cost, one can have the prover commit to the vector $(\mu_1, \dots, \mu_{\ell_2})$ and then prove (22), i.e.

$$\langle \mathbf{f}_{*j}, \pi_q^{-1}(\mathbf{v}^{(1)}) \rangle = \mu_j \quad \text{for all } j \in [\ell_2].$$

using an IOP for constrained codes. Concretely, the prover commits to the vector $\text{bits}(\mu)$ of bits forming the elements $(\mu_1, \dots, \mu_{\ell_2})$ using BitZ , and then proves (22) using [Construction 5.13](#) from [Section 5.6](#).

5.4 Soundness and completeness

For readability, we temporarily denote

$$\mathcal{R}_{\text{PESAT}} = \mathcal{R}_{\text{PESAT}}(\mathbb{F}_{2^\nu}, Q_{\text{BitZ}}, \mathcal{C}, \text{Pack}_\nu), \quad \mathcal{R}_{\text{Lin}} = \mathcal{R}_{\text{Lin}}(\mathbb{F}_{2^\nu}, \mathcal{C}, \text{Pack}_\nu).$$

Theorem 5.7 (Completeness and round-by-round knowledge soundness of [Construction 5.4](#)). *Let $\delta \in (0, 1)$ be a proximity parameter, and let $L_\delta \geq 1$ be such that $\mathcal{C}$ is (δ, L_δ) -list-decodable (cf. [Section 4.2](#)). Assume*

$$\max \mathcal{P} < (|\mathbb{F}_{2^\nu}| - 1)/\ell_1.$$

Assume Π_{red} is a perfectly complete t_{red} -round IOR from $\mathcal{R}_{\text{PESAT}}$ to $\mathcal{R}_{\text{Lin}}$ with round-by-round knowledge soundness with relaxation $(\mathcal{R}_{\text{PESAT}}^\delta, \mathcal{R}_{\text{Lin}}^\delta)$, with errors $(\kappa_{\text{red}1}, \dots, \kappa_{\text{red}t_{\text{red}}})$. Assume Π_{iopp} is a perfectly complete t_{iopp} -round IOP for $\mathcal{R}_{\text{Lin}}$ (i.e. an IOR from $\mathcal{R}_{\text{Lin}}$ to the trivial relation $\perp$) with round-by-round knowledge soundness with relaxation $(\mathcal{R}_{\text{Lin}}^\delta, \perp)$, and errors

$$\kappa_{\text{iopp}1}, \dots, \kappa_{\text{iopp}t_{\text{iopp}}}.$$

Then [Construction 5.4](#) is a perfectly complete $(2 + t_{\text{red}} + t_{\text{iopp}})$ -round IOP for $\mathcal{R}_{\text{Lin-BitsDual}}$ with round-by-round knowledge soundness with relaxation $(\mathcal{R}_{\text{Lin-BitsDual}}^\delta, \perp)$, with errors

$$\kappa_{\text{OOD}}, \kappa_0, (\kappa_{\text{red}i})_{i \in [t_{\text{red}}]}, (\kappa_{\text{iopp}i})_{i \in [t_{\text{iopp}}]},$$

where

$$\kappa_{\text{OOD}} \leq \binom{L_\delta}{2} \cdot \frac{\ell/\nu - 1}{|\mathbb{F}_{2^\nu}|}, \quad \kappa_0 = \theta(R, \ell)$$

(see [Lemmas A.1](#) and [5.3](#)). The error terms $\kappa_{\text{OOD}}, \kappa_0$ are void if either Round 0 or Round 1 are not executed, respectively. In that case [Construction 5.4](#) has one or two rounds less.

To prove [Lemma 5.3](#) we will use the following result, which generalizes [Albert: point to lemma](#) in [\[AGK⁺26\]](#).

Lemma 5.8. *Let $h \in \mathbb{Z}[X_1, \dots, X_m]$ be a nonzero polynomial of total degree $\leq d$ and with coefficients in the range $[-2^B + 1 \dots 2^B - 1]$. Let $\mathcal{P}$ be a set of primes, and let $q_{\min}$ be the smallest prime in $\mathcal{P}$. Given a prime q and $\alpha \in \mathbb{F}_q^m$, write $\pi_{q,\alpha}(h) := \pi_q(h)(\alpha) \in \mathbb{F}_q$. Then *Albert: should we have two cases in the second term? One with denom 1 and another with denom d ?**

$$\Pr_{\substack{q \leftarrow \mathcal{P}, \\ \alpha \leftarrow \mathbb{F}_q^m}} [\pi_{q,\alpha}(h) = 0] \leq \frac{B}{\lfloor \log(q_{\min}) \rfloor \cdot |\mathcal{P}|} + \frac{d}{q_{\min}}.$$

Proof. Let $\mathcal{E}_{\text{prime}}$ be the event that all coefficients of h are multiples of q . Then, by the Schwartz–Zippel lemma,

$$\Pr_{\substack{q \leftarrow \mathcal{P}, \\ \alpha \leftarrow \mathbb{F}_q^m}} [\pi_{q,\alpha}(h) = 0 \mid \neg \mathcal{E}_{\text{prime}}] \leq \frac{d}{q_{\min}}.$$

Now, every nonzero coefficient of h is divisible by at most $B/\lfloor \log(q_{\min}) \rfloor$ primes from $\mathcal{P}$. Hence, the probability that $\mathcal{E}_{\text{prime}}$ occurs is at most $(B/\lfloor \log(q_{\min}) \rfloor)/|\mathcal{P}|$, and the lemma follows. $\square$

Proof of Lemma 5.3. Let $h \in \mathbb{Z}[X_1, \dots, X_k]$ be a polynomial of total degree at most $2 \cdot D$, with coefficients in $[-2 \cdot T \cdot \ell \cdot N_R^2 \dots 2 \cdot T \cdot \ell \cdot N_R^2]$, and with $\pi_{\text{canon}}^R(h) \neq 0$ (cf. the definition of $\theta(R, \ell)$). In particular h is nonzero, and its coefficients lie in $[-2^B + 1 \dots 2^B - 1]$ for $B := \lfloor \log(T \cdot \ell \cdot N_R^2) \rfloor + 2 \leq \log(4 \cdot T \cdot \ell \cdot N_R^2)$. Since h is nonzero and has total degree at most $2 \cdot D$, Lemma 5.8 (applied with $m = k$ variables and total degree $2 \cdot D$) yields

$$\Pr_{\substack{q \leftarrow \mathcal{P}, \\ \alpha \leftarrow \mathbb{F}_q^k}} [\pi_q(h)(\alpha) = 0] \leq \frac{B}{\lfloor \log(q_{\min}) \rfloor \cdot |\mathcal{P}|} + \frac{2 \cdot D}{q_{\min}},$$

as needed. $\square$

Proof of Theorem 5.7. The completeness of Construction 5.4 is clear from its description *Albert: Review*. To prove its soundness, we first treat the case where Round 0 (OOD sampling) is not executed; the case where Round 0 is executed is treated at Section A.2 following (now standard) arguments from [?, ACFY25].

We decompose Construction 5.4 into three IORs Π_1, Π_2, Π_3 . The first one, Π_1 , is an IOR from $\mathcal{R}_{\text{Lin-BitsDual}}$ to $\mathcal{R}_{\text{PESAT}}$, and it consists of applying Rounds 1 and 2 of Construction 5.4, and outputting

$$(\mathbf{x}' = ((g^{\gamma_i})_{i \in [\ell_1]}, (g^{\mu_j})_{j \in [\ell_2]}), \mathbf{y}' = f, \mathbf{w}' = \pi_2(\mathbf{f})),$$

as described in Phase 3 of Construction 5.4. Formally, Π_1 is a 2-round IOR whose second verifier message is empty: the prover first sends μ' , the verifier replies with (q, π_{rand}) , and the prover then sends $\mu = (\mu_1, \dots, \mu_{\ell_2})$. The second IOR, Π_2 , is simply Π_{red} , reducing $\mathcal{R}_{\text{PESAT}}$ to $\mathcal{R}_{\text{Lin}}$. Finally, the third IOR, Π_3 , is simply the IOP Π_{iopp} for $\mathcal{R}_{\text{Lin}}$.

Below we show that Π_1 has round-by-round knowledge soundness with relaxation $(\mathcal{R}_{\text{Lin-BitsDual}}^\delta, \mathcal{R}_{\text{PESAT}}^\delta)$, with errors $(\kappa_0, 0)$, assuming Round 1 is not skipped. The second verifier message of Π_1 is empty, hence the last prover message of Π_1 can be sent together with the first prover message of Π_2 , and thus Construction 5.4 has $1 + t_{\text{red}} + t_{\text{iopp}}$ rounds. The theorem then follows from the composition of IORs result (cf. Remark 4.8).

At the end of the proof we discuss the case when Round 1 is skipped.

Completeness and soundness of Π_1 We describe a knowledge state function KState . Let $\mathbf{x} = (\mu, \mathbf{v}) \in R \times R^\ell$ and $\mathbf{y} = f \in \mathbb{F}_2^m$. Let tr be a partial interaction transcript for Π_1 . We start by setting KState so that the first condition in Definition 4.6 are met. Precisely, if $\text{tr} = \emptyset$, we set $\text{KState}(\mathbf{x}, \mathbf{y}, \text{tr}, \mathbf{w}) = 1$ if and only if $(\mathbf{x}, \mathbf{y}, \mathbf{w}) \in \mathcal{R}_{\text{Lin-BitsDual}}^\delta$. Further, if $\text{tr} = (\mu')$ consists of a single prover message, we set $\text{KState}(\mathbf{x}, \mathbf{y}, \mu', \mathbf{w}) = \text{KState}(\mathbf{x}, \mathbf{y}, \emptyset, \mathbf{w})$, so that the second condition in Definition 4.6 holds for the first prover message round.

After the verifier's message, tr is as follows:

$$\text{tr} = (\mu', (q, \pi_{\text{rand}})) \in \mathbb{Z}[X_1, \dots, X_k] \times (\mathcal{P} \times \mathcal{M}_q).$$

We define $\text{KState}(\mathbf{x}, \mathbf{y}, \text{tr}, \mathbf{w}) = 1$ if and only if the verifier's degree and range checks on μ' in Round 1 pass, $\mathbf{w} = \mathbf{f} \in \{0, 1\}^\ell \subseteq \mathbb{Z}^\ell$, $\mathbf{f} = (\mathbf{f}_{*j})_{j \in [\ell_2]} \in (\{0, 1\}^{\ell_1})^{\ell_2}$, and

$$\begin{aligned} \pi_{\text{canon}}(\mu') &= \mu, \\ \langle \pi_{\text{rand}} \circ \pi_{\text{canon}}^{-1}(\mathbf{v}^{(1)}) \otimes \pi_{\text{rand}} \circ \pi_{\text{canon}}^{-1}(\mathbf{v}^{(2)}), \pi_q(\mathbf{f}) \rangle &= \pi_{\text{rand}}(\mu') \quad \text{over } \mathbb{F}_q, \\ \Delta(f, \text{Enc}_C \circ \text{Pack}_\nu \circ \pi_2(\mathbf{f})) &< \delta. \end{aligned} \tag{25}$$

We continue with our definition of KState . Let $\text{tr} = (\mu', (q, \pi_{\text{rand}}), \mu)$ for a prover message $\mu = (\mu_1, \dots, \mu_{\ell_2}) \in [0..(q-1) \cdot \ell_1]^{\ell_2}$, or let tr be the full transcript obtained by appending the empty second verifier message to $(\mu', (q, \pi_{\text{rand}}), \mu)$. We set $\text{KState}(\mathbf{x}, \mathbf{y}, \text{tr}, \mathbf{w}) = 1$ if and only if the verifier accepts on tr (i.e. its checks in Rounds 1 and 2 pass) and $(\mathbf{x}', \mathbf{y}', \mathbf{w}) \in \mathcal{R}_{\text{PESAT}}^\delta$, where $(\mathbf{x}', \mathbf{y}')$ is the output of the verifier, namely

$$\mathbf{x}' = ((g^{\gamma_i})_{i \in [\ell_1]}, (g^{\mu_j})_{j \in [\ell_2]}), \quad \mathbf{y}' = f, \quad \text{with } \gamma = \pi_q^{-1} \left(\pi_{\text{rand}} \circ \pi_{\text{canon}}^{-1}(\mathbf{v}^{(1)}) \right) \in [0..q-1]^{\ell_1} \subseteq \mathbb{Z}^{\ell_1}$$

(here and below we identify $\mathbf{w} = \mathbf{f} \in \{0, 1\}^\ell \subseteq \mathbb{Z}^\ell$ with $\pi_2(\mathbf{f}) \in \mathbb{F}_2^\ell$ **Albert: remove?**). Thus KState satisfies the third condition in Definition 4.6.

We define an extractor as $\mathcal{E}_{\text{rbr}}(\mathbf{x}, \mathbf{y}, \text{tr}, \mathbf{w}) = \mathbf{w}$. It remains to prove two statements: first, that

$$\Pr_{q, \pi_{\text{rand}}} \left[\exists \mathbf{w} \left| \begin{array}{l} \text{KState}(\mathbf{x}, \mathbf{y}, \mu', \mathcal{E}_{\text{rbr}}(\mathbf{x}, \mathbf{y}, (\mu', (q, \pi_{\text{rand}})), \mathbf{w})) = 0 \\ \wedge \text{KState}(\mathbf{x}, \mathbf{y}, (\mu', (q, \pi_{\text{rand}})), \mathbf{w}) = 1 \end{array} \right. \right] \leq \kappa_0;$$

and second, that KState satisfies the second condition in Definition 4.6 for transcripts where the prover is about to send μ (its second message).

We start with the first statement. Assume

$$\text{KState}(\mathbf{x}, \mathbf{y}, (\mu', (q, \pi_{\text{rand}})), \mathbf{w}) = 1 \quad \text{and} \quad \text{KState}(\mathbf{x}, \mathbf{y}, \mu', \mathbf{w}) = \text{KState}(\mathbf{x}, \mathbf{y}, \mathbf{w}) = 0.$$

The latter occurs if and only if $(\mathbf{x}, \mathbf{y}, \mathbf{w}) \notin \mathcal{R}_{\text{Lin-BitsDual}}^\delta$. The former implies that $\mathbf{w} = \mathbf{f} = (\mathbf{f}_{*j})_{j \in [\ell_2]} \in \{0, 1\}^\ell$ satisfies (25); in particular $\Delta(f, \text{Enc}_C \circ \text{Pack}_\nu \circ \pi_2(\mathbf{f})) < \delta$, and hence, since $(\mathbf{x}, \mathbf{y}, \mathbf{w}) \notin \mathcal{R}_{\text{Lin-BitsDual}}^\delta$, necessarily

$$\langle \pi(\mathbf{f}), \mathbf{v} \rangle \neq \mu \quad \text{over } R. \tag{26}$$

Define the polynomial

$$h := \langle \pi_{\text{canon}}^{-1}(\mathbf{v}^{(1)}) \otimes \pi_{\text{canon}}^{-1}(\mathbf{v}^{(2)}), \mathbf{f} \rangle - \mu' \in \mathbb{Z}[X_1, \dots, X_k].$$

Since π_{canon} is a ring homomorphism satisfying $\pi_{\text{canon}} \circ \pi_{\text{canon}}^{-1} = \text{id}$ and restrict to π on $\mathbb{Z}$ (so $\pi_{\text{canon}}(\mathbf{f}) = \pi(\mathbf{f})$), the first condition in (25) gives

$$\pi_{\text{canon}}(h) = \langle \mathbf{v}^{(1)} \otimes \mathbf{v}^{(2)}, \pi(\mathbf{f}) \rangle - \pi_{\text{canon}}(\mu') = \langle \pi(\mathbf{f}), \mathbf{v} \rangle - \mu,$$

which is nonzero by (26). In particular $h \neq 0$. On the other hand, since π_{rand} is a ring homomorphism restricting to π_q on $\mathbb{Z}$, the second condition in (25) states precisely that $\pi_{\text{rand}}(h) = 0$. Moreover, h has total degree at most $2 \cdot D$ and coefficients in $[-2 \cdot T \cdot \ell \cdot N_R^2 .. 2 \cdot T \cdot \ell \cdot N_R^2]$: indeed, the lifts $\pi_{\text{canon}}^{-1}(\mathbf{v}_i^{(1)})$ and $\pi_{\text{canon}}^{-1}(\mathbf{v}_j^{(2)})$ have total degree at most D and coefficients of absolute value at most N_R ,

Since π_{canon} is a ring homomorphism satisfying $\pi_{\text{canon}} \circ \pi_{\text{canon}}^{-1} = \text{id}$ and restricting to π on $\mathbb{Z}$, the first condition in (25) gives

$$\pi_{\text{canon}}(h) = \langle \pi(\mathbf{f}), \mathbf{v}^{(1)} \otimes \mathbf{v}^{(2)} \rangle - \pi_{\text{canon}}(\mu') = \langle \pi(\mathbf{f}), \mathbf{v} \rangle - \mu,$$

which is nonzero by (26). In particular $h \neq 0$. On the other hand, since π_{rand} is a ring homomorphism restricting to π_q on $\mathbb{Z}$, the second condition in (25) states precisely that $\pi_{\text{rand}}(h) = 0$. We claim that h has total degree at most $2 \cdot D$ and satisfies

$$\|h\|_{\infty} \leq 2 \cdot T \cdot \ell \cdot N_R^2.$$

Indeed, the lifts $\pi_{\text{canon}}^{-1}(\mathbf{v}_i^{(1)})$ and $\pi_{\text{canon}}^{-1}(\mathbf{v}_j^{(2)})$ have total degree at most D and coefficients of absolute value at most N_R , so each product $\pi_{\text{canon}}^{-1}(\mathbf{v}_i^{(1)}) \cdot \pi_{\text{canon}}^{-1}(\mathbf{v}_j^{(2)})$ has total degree at most $2 \cdot D$ and coefficients of absolute value at most $T \cdot N_R^2$ by the definition of T . Further, by the verifier's checks in Round 1,

$$\|\mu'\|_{\infty} \leq T \cdot \ell \cdot N_R^2$$

and the total degree of μ' is at most $2 \cdot D$. The claim follows now from the fact that the entries of $\mathbf{f}$ are bits and there are ℓ of them.

Since additionally $\pi_{\text{canon}}(h) \neq 0$, the polynomial h belongs to the class of polynomials over which $\theta(R, \ell)$ is defined. Finally, since we have assumed (for now) that $\mathcal{C}$ is $(\delta, 1)$ -list-decodable, there is at most one $\mathbf{w}$ such that $\Delta(f, \text{Enc}_{\mathcal{C}} \circ \text{Pack}_{\nu} \circ \pi_2(\mathbf{w})) < \delta$. Therefore [Albert: review](#)

$$\begin{aligned} & \Pr_{q, \pi_{\text{rand}}} [\exists \mathbf{w} \text{ such that } (\mathbf{x}, \mathbf{y}, \mathbf{w}) \notin \mathcal{R}_{\text{Lin-BitsDual}}^{\delta}, \text{ and } \mathbf{w} = \mathbf{f} \text{ satisfies (25)}] \\ & \leq \Pr_{q \leftarrow \mathcal{P}, \pi_{\text{rand}} \leftarrow \mathcal{M}_q} [\pi_{\text{rand}}(h) = 0] \leq \theta(R, \ell) = \kappa_0, \end{aligned}$$

where the last inequality holds by the definition of $\theta(R, \ell)$.

It remains to prove the second statement, namely: if $\text{KState}(\mathbf{x}, \mathbf{y}, (\mu', (q, \pi_{\text{rand}})), \mathbf{w}) = 0$, then $\text{KState}(\mathbf{x}, \mathbf{y}, (\mu', (q, \pi_{\text{rand}}), \boldsymbol{\mu}), \mathbf{w}) = 0$ for every prover message $\boldsymbol{\mu} = (\mu_1, \dots, \mu_{\ell_2}) \in [0 .. (q-1) \cdot \ell_1]^{\ell_2}$. Suppose towards a contradiction that $\text{KState}(\mathbf{x}, \mathbf{y}, (\mu', (q, \pi_{\text{rand}}), \boldsymbol{\mu}), \mathbf{w}) = 1$, i.e. that the verifier accepts and $(\mathbf{x}', \mathbf{y}', \mathbf{w}) \in \mathcal{R}_{\text{PESAT}}^{\delta}$. Then $\mathbf{w} = \mathbf{f} = (\mathbf{f}_{*j})_{j \in [\ell_2]} \in \{0, 1\}^{\ell}$, $\Delta(f, \text{Enc}_{\mathcal{C}} \circ \text{Pack}_{\nu} \circ \pi_2(\mathbf{f})) < \delta$, and, for all $j \in [\ell_2]$,

$$g^{\langle \mathbf{f}_{*j}, \gamma \rangle} = \prod_{i \in [\ell_1]} ((g^{\gamma_i} - 1) \cdot \pi_2(\mathbf{f}_{ij}) + 1) = g^{\mu_j},$$

and hence

$$\langle \mathbf{f}_{*j}, \gamma \rangle = \mu_j \pmod{\text{ord}(g)},$$

where $\text{ord}(g)$ denotes the multiplicative order of g . Now, since $\gamma_i \in [0..q-1]$, $\mu_j \in [0..(q-1) \cdot \ell_1]$, and $\mathbf{f}_{*j}$ contains bits, we have $|\langle \mathbf{f}_{*j}, \gamma \rangle - \mu_j| \leq \ell_1 \cdot (q-1)$. Since g is a generator of $\mathbb{F}_{2^\nu}^*$ and $q \leq \max \mathcal{P} < (|\mathbb{F}_{2^\nu}| - 1)/\ell_1$, we have

$$\text{ord}(g) = |\mathbb{F}_{2^\nu}| - 1 > \ell_1 \cdot \max \mathcal{P} \geq \ell_1 \cdot q > \ell_1 \cdot (q-1),$$

and hence

$$\langle \mathbf{f}_{*j}, \gamma \rangle = \mu_j \quad \text{over } \mathbb{Z}, \quad \text{for all } j \in [\ell_2]. \quad (27)$$

Applying π_q to (27), and using that $\pi_q(\gamma) = \pi_{\text{rand}} \circ \pi_{\text{canon}}^{-1}(\mathbf{v}^{(1)})$, we obtain

$$\langle \pi_q(\mathbf{f}_{*j}), \pi_{\text{rand}} \circ \pi_{\text{canon}}^{-1}(\mathbf{v}^{(1)}) \rangle = \pi_q(\mu_j) \quad \text{over } \mathbb{F}_q, \quad \text{for all } j \in [\ell_2].$$

Since the verifier accepts, its check (22) in Round 2 passes, i.e. $\sum_{j \in [\ell_2]} \pi_q(\mu_j) \cdot (\pi_{\text{rand}} \circ \pi_{\text{canon}}^{-1}(\mathbf{v}^{(2)}))_j = \pi_{\text{rand}}(\mu')$. Now

$$\begin{aligned} & \langle \pi_{\text{rand}} \circ \pi_{\text{canon}}^{-1}(\mathbf{v}^{(1)}) \otimes \pi_{\text{rand}} \circ \pi_{\text{canon}}^{-1}(\mathbf{v}^{(2)}), \pi_q(\mathbf{f}) \rangle \\ &= \sum_{j \in [\ell_2]} \langle \pi_q(\mathbf{f}_{*j}), \pi_{\text{rand}} \circ \pi_{\text{canon}}^{-1}(\mathbf{v}^{(1)}) \rangle \cdot \left(\pi_{\text{rand}} \circ \pi_{\text{canon}}^{-1}(\mathbf{v}^{(2)}) \right)_j \\ &= \sum_{j \in [\ell_2]} \pi_q(\mu_j) \cdot \left(\pi_{\text{rand}} \circ \pi_{\text{canon}}^{-1}(\mathbf{v}^{(2)}) \right)_j = \pi_{\text{rand}}(\mu'). \end{aligned}$$

Hence the second condition in (25) holds. The first condition in (25), i.e. $\pi_{\text{canon}}(\mu') = \mu$, holds because the verifier accepts, and the third one, i.e. $\Delta(f, \text{Enc}_{\mathcal{C}} \circ \text{Pack}_{\nu} \circ \pi_2(\mathbf{f})) < \delta$, holds because $(\mathbf{x}', \mathbf{y}', \mathbf{w}) \in \mathcal{R}_{\text{PESAT}}^\delta$. Hence $\text{KState}(\mathbf{x}, \mathbf{y}, (\mu', (q, \pi_{\text{rand}})), \mathbf{w}) = 1$, a contradiction.

The case where Round 1 is skipped. Albert: Review: AI written In this case, the messages μ' and (q, π_{rand}) and the error κ_0 disappear, and the arguments above apply after replacing $\pi_{\text{rand}} \circ \pi_{\text{canon}}^{-1}$ with the identity map on $\mathbb{F}_q$ or on $\mathbb{Z}$. In the latter case, π_q is also replaced by the identity map, cf. Remark 5.5. $\square$

5.5 The general LinBitsRings relation

We now proceed to describe the full relation we seek to work with in this section. Compared to $\mathcal{R}_{\text{Lin-BitsDual}}(\{0, 1\}, \pi, R, \mathcal{C})$, the witness $\mathbf{f}$ will now entries in an arbitrary finitely generated ring S , instead of in $\{0, 1\} \subseteq \mathbb{Z}$.

The witness entries will be restricted to a bounded subset of S , so that each witness can be expressed as a pre-specified linear expression (over S) of a string of bits.

We set the following additional notation:

- Let S be a ring, possibly different than R , let $\psi : S \rightarrow R$ be a ring homomorphism.
- Let $\gamma_1, \dots, \gamma_d \in S$ be a fixed generating set of S (??), for some $d \geq 1$.
- Recall from Section 4.1 that $S_{\leq B}$ denotes the set of elements of S of the form $h = \sum_{i \in [d]} h_i \cdot \gamma_i$ for

$$h_i \in [0..2^B - 1]_S \subseteq S,$$

and that for a vector of integers $\mathbf{w}$, we write $\mathbf{w}_S$ for its projection to S via reduction modulo $\text{char}(S)$ (see (19)).

- Let ℓ_0 be a vector witness length, so that our witnesses are $\mathbf{f} \in S^{\ell_0}$.

We will use the natural homomorphism from $\pi_{\text{char}(S)} : \mathbb{Z} \rightarrow S$ consisting in reducing an element modulo the characteristic of S , see (18) in Section 4.1.

Definition 5.9 (Bitification of the witness). Let $\mathbf{f} \in S_{\leq B}^{\ell_0}$. Let ℓ be a vector length, called *bitification length*. Let $L \in S^{\ell_0 \times \ell}$ be a matrix, called *bitification matrix*. Let $\mathbf{g} \in \{0, 1\}^\ell \subseteq \mathbb{Z}^\ell$ be a vector filled with integer bits. We say $\mathbf{g}$ is a *bitification* of $\mathbf{f}$ (with respect to L) if

$$\mathbf{f} = L \cdot \mathbf{g}_S.$$

where for a vector $\mathbf{w}$ of integer bits, $\mathbf{w}_S$ denotes its natural projection to a vector with entries the elements $\{0, 1\}_S \subseteq S$. The choice of L and a method for building a bitification of $\mathbf{f}$ can be customized. Below, we describe what we call the *standard bitification* procedure (see below). This can be used for arbitrary finitely generated rings S and bit bounds B . In some scenarios (see ??), one may choose different more convenient approaches.

For a fixed L and $\mathbf{f}$, bitifications of $\mathbf{f}$ are NOT unique in general. Nevertheless, abusing the notation, we often denote them as

$$\text{bits}(\mathbf{f}).$$

A *standard bitification* of $\mathbf{f} \in S_{\leq B}^{\ell_0}$, denoted $\text{bits}(\mathbf{f})$, is obtained as follows: write $\mathbf{f} = \sum_{j \in [d]} \mathbf{f}_{*j} \cdot \gamma_j$ for some $\mathbf{f}_{*j} \in \mathbb{Z}_{\text{char}(S)}^{\ell_0}$, $j \in [d]$ (this expression is not unique). Then write

$$\mathbf{f}_{*j} = \sum_{k=1}^B \mathbf{f}_{*jk} \cdot \pi_{\text{char}(S)}(2^{k-1})$$

for some vectors

$$\mathbf{f}_{*jk} \in \{0, 1\}_S^{\ell_0} \subseteq S^{\ell_0},$$

for all j, k . We define

$$\begin{aligned} \ell &:= \ell_0 \cdot d \cdot B, \\ \text{bits}(\mathbf{f}) &\in \{0, 1\}^\ell \subseteq \mathbb{Z}^\ell \end{aligned}$$

as the vector with integer bit entries obtained by concatenating the vectors $\mathbf{f}_{*jk}$ and lifting their entries to $\{0, 1\} \subseteq \mathbb{Z}$ in a natural way, i.e.

$$\text{bits}(\mathbf{f}) := (\text{lift}(\mathbf{f}_{*jk}) \mid j \in [d], k \in [B]) \in \{0, 1\}^\ell \subseteq \mathbb{Z}^\ell, \quad (28)$$

where lift is the identity function if $\text{char}(S) = 0$, and $\text{lift} = \pi_{\text{char}(S)}^{-1}$ if $\text{char}(S) > 0$. Defining

$$L = \text{Id}_{\ell_0 \times \ell_0} \otimes (1, 2, \dots, 2^{B-1})_S \otimes (\gamma_1, \dots, \gamma_d) \in S^{\ell_0 \times \ell},$$

we have

$$\mathbf{f} = L \cdot \text{bits}(\mathbf{f})_S.$$

Assume $\mathcal{C}$ is a linear code over $\mathbb{F}_{2^\nu}$ of dimension ℓ/ν and blocklength m .

Definition 5.10. The *general LinBitsRings* relation is defined as **Albert:** Here we should allow for $\psi = \phi \circ \pi_{\text{canon}}^S$ for $\phi : \mathbb{Z}[X_1, \dots, X_k] \rightarrow R$ a morphism. This is what is actually happening and it allows us to build later full IOPs for constraints over arbitrary f.g. rings

$$\mathcal{R}_{\text{Lin-BitsDual}}(S_{\leq B}, \psi, R, \mathcal{C}, L) := \left\{ \left(\begin{array}{l} \mathbf{x} = (\mu, \mathbf{v}) \in R \times R^{\ell_0}, \\ \mathbf{y} = f \in \mathbb{F}_{2^\nu}^m, \\ \mathbf{w} = \mathbf{f} \in S_{\leq B}^{\ell_0} \end{array} \right) \mid \begin{array}{l} f = \text{Enc}_{\mathcal{C}} \circ \text{Pack}_{\nu} \circ \pi_2(\text{bits}(\mathbf{f})), \\ \langle \psi(\mathbf{f}), \mathbf{v} \rangle = \mu \quad \text{in } R, \\ \mathbf{f} = L \cdot (\text{bits}(\mathbf{f}))_S \text{ for some } \text{bits}(\mathbf{f}) \in \{0, 1\}^\ell \subseteq \mathbb{Z}^\ell \end{array} \right\}.$$

When there is no risk of confusion, we denote $\mathcal{R}_{\text{Lin-BitsDual}}(S_{\leq B}, \psi, R, \mathcal{C})$ simply by $\mathcal{R}_{\text{Lin-BitsDual}}$. The Core LinBitsRings relation is precisely $\mathcal{R}_{\text{Lin-BitsDual}}(S_{\leq B}, \psi, R, \mathcal{C})$ after taking $S = \mathbb{Z}$, $B = 1$, and $\psi = \pi_{\text{char}(R)}$.

For $\delta \in (0, 1)$, we let $\mathcal{R}_{\text{Lin-BitsDual}}^\delta$ be the relation $\mathcal{R}_{\text{Lin-BitsDual}}$ after replacing the constraint $f = \text{Enc}_{\mathcal{C}} \circ \text{Pack}_{\nu} \circ \pi_2(\text{bits}(\mathbf{f}))$ by $\Delta(f, \text{Enc}_{\mathcal{C}} \circ \text{Pack}_{\nu} \circ \pi_2(\text{bits}(\mathbf{f}))) < \delta$, as in [Section 5.1](#).

Remark 5.11 (Non-uniform bit-decomposition sizes). In practical settings, it is possible that not all entries of $\mathbf{f}$ have the same bit size. For example, some chunks of $\mathbf{f}$ may be comprised of elements from $\{0, 1\}_S$, while others may be comprised of generic ring elements. However, the standard bitifaction procedure from [Definition 5.9](#) allocates $d \cdot B$ bits for every entry of $\mathbf{f}$, no matter its size. To avoid this, one can find alternative approaches to defining L and $\text{bits}(\mathbf{f})$.

For efficiency purposes, to make sure L has enough structure to later support tensor decomposing certain vectors, as discussed in [Remark 5.14](#). In any case, one can always use the matrix L above, which in general does possess enough structure as argued in [Remark 5.14](#).

5.6 The Bit $\mathbb{Z}$ IOP for the general LinBitsRings relation

Albert: We could add asymptotic/concrete operation numbers per protocol?

Next we describe our IOP for the general relation $\mathcal{R}_{\text{Lin-BitsDual}}$. We assume the notation of [Section 5.5](#). Let L be a *bitifaction* matrix as in [Definition 5.9](#), with $L \in S^{\ell_0 \times \ell}$ so that ℓ is the vector length of the vector $\text{bits}(\mathbf{f}_0)$. For the standard bitifaction matrix $L = \text{Id}_{\ell_0 \times \ell_0} \otimes (1, \dots, 2^{B-1})_S \otimes (\gamma_1, \dots, \gamma_d)$, we have $\ell = \ell_0 \cdot d \cdot B$. Let further ℓ_1, ℓ_2 be tensor decomposition parameters (to be used in [Construction 5.4](#)) such that $\ell = \ell_1 \cdot \ell_2$. We assume that $\mathcal{C}$ is a linear code over $\mathbb{F}_{2^\nu}$ of dimension ℓ/ν .

We will need the following result.

Lemma 5.12. Let $\mathbf{f} \in S_{\leq B}^{\ell_0}$, let $\mathbf{v} \in R^{\ell_0}$, and let $\text{bits}(\mathbf{f})$ be a bitifaction of $\mathbf{f}$ with respect to L . Then **Albert:** General comment: I need to discuss how the verifier handles these vectors $\mathbf{v}$ and $\mathbf{u}$. Reviewers may complain that verifier could be non-succinct. **Albert:** Answer: write the tables mentioned above

$$\begin{aligned} \langle \psi(\mathbf{f}), \mathbf{v} \rangle &= \langle \text{bits}(\mathbf{f})_R, \mathbf{u} \rangle \in R \\ \mathbf{u} &:= \psi(L)^\top \cdot \mathbf{v} \in R^\ell, \end{aligned}$$

where $\top$ denotes transposition. In particular, if $L = \text{Id}_{\ell_0 \times \ell_0} \otimes (1, \dots, 2^{B-1})_S \otimes (\gamma_1, \dots, \gamma_d) \in S^{\ell_0 \times (\ell_0 \cdot B \cdot d)} = S^{\ell_0 \times \ell}$ (cf. [Definition 5.9](#)), then

$$\mathbf{u} = (\mathbf{v} \otimes (1, 2, \dots, 2^{B-1})_R \otimes (\psi(\gamma_1), \dots, \psi(\gamma_d))) \in R^\ell \quad (29)$$

Proof. Rewriting $\langle \psi(\mathbf{f}), \mathbf{v} \rangle$, we obtain by Remark 5.11 and Eq. (17)

$$\langle \psi(\mathbf{f}), \mathbf{v} \rangle = \langle \psi(L \cdot \text{bits}(\mathbf{f})_S), \mathbf{v} \rangle = \langle \psi(\text{bits}(\mathbf{f})_S), \psi(L)^\top \cdot \mathbf{v} \rangle$$

Now $\psi(\text{bits}(\mathbf{f})_S) = \text{bits}(\mathbf{f})_R$ because every entry of $\text{bits}(\mathbf{f})$ consists of bits and ψ is a ring homomorphism, so it sends the 0 and 1 elements of S to the 0 and 1 elements of R . Further,

$$\psi(L) = \text{Id}_{\ell_0 \times \ell_0} \otimes \psi((1, \dots, 2^{B-1})_S) \otimes (\psi(\gamma_1), \dots, \psi(\gamma_d)),$$

and $\psi \circ \pi_{\text{char}(S)}(2^i) = \pi_{\text{char}(R)}(2^i)$ because since $\psi \circ \pi_{\text{char}(S)}$ is a ring homomorphism, $\psi \circ \pi_{\text{char}(S)}(2^i) = 2^i \cdot (\psi \circ \pi_{\text{char}(S)}(1))$, hence $\psi \circ \pi_{\text{char}(S)}(2^i)$ is 2^i times the identity element of R , and that is precisely the definition of $\pi_{\text{char}(R)}(2^i)$.

The specific form of $\mathbf{u}$ in (29) is now obtained by direct calculation of $\psi(L)^\top \cdot \mathbf{v}$. $\square$

We assume that the input vector $\mathbf{v}$ for $\mathcal{R}_{\text{Lin-BitsDual}}(S_{\leq B}, \psi, R, \mathcal{C}, L)$, the matrix L , and the tensor decomposition dimensions are such that $\mathbf{u} = \psi(L)^\top \cdot \mathbf{v}$ admits a tensor decomposition $\mathbf{u} = \mathbf{u}^{(1)} \otimes \mathbf{u}^{(2)}$ with $\mathbf{u}^{(1)} \in R^{\ell_1}$, $\mathbf{u}^{(2)} \in R^{\ell_2}$, see Remark 5.14 for an explanation that this can be achieved easily in applications, and what to do otherwise.

Construction 5.13 (IOP for $\mathcal{R}_{\text{Lin-BitsDual}}$). **Parameters.** $(\ell_0, \ell, \ell_1, \ell_2, \Pi_{\text{iopp}}, \mathcal{P}, (\mathcal{M}_q)_{q \in \mathcal{P}})$ comprised of: a witness vector length ℓ_0 ; a *bitified witness* vector length ℓ ; tensor decomposition dimensions ℓ_1, ℓ_2 for Construction 5.4 satisfying $\ell_1 \cdot \ell_2 = \ell$; IORs and IOPs Π_{iopp} and Π_{red} as in Construction 5.4; and sets of primes and homomorphisms $\mathcal{P}, (\mathcal{M}_q)_{q \in \mathcal{P}}$ again as in Construction 5.4.

Inputs. The prover and verifier receive $\mathbf{x} = (\mu, \mathbf{v}) \in R \times R^{\ell_0}$ and $y = f \in \mathbb{F}_2^m$. The prover receives $\mathbf{w} = \mathbf{f} \in S_{\leq B}^{\ell_0}$ such that $(\mathbf{x}, y, \mathbf{w}) \in \mathcal{R}_{\text{Lin-BitsDual}}(S_{\leq B}, \psi, R, \mathcal{C}, L)$, i.e.,

$$f = \text{Enc}_{\mathcal{C}} \circ \text{Pack}_{\nu} \circ \pi_2(\text{bits}(\mathbf{f})), \quad \langle \psi(\mathbf{f}), \mathbf{v} \rangle = \mu \quad \text{in } R.$$

for some $\text{bits}(\mathbf{f})$ such that $L \cdot \text{bits}(\mathbf{f})_S = \mathbf{f}$.

1. By Lemma 5.12, the claim $(\mathbf{x}, y, \mathbf{w}) \in \mathcal{R}_{\text{Lin-BitsDual}}(S_{\leq B}, \psi, R, \mathcal{C})$ is equivalent to the claim

$$(\mathbf{x}' = (\mu, \mathbf{u}), y' = f = \text{Enc}_{\mathcal{C}} \circ \text{Pack}_{\nu} \circ \pi_2(\text{bits}(\mathbf{f})), \mathbf{w}' = \text{bits}(\mathbf{f})) \in \mathcal{R}_{\text{Lin-BitsDual}}(\{0, 1\}, \pi, R, \mathcal{C}). \quad (30)$$

where $\mathbf{u} = \psi(L)^\top \cdot \mathbf{v}$.

By assumption, $\mathbf{u} = \mathbf{u}^{(1)} \otimes \mathbf{u}^{(2)}$ with $\mathbf{u}^{(1)} \in R^{\ell_1}$, $\mathbf{u}^{(2)} \in R^{\ell_2}$. Then the prover and the verifier use Construction 5.4 to prove (30). They use parameters $(\ell, \ell_1, \ell_2, \Pi_{\text{iopp}}, \Pi_{\text{red}}, \mathcal{P}, (\mathcal{M}_q)_{q \in \mathcal{P}})$. See Remark 5.14 for comments on how to obtain tensor decompositions of $\mathbf{u}$ in Construction 5.4 assuming $\mathbf{v}$ is appropriately tensor-decomposable.

Remark 5.14 (Tensor decompositions of $\mathbf{u}$). Recall that Construction 5.4 considers a (possibly trivial) tensor decomposition of its public input vector $\mathbf{u}$ (denoted $\mathbf{v}$ in Construction 5.4). Moreover, as argued in Sections 3.3 and 6, this tensor decomposition is crucial for performance of BitZ (concretely, we seek $\ell^{0.5} \leq \ell_1 \leq \ell^{0.6}$, cf. Section 6).

In many applications, the input vector $\mathbf{v}$ for $\mathcal{R}_{\text{Lin-BitsDual}}(S_{\leq B}, \psi, R, \mathcal{C})$ is an “eq” vector, i.e. a vector of the form $(\text{eq}(\mathbf{b}, \mathbf{r}))_{\mathbf{b} \in \{0, 1\}^{\log \ell_0}}$ for some $\mathbf{r} \in R^{\log \ell_0}$. In this case $\mathbf{v} = \mathbf{v}^{(1)} \otimes \mathbf{v}^{(2)}$ with the dimensions of $\mathbf{v}^{(1)}, \mathbf{v}^{(2)}$ being arbitrary as long as $|\mathbf{v}^{(1)}| \cdot |\mathbf{v}^{(2)}| = \ell_0$. In that case, if

$$L = \text{Id}_{\ell_0 \times \ell_0} \otimes (1, \dots, 2^{B-1})_S \otimes (\gamma_1, \dots, \gamma_d) \quad (31)$$

(cf. [Definition 5.9](#)), then $\mathbf{u}$ has the form (29), i.e.

$$\mathbf{u} = (\mathbf{v} \otimes (1, 2, \dots, 2^{B-1})_R \otimes (\psi(\gamma_1), \dots, \psi(\gamma_d)))$$

and we may use a tensor decomposition for $\mathbf{u}$ in [Construction 5.4](#) of the form, say, $\mathbf{u} = \mathbf{u}^{(1)} \otimes \mathbf{u}^{(2)}$, where

$$\mathbf{u}^{(1)} := \mathbf{v}^{(1)}, \quad \mathbf{u}^{(2)} := \mathbf{v}^{(2)} \otimes (1, 2, \dots, 2^{B-1})_R \otimes (\psi(\gamma_1), \dots, \psi(\gamma_d)).$$

Other decompositions apply. For example one of the tensor factors in $\mathbf{u}^{(2)}$ above could be placed in $\mathbf{u}^{(1)}$, and the vector $(1, 2, \dots, 2^{B-1})_R$ can admit further tensor decomposition, and some of the factors can be placed in $\mathbf{u}^{(1)}$.

For more general matrices L , one needs to make sure that L has enough structure to allow for a favorable tensor decomposition of $\mathbf{u}$. If that is not possible, one can either:

- Fall back to using the matrix (31), which is always available for arbitrary finitely generated rings S with bit-bound B : or
- Alternatively, if the ring R supports it, one can run a sumcheck to reduce the claim $\langle \mathbf{u}, \psi(\mathbf{f}) \rangle = \mu$ to an MLE evaluation claim over R , thereby “replacing” the vector $\mathbf{u}$ by an “eq vector” that, as such, admits tensor decompositions of arbitrary dimensions.

Theorem 5.15. *Under the same hypotheses and notation as in [Theorem 5.7](#) for the parameters $(\ell, \ell_1, \ell_2, \Pi_{\text{iopp}}, \Pi_{\text{red}}, \mathcal{P}, (\mathcal{M}_q)_{q \in \mathcal{P}})$, the ring R , the code $\mathcal{C}$, and the binary field $\mathbb{F}_{2^\nu}$ with packing map Pack_ν , [Construction 5.13](#) is a perfectly complete $(2+t_{\text{red}}+t_{\text{iopp}})$ -round IOP for $\mathcal{R}_{\text{Lin-BitsDual}}(S_{\leq B}, \psi, R, \mathcal{C}, L)$ with round-by-round knowledge soundness with relaxation $(\mathcal{R}_{\text{Lin-BitsDual}}^\delta, \perp)$ and errors*

$$\kappa_{\text{OOD}}, \kappa_0, (\kappa_{\text{red}i})_{i \in [t_{\text{red}}]}, (\kappa_{\text{iopp}i})_{i \in [t_{\text{iopp}}]},$$

where κ_{OOD} is as in [Theorem 5.7](#) and $\kappa_0 = \theta(R, \ell)$. (see [Theorem 5.7](#); here $\theta(R, \ell)$ is computed with respect to the tensor decomposition $\mathbf{u} = \mathbf{u}^{(1)} \otimes \mathbf{u}^{(2)}$ of [Remark 5.14](#) used in [Construction 5.4](#), i.e. the quantities D, T, N_R are determined by the lifts of the entries of $\mathbf{u}^{(1)}$ and $\mathbf{u}^{(2)}$).

The error terms $\kappa_{\text{OOD}}, \kappa_0$ are void if Round 0 or Round 1 of [Construction 5.4](#) are not executed, respectively. In that case [Construction 5.13](#) has one or two rounds less.

Proof. Deferred to [Section A.1](#). In short, the result is a consequence of [Lemma 5.12](#), which establishes an equivalence between triples for $\mathcal{R}_{\text{Lin-BitsDual}}(S_{\leq B}, \psi, R, \mathcal{C}, L)$ and $\mathcal{R}_{\text{Lin-BitsDual}}(\{0, 1\}, \pi, R, \mathcal{C})$. [Lemma 5.12](#) further provides a method for efficiently computing one triple from the other. This allows to build a round-by-round knowledge soundness state and an extractor for [Construction 5.13](#) from those of [Construction 5.4](#). $\square$

5.7 Instantiation

We instantiate [Constructions 5.4](#) and [5.13](#) by letting Π_{red} be an optimized GKR protocol for multiple grand product claims of certain form (cf. [Section 6](#)). The IOP Π_{iopp} is taken to be WHIR/Ligerito [[ACFY25](#), ?] with the ring switching step from [[DP23](#)]. Our implementation of WHIR is taken from Flock’s [[BRW26](#)] implementation. The inner product claims when Π_{iopp} is called are not MLE evaluation claims. Since the ring-switch implementation in [[BRW26](#)] does not support arbitrary inner product claims, we run a sumcheck to reduce to those to MLE evaluation claims. Alternatively, one could avoid the sumcheck and use the approach from [Section B](#). We instantiate them solely with the following parameters:

Parameter	Choices
R	A finite field (possibly non-prime)
S	Integers $\mathbb{Z}$
π	Reduction mod $\text{char}(R)$
ψ	Reduction mod $\text{char}(R)$
ν	128
$\mathcal{C}$	Reed-Solomon code over $\mathbb{F}_{2^\nu}$ of rate $1/8$
Proximity parameter δ	Close to the Johnson bound
L	The standard bitification matrix (??)
$\mathcal{P}$	Discussed below
$\mathcal{M}_q$	Reduction modulo q
ℓ_1, ℓ_2	Discussed below

Table 3: Parameter choices in our deployment of [Constructions 5.4](#) and [5.13](#)

We next provide example instantiations of [Constructions 5.4](#) and [5.13](#) so as to achieve $\lambda = 128$ bits of security, while keeping the protocols as efficient as possible, with the main challenge being to use $\mathbb{F}_{2^\nu}$ of size 2^{128} for efficiency purpose. We remark that in ?? we provide experimental results both for 128 and 100 bits of security. Achieving $\lambda = 100$ bits of security is relatively straightforward and we do not discuss it. We use a careful selection of parameters in combination with grinding. In [Remark 5.17](#) we discuss other instantiations and techniques that do not require grinding.

The contents of this section should be taken as examples of how to instantiate [Constructions 5.4](#) and [5.13](#), not as strict rules. Other configurations may be perfectly plausible and practical.

In terms of security, we need to ensure the following:

1. The round-by-round knowledge soundness errors of [Construction 5.4](#) $\kappa_{\text{OOD}}, \kappa_0, (\kappa_{\text{red}_i})_i, (\kappa_{\text{iopp}_i})_i$ are all at most $2^{-\lambda}$.
2. The round-by-round knowledge soundness errors of [Construction 5.4](#) $\kappa_0, (\kappa_{\text{red}_i})_i, (\kappa_{\text{iopp}_i})_i$ are all at most $2^{-\lambda}$.

In terms of efficiency, the most important step of [Construction 5.4](#) to be aware of is Phase 3, where one uses an IOR Π_{red} to reduce the claim

$$\prod_{i \in [\ell_1]} ((g^{\gamma_i} - 1) \cdot \pi_2(\mathbf{f}_{ij}) + 1) = g^{\mu_j} \quad \in \mathbb{F}_{2^\nu}, \quad \text{for all } j \in [\ell_2], \quad (32)$$

into a linear claim on $\pi(\mathbf{f})$. As mentioned, in our instantiation, Π_{red} is the GKR protocol for multiple grand product claims, equipped with the optimizations discuss in [Section 6](#). This is the most expensive prover part of [Constructions 5.4](#) and [5.13](#), constituting between 50% to 66% of prover costs, cf. ?? [Albert: show in experiments](#). Because of this, it is important for efficiency that operating in $\mathbb{F}_{2^\nu}$ is as efficient as possible.

With this in mind, we target using $\mathbb{F}_{2^\nu}$ with $\nu \approx 128$. A natural candidate is the 2^{128} sized binary field in GHASH representation

$$\mathbb{F}_{2^\nu} = \mathbb{F}_2[X]/(X^{128} + X^7 + X^2 + X + 1).$$

Before anything else, we discuss the constraint $\max \mathcal{P} < (|\mathbb{F}_{2^\nu}| - 1)/\ell_1$. First of all, we set ℓ_1, ℓ_2 with $\ell_1 \cdot \ell_2 = \ell$ and

$$\ell_1 \in [\ell^{0.5}, \ell^{0.6}]. \quad (33)$$

In general, choosing $\ell_1 \approx \ell^{0.6}$ can lead to better prover time and proof size for the GKR grand product step, as observed in our experiments. However choosing $\ell_1 \approx \ell^{0.5}$ allows to use slightly smaller soundness errors, which reduces grinding times if targeting 128 bits of security.

We now discuss how to make the errors $\kappa_{\text{OOD}}, \kappa_0, (\kappa_{\text{red}_i})_i, (\kappa_{\text{iopp}_i})_i$ be all at most $2^{-\lambda}$.

The errors $\kappa_{i,\text{iopp}}$. The ring switching step adds an error term of the form $7/\mathbb{F}_{2^\nu}$. If targeting 128 bits of security, we use grinding (Section 4.3) of 3 bits in this round. The sumcheck before the ring switch adds similarly small error terms, which are handled with a few bits of grinding. WHIR is configured at any desired $\lambda \leq 128$ bits of security using $|\mathbb{F}_{2^\nu}| = 2^{128}$ following [?, ?]. We do not discuss this further since WHIR is a well-known IOP, and it can achieve 128 bits of security (in the Unique Decoding Radius) in 128 bit fields when codewords have dimension approximately 2^{30} (assuming an interleaved RS code) [Albert: Review. Why do we need interleaving?](#).

The error κ_{OOD} . As mentioned, we instantiate Π_{iopp} as WHIR with the ring-switching techniques of [DP23] and Section B. For $\lambda = 100$ we use a proximity parameter δ up to the Johnson bound. For $\lambda = 128$ we set δ up to the unique decoding radius, so that the Out Of Domain sampling step (Round 0) is skipped).

The errors $\kappa_{i,\text{red}}$. As mentioned, our Π_{red} IOR is the GKR protocol for multiple grand product claims from Eq. (32), see Section 6. Each round of the GKR protocol is a round of either a sumcheck for a polynomial of degree 3, or a unification of two evaluation claims of the same multilinear polynomial. Accordingly we have $\kappa_{\text{red}_i} \leq 3/|\mathbb{F}_{2^\nu}|$ for all i . This immediately guarantees $\lambda = 100$ bits of security. To achieve $\lambda = 128$ bits of security, one can do two bits of grinding (Section 4.3) per round.

The error κ_0 . The error κ_0 is perhaps the most delicate, as it demands at the same time that $|\mathcal{P}|$ is large, while still satisfying $\max \mathcal{P} < (|\mathbb{F}_{2^\nu}| - 1)/\ell_1$.

Recall that in many scenarios κ_0 does not exist in the IOP (i.e. sometimes our IOP skips Round 1). Concretely, if our initial inner product claim is over $R = \mathbb{F}_q$ a prime field of size $< (|\mathbb{F}_{2^\nu}| - 1)/\ell_1$, then κ_0 does not appear as an error term, and we do not need to do anything to adjust for it. The term κ_0 does appear in all other scenarios ($R = \mathbb{Z}$ with large or negative entries in $\mathbf{v}^{(1)}$ [Albert: negative?](#) (cf. Remark 5.5), or R a prime field of larger size, or R a proper field extension). We divide our approaches by whether this first step is skipped or not.

For the purposes of this discussion, we assume BitZ is used within a bigger proof system, as described in Section 3.1 or ??, so that BitZ is used to prove an inner product claim for a committed witness, as the last phase of a bigger proving system where [Albert: review what the technical overview exactly says](#) 1) first the initial constraint is projected to a finite field $\mathbb{F}$, or, alternatively, this step is skipped if the constraint is already over a finite field; and 2) a PIOP is run over $\mathbb{F}$ for the projected constraint, outputting an inner product claim on the projected witness. Now, this inner product claim is the input of Construction 5.4, and we aim to configure Construction 5.4 at 128 bits of security for this setting.

Strategy 1: the PIOP achieves 128 bits of security through grinding and we avoid the error term κ_0 . If allowing, say, 18 – 19 bits of grinding during each round of the PIOP (which adds 1.8 – 3.5ms per round, cf. [Table 2](#)), then, say, a Spartan-like PIOP will achieve 128 bits of security if run on a field $\mathbb{F}$ of size $\approx 2^{112}$ to 2^{113} (since each one of its rounds is either a sumcheck round for a degree ≤ 3 polynomial, or a step that batches some claims with error $1/|\mathbb{F}|$). Hence, if the initial constraint is over $\mathbb{Z}$ or $\mathbb{Z}[X]$ [Albert: do we mention \$\mathbb{Z}\[X\]\$? Why not mention other rings here? Again, wait for the TO](#), then we may project it to a 112-113 bit random prime field $\mathbb{F}$ and run a PIOP over $\mathbb{F}$. We study this case next. Our analysis also applies if the original constraint was already expressed over a prime field of such size.

Note that the projection of the constraint to $\mathbb{F}$ also requires to be set up to 128 bits of security. We discuss this in [Remark 5.16](#).

Let $q = |\mathbb{F}|$. Then if

$$q < (|\mathbb{F}_{2^\nu}| - 1)/\ell_1,$$

then κ_0 is not present as an error term, and there is nothing to adjust. Since $|\mathbb{F}_{2^\nu}| = 2^{128}$ and $q \approx 2^{113}, 2^{112}$, this occurs as long as $\ell_1 \lesssim 2^{15}, 2^{16}$, and using [\(33\)](#), as long as

$$\ell \lesssim 2^{25}, 2^{26} \quad \text{if } \ell_1 \approx \ell^{0.6}$$

and

$$\ell \lesssim 2^{30}, 2^{32} \quad \text{if } \ell_1 \approx \ell^{0.5}.$$

The specific upper bounds depend on how much grinding one is doing at the PIOP level.

Remark 5.16 (Projecting polynomial constraints to a 112-113 bit prime field with 128 bits of security). [Albert: Need to clean up the UPESAT section first](#)

Strategy 2: No or essentially no grinding during the PIOP, and κ_0 is present and grinding is used during the first projection step (Round 1) of [Construction 5.4](#). We now assume $\mathbb{F}$ is either a prime field with $|\mathbb{F}| \geq (|\mathbb{F}_{2^\nu}| - 1)/\ell_1$, or a proper extension field. In this case [Construction 5.4](#) goes through an initial projection step with soundness error κ_0 , i.e. Round 1 of [Construction 5.4](#).

Recall ([Theorem 5.7](#)) that $\kappa_0 = \theta(\mathbb{F}, \ell)$. We use $\pi_{\text{canon}} : \mathbb{Z}[X] \rightarrow \mathbb{F}$ as in [??](#), so that $\pi_{\text{canon}}(h(X)) = \pi_{\text{char}(F)}(h)(x)$ where $\pi_{\text{char}(F)}(h)$ is the polynomial h after reducing its coefficients modulo $\text{char}(F)$, and x is a root of an irreducible polynomial defining $\mathbb{F}_{q^d}$ (if $d = 1$ and so $\mathbb{F}_{q^d}$ is a prime field, then we can use a morphism $\pi_{\text{canon}} : \mathbb{Z} \rightarrow \mathbb{F}_q$). We have $D = d - 1$, $T \leq d$, and $N_{\mathbb{F}} \leq q - 1$ (see the [??](#) for a definition of these quantities). We instantiate the morphism sets $\mathcal{M}_q$ as in [Lemma 5.3](#), i.e. π_{rand} reduces coefficients modulo a sampled prime and evaluates X_1 at a random $\alpha \in \mathbb{F}_q$. In this case, by [Lemma 5.3](#),

$$\kappa_0 \leq \frac{2 \cdot (d - 1)}{q_{\min}} + \frac{\log(4 \cdot d \cdot \ell \cdot (q - 1)^2)}{|\mathcal{P}| \cdot \lfloor \log(q_{\min}) \rfloor},$$

where q is the characteristic of $\mathbb{F}$.

We select $\mathcal{P}$ to be the set of all primes in a range $[2^a, 2^b]$. There are approximately

$$|\mathcal{P}| \approx (2^b/b - 2^a/a)/\log_e(2)$$

such primes. Using $b = 128 - \tau \cdot \log(\ell)$, $\ell_1 = 2^{\tau \cdot \log(\ell)}$, for a parameter $\tau \in [0.5..0.6]$ and plugging these values into κ_0 we obtain

$$\kappa_0 \lesssim \frac{2 \cdot (d-1)}{2^a} + \frac{\log(4 \cdot d \cdot \ell \cdot (q-1)^2) \cdot \log_e(2)}{\frac{a}{b} \cdot 2^b - 2^a}, \quad (34)$$

$$b := 128 - r_1, \quad r_1 := \tau \cdot \log(\ell).$$

We calculate several minima for the right hand side of (34) when $\mathbb{F}$ is a field of size $\approx 2^{128}$ (not necessarily a prime field), for varying values of ℓ and $\tau \in \{0.5, 0.6\}$. We include the corresponding grinding times in order to achieve 128 bits of security for this initial step of [Construction 5.4](#). **Albert:** [Review grinding times](#).

$\log \mathbb{F} $	d	$\log q$	$\log \ell$	τ	a^*	$\kappa_0 \leq$	Bit-deficit to 128	Grinding time (ms)
128	1	128	24	0.5	110	$2^{-108.3}$	20	8.9
128	1	128	24	0.6	107	$2^{-105.9}$	23	84.6
128	1	128	28	0.5	108	$2^{-106.3}$	22	35.5
128	1	128	28	0.6	105	$2^{-103.5}$	25	240.8
128	1	128	30	0.5	107	$2^{-105.3}$	23	84.6
128	1	128	32	0.5	106	$2^{-104.2}$	24	148.6
128	4	32	24	0.5	114	$2^{-109.2}$	19	3.5
128	4	32	24	0.6	111	$2^{-106.8}$	22	35.5
128	4	32	28	0.5	112	$2^{-107.1}$	21	14.3
128	4	32	28	0.6	109	$2^{-104.4}$	24	148.6
128	4	32	30	0.5	111	$2^{-106.1}$	22	35.5
128	4	32	30	0.6	108	$2^{-103.1}$	25	240.8
128	4	32	32	0.5	110	$2^{-105.1}$	23	84.6

Table 4: Minima for the upper bound of κ_0 in (34) for different configurations of $\mathbb{F}$ (of characteristic q and degree d over its prime field), witness sizes ℓ , and τ such that $\ell_1 = \ell^\tau$. By choosing primes in the interval $[2^{a^*}, 2^{128-r_1}]$, we have that the first error term κ_0 satisfies is bounded by the quantity in the $\kappa_0 \leq$ column. We include the bits left required to achieve $\kappa_0 < 1/2^{128}$ and the cost of grinding such amount of bits in a MacBook Air M4 16GB, multi-threaded, using Blake3 as hash function.

Remark 5.17 (128 bits of security for larger witness sizes ℓ , without or with minimal grinding). Grinding costs scale exponentially in the number of grinding bits. Because of this, we cannot resort to grinding for very large witness sizes ℓ . There are at least two ways to achieve 128 bits of security (or more), in this case.

Method 1: Use a larger binary field $\mathbb{F}_{2^\nu}$, i.e. simply take $\mathbb{F}_{2^\nu}$ with $|\mathbb{F}_{2^\nu}|$ comfortably larger, rather than $|\mathbb{F}_{2^\nu}| = 2^{128}$.

Method 2: We generalize an idea from [Bin26]. In this setting we make the assumption that the entries in γ (cf. Phase 3 of [Construction 5.4](#)) are smaller than $|\mathbb{F}_{2^\nu}| - 1$. This means we require $\max \mathcal{P} < |\mathbb{F}_{2^\nu}| - 1$ instead of $\max \mathcal{P} < (|\mathbb{F}_{2^\nu}| - 1)/\ell_1$ (and similarly for the conditions that would allow to skip Round 1 in [Construction 5.4](#)). Hence now the bound on the primes in $\mathcal{P}$ does not depend on the witness size.

Note that when we attempt to prove the integer equalities $\langle \gamma, \mathbf{f}_{*j} \rangle = \mu_j$ ($j \in [\ell_2]$) “in the exponent” during Phase 3 of [Construction 5.4](#), what we are really doing is proving the equalities modulo $|\mathbb{F}_{2^\nu}| - 1$. So far, we have used $\mathbb{F}_{2^\nu}$ large enough hat this distinction does not matter, in that, because of the total possible magnitudes of $\langle \gamma, \mathbf{f}_{*j} \rangle$ does not exceed $|\mathbb{F}_{2^\nu}| - 2$.

If that is not the case, one can, in addition to proving $\langle \gamma, \mathbf{f}_{*j} \rangle = \mu_j$ modulo $|\mathbb{F}_{2^\nu}| - 1$, prove the same equality modulo ℓ_1 . This works because, since ℓ_1 and $|\mathbb{F}_{2^\nu}| - 1$ are coprime (because ℓ_1 is a power of two), by the CRT, if both equalities are true, then $\langle \gamma, \mathbf{f}_{*j} \rangle - \mu_j$ is an integer multiple of $\ell_1 \cdot (|\mathbb{F}_{2^\nu}| - 1)$. However, because the entries of $\mathbf{f}$ are bits and the entries of γ are at most $|\mathbb{F}_{2^\nu}| - 2$, this means that $\langle \gamma, \mathbf{f}_{*j} \rangle = \mu_j$ as integers.

To prove that $\langle \gamma, \mathbf{f}_{*j} \rangle = \mu_j$ modulo ℓ_1 , one can prove the integer equality $\langle \gamma', \mathbf{f}_{*j} \rangle = \mu'_j + a_j \cdot \ell_1$ for some $a_j \in \mathbb{Z}$ (provided by the prover), where γ' contains the entries of γ reduced modulo ℓ_1 , and μ'_j is μ_j reduced modulo ℓ_1 . The verifier checks that a_j is not too large, namely it checks that $|a_j| \cdot \ell_1 \leq (\ell_1 - 1) \cdot \ell_1$, which is the maximum value $|a_j| \cdot \ell_1$ can achieve when $\mathbf{f}$ is filled with bits (which is guaranteed by our commitment via WHIR). Hence the value $|\langle \gamma', \mathbf{f}_{*j} \rangle - \mu'_j - a_j \cdot \ell_1|$ can be at most $2 \cdot (\ell_1 + 1) \cdot \ell_1$. As long as $2 \cdot (\ell_1 + 1) \cdot \ell_1 < |\mathbb{F}_{2^\nu}| - 1$, this equality can be proved “in the exponent” without causing overflow.

Note that if γ admits a tensor decomposition $\gamma = \gamma^{(1)} \otimes \gamma^{(2)}$, then we can take, instead of γ' , the tensor-product $\gamma^{(1)'} \otimes \gamma^{(2)'}$, where $\gamma^{(i)'}$ denotes $\gamma^{(i)}$ reduced modulo ℓ_1 . This allows to use all of our “low-entropy” grand product optimizations from [Section 6](#), though in this case the required bound becomes $2 \cdot (\ell_1^2 + 1) \cdot \ell_1 < |\mathbb{F}_{2^\nu}| - 1$.

Overall, this method allows to use $\max \mathcal{P} < |\mathbb{F}_{2^\nu}| - 1$, meaning we can choose primes that are $\log(\ell_1)$ bits larger than we otherwise would. This makes soundness errors be smaller by this same amount of bits, which leads to small grinding requirements. The cost is an extra batch of low entropy grand products.

6 A GKR protocol for low entropy batched grand products via lookup tables

The bulk of the work in $\text{Bit}\mathbb{Z}$ ([Construction 5.4](#)) is to reduce the claim over the binary field $\mathbb{F}_{2^\nu}$

$$\prod_{i \in [\ell_1]} ((g^{\gamma_i} - 1) \cdot \pi_2(\mathbf{f}_{ij}) + 1) = g^{\mu_j} \quad \in \mathbb{F}_{2^\nu}, \quad \text{for all } j \in [\ell_2] \quad (35)$$

to an inner product claim of the form $\langle \mathbf{v}, \pi_2(\mathbf{f}) \rangle = \mu$ over $\mathbb{F}_{2^\nu}$, for some $\mathbf{v} \in \mathbb{F}_{2^\nu}^\ell$, where here $\mathbf{f} \in \{0, 1\}^\ell \subseteq \mathbb{Z}^\ell$. Recall that

$$\ell = \ell_1 \cdot \ell_2.$$

To fix intuition, in our instantiation (cf. [Section 5.7](#)) we set $\ell_1 \approx \ell^{0.6}, \ell_2 = \ell^{0.4}$.

Cryptographically speaking, our protocol is simply a standard batched version of the GKR protocol applied to grand products from [\[?\]](#). However, our implementation uses lookup tables (i.e. preprocessed data) to optimize the prover time of such protocol. In particular, our techniques leverage the fact that the factors in (35) have *low entropy*, in the sense that for every $i \in [\ell_1]$ and $j \in [\ell_2]$, the i -th factor in the j -th grand product of (35) can take only two values, depending on whether $\mathbf{f}_{ij} = 1$ or $\mathbf{f}_{ij} = 0$.

The optimizations we introduce apply to any (batched) grand product with low entropy factors, and not only to the grand product in (35). They also work on any field. Because of this, we describe

a general protocol IOR that, given a finite field $\mathbb{K}$; a small subset $S \subseteq \mathbb{K}$; and dimensions $\ell = \ell_1 \cdot \ell_2$, reduces triples from

$$\mathcal{R}_{\text{PESAT}}(\mathbb{K}, p) = \left\{ \left(\begin{array}{l} \mathbf{x} = (\mathbf{y}, \boldsymbol{\mu}) \in \mathbb{K}^{\ell_1} \times \mathbb{K}^{\ell_2}, \\ \mathbf{y} = \perp, \\ \mathbf{w} = \mathbf{f} = (\mathbf{f}_{*j})_{j \in [\ell_2]} \in S^{\ell_1 \times \ell_2} \subseteq \mathbb{K}^{\ell_1 \times \ell_2} \end{array} \right) \left| \prod_{i \in [\ell_1]} p(\mathbf{y}_i, \mathbf{f}_{ij}) = \boldsymbol{\mu}_j \text{ for all } j \in [\ell_2] \right. \right\},$$

to triples in

$$\mathcal{R}_{\text{Lin}}(\mathbb{K}) = \left\{ \left(\begin{array}{l} \mathbf{x} = (\mathbf{u}, \mu) \in \mathbb{K}^\ell \times \mathbb{K}, \\ \mathbf{y} = \perp, \\ \mathbf{w} = \mathbf{f} \in S^{\ell_1 \times \ell_2} \subseteq \mathbb{K}^{\ell_1 \times \ell_2} \end{array} \right) \left| \langle \mathbf{u}, \mathbf{f} \rangle = \mu \right. \right\}.$$

One can additionally add an encoding $f = \text{Enc}_C(\mathbf{f})$ as an oracle in the above relations (matching the context in which we use the IOR), but our IOR does not make any use of it whatsoever, hence we omit it.

We describe Π_{red} now. We will describe our implementation-level optimizations afterwards. As we mentioned, cryptographically/protocol-wise, our protocol is a standard batched GKR for grand products [?]. We assume ℓ_1, ℓ_2 are powers of two:

$$\ell_1 = 2^{r_1}, \quad \ell_2 = 2^{r_2}.$$

We index $[\ell_1], [\ell_2]$ via elements of the hypercubes $\{0, 1\}^{r_1}, \{0, 1\}^{r_2}$, respectively.

Construction 6.1 (An IOR Π_{red} from $\mathcal{R}_{\text{PESAT}}(\mathbb{K}, p)$ to $\mathcal{R}_{\text{Lin}}(\mathbb{K})$). **Input.** The prover and verifier receive $(\mathbf{y}, \boldsymbol{\mu}) \in \mathbb{K}^{\ell_1} \times \mathbb{K}^{\ell_2}$. The prover receives also $\mathbf{f} \in S^{\ell_1 \times \ell_2} \subseteq \mathbb{K}^{\ell_1 \times \ell_2}$.

Let L_{r_1} be a *multilinear* polynomial over $\mathbb{K}$ on $r_1 + r_2$ variables such that

$$L_{r_1}(\mathbf{b}, \mathbf{c}) = p(\mathbf{y}_{\mathbf{b}}, \mathbf{f}_{\mathbf{b}, \mathbf{c}}) \quad \text{for all } (\mathbf{b}, \mathbf{c}) \in \{0, 1\}^{r_1} \times \{0, 1\}^{r_2}.$$

(note that p may not be multilinear). For $i = 0, \dots, r_1 - 1$, let L_i be a multilinear polynomial on $r_2 + i$ variables such that

$$L_i(\mathbf{b}, \mathbf{c}) = L_{i+1}(\mathbf{b}, 0, \mathbf{c}) \cdot L_{i+1}(\mathbf{b}, 1, \mathbf{c}) \quad \text{for all } (\mathbf{b}, \mathbf{c}) \in \{0, 1\}^i \times \{0, 1\}^{r_2}.$$

1. The verifier sends $\boldsymbol{\alpha}^{(0)} \in \mathbb{K}^{r_2}$. Both prover and verifier compute $C^{(0)} := L_0(\boldsymbol{\alpha}^{(0)})$ (note that L_0 is the multilinear extension of $\boldsymbol{\mu}$ over $\mathbb{K}$).
2. For $i = 0, \dots, r_1 - 1$:

- (a) The prover and verifier run the sumcheck protocol for the following claim:

$$C^{(i)} = \sum_{(\mathbf{b}, \mathbf{c}) \in \{0, 1\}^i \times \{0, 1\}^{r_2}} L_{i+1}(\mathbf{b}, 0, \mathbf{c}) \cdot L_{i+1}(\mathbf{b}, 1, \mathbf{c}) \cdot \text{eq}(\mathbf{b}, \mathbf{c}; \boldsymbol{\alpha}^{(i)}).$$

At the end of the sumcheck protocol, the prover and verifier are left with a claim of the form

$$L_{i+1}(\mathbf{r}_{\mathbf{b}}^{(i)}, 0, \mathbf{r}_{\mathbf{c}}^{(i)}) = a_0^{(i)}, \quad L_{i+1}(\mathbf{r}_{\mathbf{b}}^{(i)}, 1, \mathbf{r}_{\mathbf{c}}^{(i)}) = a_1^{(i)} \tag{36}$$

for some $\mathbf{r}_{\mathbf{b}}^{(i)} \in \mathbb{K}^i$, $\mathbf{r}_{\mathbf{c}}^{(i)} \in \mathbb{K}^{r_2}$ and $a_0^{(i)}, a_1^{(i)} \in \mathbb{K}$.

(b) The prover and verifier unify the two claims from (36) into a single one via the following standard procedure:

- i. The verifier sends a random element $\beta^{(i)} \in \mathbb{K}$.
- ii. The claim (36) is replaced by

$$L_{i+1}(\mathbf{r}_{\mathbf{b}}^{(i)}, \beta^{(i)}, \mathbf{r}_{\mathbf{c}}^{(i)}) = C^{(i+1)} := a_0^{(i)} \cdot (1 - \beta^{(i)}) + a_1^{(i)} \cdot \beta^{(i)}.$$

Let

$$\boldsymbol{\alpha}^{(i+1)} := (\mathbf{r}_{\mathbf{b}}^{(i)}, \beta^{(i)}, \mathbf{r}_{\mathbf{c}}^{(i)}).$$

3. At this stage, the prover and verifier are left with the claim

$$L_{r_1}(\boldsymbol{\alpha}^{(r_1)}) = C^{(r_1)}.$$

This claim is equivalent to

$$\sum_{(\mathbf{b}, \mathbf{c}) \in \{0,1\}^{r_1} \times \{0,1\}^{r_2}} p(\mathbf{y}_{\mathbf{b}}, \mathbf{f}_{\mathbf{b}, \mathbf{c}}) \cdot \text{eq}((\mathbf{b}, \mathbf{c}), \boldsymbol{\alpha}^{(r_1)}) = C^{(r_1)}, \quad (37)$$

If p is multilinear, then (37) is a linear claim on $\mathbf{f}$. Namely, it is equivalent to

$$\langle \mathbf{u}, \mathbf{f} \rangle = C^{(r_1)} + c$$

for some $\mathbf{u} \in \mathbb{K}^\ell, c \in \mathbb{K}$. In this case, prover and verifier output

$$(\mathbf{x}' = (\mathbf{u}, C^{(r_1)} + c), \perp, \mathbf{w}' = \mathbf{f}) \in \mathcal{R}_{\text{Lin}}(\mathbb{K}).$$

Otherwise, i.e. if p is not multilinear, prover and verifier run a sumcheck for the claim (37), which results in MLE claims for $\mathbf{y}$ and $\mathbf{f}$. The verifier checks the claim for $\mathbf{y}$ itself, and prover and verifier output the MLE claim for $\mathbf{f}$.

6.1 Optimization via lookup tables

We optimize Construction 6.1 as follows. For simplicity we use the subset

$$S = \{0, 1\} \subseteq \mathbb{K},$$

but any small subset of $\mathbb{K}$ is amenable to the techniques discussed here.

Setup We start with the following observation: for every $(\mathbf{b}, \mathbf{c}) \in \{0, 1\}^{r_1} \times \{0, 1\}^{r_2}$, we have $\mathbf{f}_{\mathbf{b}, \mathbf{c}} \in S = \{0, 1\}$, and so

$$L_{r_1}(\mathbf{b}, \mathbf{c}) = p(\mathbf{z}_{\mathbf{b}}, \mathbf{f}_{\mathbf{b}, \mathbf{c}}) = \begin{cases} y_{\mathbf{b}}^{(1)} := p(\mathbf{z}_{\mathbf{b}}, 1) & \text{if } \mathbf{f}_{\mathbf{b}, \mathbf{c}} = 1 \\ y_{\mathbf{b}}^{(0)} := p(\mathbf{z}_{\mathbf{b}}, 0) & \text{if } \mathbf{f}_{\mathbf{b}, \mathbf{c}} = 0. \end{cases}$$

The values $y_{\mathbf{b}}^{(0)}, y_{\mathbf{b}}^{(1)}$ do not depend on $\mathbf{c}$. Hence the prover can set up a lookup table $T_{r_1} \subseteq (\mathbb{K} \times \mathbb{K})^{\{0,1\}^{r_1}}$ defined as

$$T_{r_1}[\mathbf{b}] = (y_{\mathbf{b}}^{(1)}, y_{\mathbf{b}}^{(0)}) \quad \text{for all } \mathbf{b} \in \{0, 1\}^{r_1}$$

at the beginning of the protocol, and, whenever required, use it to obtain the values $L_{r_1}(\mathbf{b}, \mathbf{c})$. Concretely

$$L_{r_1}(\mathbf{b}, \mathbf{c}) = T_{r_1}[\mathbf{b}][\mathbf{f}_{\mathbf{b}, \mathbf{c}}] \quad \text{for all } (\mathbf{b}, \mathbf{c}) \in \{0, 1\}^{r_1} \times \{0, 1\}^{r_2}$$

In our instantiation, $\mathbf{y}_{\mathbf{b}}^{(1)} = g^{\gamma_{\mathbf{b}}}$ and $\mathbf{y}_{\mathbf{b}}^{(0)} = 1$ (cf. ??). Hence additionally the prover precomputes the elements $g^{\gamma_{\mathbf{b}}}$ at the beginning of the protocol, before computing T_{r_1} .

As we will explain later, the prover will also precompute similar tables for the values $L_{r_1-j}(\mathbf{b}, \mathbf{c})$ of the levels close to the leaves (i.e. for small j),

$$T_{r_1-j} \in \left(\mathbb{K}^{2^{2^j}} \right)^{\{0, 1\}^{r_1-j}}, \quad (38)$$

as well as other tables relevant for the different sumcheck rounds of [Construction 6.1](#), constructed from the tables T_{r_1-j} themselves. The superindex $\{0, 1\}^{r_1-j}$ in (38) indicates that the table T_{r_1-j} has one entry per element in the hypercube $\{0, 1\}^{r_1-j}$; and this entry is a vector of 2^{2^j} field elements. Concretely, for $\mathbf{b} \in \{0, 1\}^{r_1-j}$, $T_{r_1-j}[\mathbf{b}]$ consists of all possible products of 2^j consecutive factors of the grand product, namely the factors indexed by $(\mathbf{b}, \mathbf{b}')$ with $\mathbf{b}$ fixed and $\mathbf{b}'$ ranging over $\{0, 1\}^j$. Each factor can take two values, depending on whether $\mathbf{f}_{\mathbf{b}, \mathbf{b}', \mathbf{c}}$ is 1 or 0. Hence $T_{r_1-j}[\mathbf{b}]$ consists of 2^{2^j} elements. Hence the size of T_{r_1-j} in (38).

Because $|T_{r_1-j}|$ grows doubly-exponentially in j , these tables are viable only for the bottom few levels of the tree ($j = 0, 1, 2$ in our implementation). In higher levels we use standard sumchecks.

Importantly, all these tables have size independent of ℓ_2 . This (among others, cf. ??), is one of the reasons why we use the tensor decomposition structure, i.e. the split given by $\ell = \ell_1 \cdot \ell_2$, in [Construction 5.4](#).

Remark 6.2. All precomputed data in this section depends on the input vector $\mathbf{y}$ of $\mathcal{R}_{\text{PESAT}}(\mathbb{K}, p)$, and possibly on further verifier challenges. Since, in a wider context where [Construction 6.1](#) is a subprocess of a larger proof system, $\mathbf{y}$ typically depends on verifier's challenges, these precomputations must be done at *runtime*.

First prover message in the last sumcheck in Step 2 of [Construction 6.1](#) We start by analyzing Step 2 with $i = r_1 - 1$. In this step prover and verifier run a sumcheck for the claim

$$C^{(r_1-1)} = \sum_{(\mathbf{b}, \mathbf{c}) \in \{0, 1\}^{r_1-1} \times \{0, 1\}^{r_2}} L_{r_1}(\mathbf{b}, 0, \mathbf{c}) \cdot L_{r_1}(\mathbf{b}, 1, \mathbf{c}) \cdot \text{eq}(\mathbf{b}, \mathbf{c}; \boldsymbol{\alpha}^{(i)})$$

This is the most expensive sumcheck of [Construction 6.1](#) for the prover (if done naively), since it is the sumcheck over the largest number of variables throughout [Construction 6.1](#).

For $1 \leq j \leq r_1 - 1$, the j -th prover message in this sumcheck is, if $\boldsymbol{\rho}^{(j)} = (\rho_1, \dots, \rho_{j-1})$ is the vector of verifier challenges so far in this sumcheck,

$$\begin{aligned} h_j(X) &:= \sum_{(\mathbf{b}, \mathbf{c}) \in \{0, 1\}^{r_1-1-j} \times \{0, 1\}^{r_2}} L_{r_1}(\boldsymbol{\rho}^{(j)}, X, \mathbf{b}, 0, \mathbf{c}) \cdot L_{r_1}(\boldsymbol{\rho}^{(j)}, X, \mathbf{b}, 1, \mathbf{c}) \cdot \text{eq}(\boldsymbol{\rho}^{(j)}, X, \mathbf{b}, \mathbf{c}; \boldsymbol{\alpha}^{(i)}) \\ &= \sum_{\mathbf{c} \in \{0, 1\}^{r_2}} \left(\underbrace{\sum_{\mathbf{b} \in \{0, 1\}^{r_1-1-j}} L_{r_1}(\boldsymbol{\rho}^{(j)}, X, \mathbf{b}, 0, \mathbf{c}) \cdot L_{r_1}(\boldsymbol{\rho}^{(j)}, X, \mathbf{b}, 1, \mathbf{c}) \cdot \text{eq}(\boldsymbol{\rho}^{(j)}, \mathbf{b}; \boldsymbol{\alpha}^{(i,1)})}_{:= h_{j, \mathbf{c}}(X)} \right) \cdot \text{eq}(X, \mathbf{c}; \boldsymbol{\alpha}^{(i,2)}) \end{aligned}$$

$$= \sum_{\mathbf{c} \in \{0,1\}^{r_2}} h_{j,\mathbf{c}}(X) \cdot \text{eq}(X, \mathbf{c}; \boldsymbol{\alpha}^{(i,2)}),$$

where $\boldsymbol{\alpha}^{(i)} = (\boldsymbol{\alpha}^{(i,1)}, \boldsymbol{\alpha}^{(i,2)}) \in \mathbb{K}^{r_1-j-1} \times \mathbb{K}^{r_2+1}$. We claim that

$$L_{r_1}(\boldsymbol{\rho}^{(j)}, X, \mathbf{b}, 0, \mathbf{c}) = \left(L_{r_1}(\boldsymbol{\rho}^{(j)}, 1, \mathbf{b}, 0, \mathbf{c}) - L_{r_1}(\boldsymbol{\rho}^{(j)}, 0, \mathbf{b}, 0, \mathbf{c}) \right) \cdot X + L_{r_1}(\boldsymbol{\rho}^{(j)}, 0, \mathbf{b}, 0, \mathbf{c}) \quad (39)$$

Indeed, $L_{r_1}(\boldsymbol{\rho}^{(j)}, X, \mathbf{b}, 0, \mathbf{c})$ is a linear polynomial, and the right-hand side in the above expression is the line passing through the two evaluations of $L_{r_1}(\boldsymbol{\rho}^{(j)}, X, \mathbf{b}, 0, \mathbf{c})$ at $X = 0$ and $X = 1$. An analogous expression holds for $L_{r_1}(\boldsymbol{\rho}^{(j)}, X, \mathbf{b}, 1, \mathbf{c})$:

$$L_{r_1}(\boldsymbol{\rho}^{(j)}, X, \mathbf{b}, 1, \mathbf{c}) = \left(L_{r_1}(\boldsymbol{\rho}^{(j)}, 1, \mathbf{b}, 1, \mathbf{c}) - L_{r_1}(\boldsymbol{\rho}^{(j)}, 0, \mathbf{b}, 1, \mathbf{c}) \right) \cdot X + L_{r_1}(\boldsymbol{\rho}^{(j)}, 0, \mathbf{b}, 1, \mathbf{c}). \quad (40)$$

Renaming

$$E_{\varepsilon_1 \varepsilon_2}^{(j)}[\mathbf{b}, \mathbf{c}] := L_{r_1}(\boldsymbol{\rho}^{(j)}, \varepsilon_1, \mathbf{b}, \varepsilon_2, \mathbf{c}) \quad \varepsilon_1, \varepsilon_2 \in \{0, 1\}, \quad \omega[\mathbf{b}] := \text{eq}(\boldsymbol{\rho}^{(j)}, \mathbf{b}; \boldsymbol{\alpha}^{(i,1)}),$$

we obtain

$$h_{j,\mathbf{c}}(X) = \sum_{\mathbf{b} \in \{0,1\}^{r_1-1-j}} X^2 \cdot A_2^{(j)}[\mathbf{b}, \mathbf{c}] + X \cdot A_1^{(j)}[\mathbf{b}, \mathbf{c}] + A_0^{(j)}[\mathbf{b}, \mathbf{c}]$$

where,

$$\begin{aligned} A_0^{(j)}[\mathbf{b}, \mathbf{c}] &:= E_{00}^{(j)}[\mathbf{b}, \mathbf{c}] \cdot E_{01}^{(j)}[\mathbf{b}, \mathbf{c}] \cdot \omega[\mathbf{b}], \\ A_1^{(j)}[\mathbf{b}, \mathbf{c}] &:= \left((E_{10}^{(j)}[\mathbf{b}, \mathbf{c}] - E_{00}^{(j)}[\mathbf{b}, \mathbf{c}]) \cdot E_{01}^{(j)}[\mathbf{b}, \mathbf{c}] + (E_{11}^{(j)}[\mathbf{b}, \mathbf{c}] - E_{01}^{(j)}[\mathbf{b}, \mathbf{c}]) \cdot E_{00}^{(j)}[\mathbf{b}, \mathbf{c}] \right) \cdot \omega[\mathbf{b}] \\ A_2^{(j)}[\mathbf{b}, \mathbf{c}] &:= (E_{10}^{(j)}[\mathbf{b}, \mathbf{c}] - E_{00}^{(j)}[\mathbf{b}, \mathbf{c}]) \cdot (E_{11}^{(j)}[\mathbf{b}, \mathbf{c}] - E_{01}^{(j)}[\mathbf{b}, \mathbf{c}]) \cdot \omega[\mathbf{b}]. \end{aligned}$$

Now, for $j = 1$, we have $\boldsymbol{\rho}^{(j)} = \perp$, and hence each value $E_{\varepsilon_1 \varepsilon_2}^{(1)}[\mathbf{b}, \mathbf{c}]$ can be obtained, whenever needed, by querying $T_{r_1}[\mathbf{b}]$ and selecting $T_{r_1}[\mathbf{b}][0]$ or $T_{r_1}[\mathbf{b}][1]$ depending on whether $\mathbf{f}_{\mathbf{b},\mathbf{c}} = 1$ or 0. A consequence is that the coefficients themselves $A_k^{(1)}[\mathbf{b}, \mathbf{c}]$ ($k \in [3]$) can be obtained through lookup tables, as follows: we have

$$A_0^{(1)}[\mathbf{b}, \mathbf{c}] = T_{r_1}[0, \mathbf{b}, 0][\mathbf{f}_{(0,\mathbf{b},0),\mathbf{c}}] \cdot T_{r_1}[0, \mathbf{b}, 1][\mathbf{f}_{(0,\mathbf{b},1),\mathbf{c}}] \cdot \omega[\mathbf{b}]$$

Hence

$$A_0^{(1)}[\mathbf{b}, \mathbf{c}] = \begin{cases} \mathbf{y}_{0,\mathbf{b},0}^{(1)} \cdot \mathbf{y}_{0,\mathbf{b},1}^{(1)} \cdot \omega[\mathbf{b}] & \text{if } \mathbf{f}_{(0,\mathbf{b},0),\mathbf{c}} = 1, \mathbf{f}_{(0,\mathbf{b},1),\mathbf{c}} = 1 \\ \mathbf{y}_{0,\mathbf{b},0}^{(1)} \cdot \mathbf{y}_{0,\mathbf{b},1}^{(0)} \cdot \omega[\mathbf{b}] & \text{if } \mathbf{f}_{(0,\mathbf{b},0),\mathbf{c}} = 1, \mathbf{f}_{(0,\mathbf{b},1),\mathbf{c}} = 0 \\ \mathbf{y}_{0,\mathbf{b},0}^{(0)} \cdot \mathbf{y}_{0,\mathbf{b},1}^{(1)} \cdot \omega[\mathbf{b}] & \text{if } \mathbf{f}_{(0,\mathbf{b},0),\mathbf{c}} = 0, \mathbf{f}_{(0,\mathbf{b},1),\mathbf{c}} = 1 \\ \mathbf{y}_{0,\mathbf{b},0}^{(0)} \cdot \mathbf{y}_{0,\mathbf{b},1}^{(0)} \cdot \omega[\mathbf{b}] & \text{if } \mathbf{f}_{(0,\mathbf{b},0),\mathbf{c}} = 0, \mathbf{f}_{(0,\mathbf{b},1),\mathbf{c}} = 0 \end{cases}$$

Thus the prover can precompute the table

$$T_{A_0} \in (\mathbb{K}^4)^{\{0,1\}^{r_1-2}} = \mathbb{K}^{\ell_1}$$

given by

$$T_{A_0}[\mathbf{b}] = (\mathbf{y}_{0,\mathbf{b},0}^{(1)} \cdot \mathbf{y}_{0,\mathbf{b},1}^{(1)} \cdot \omega[\mathbf{b}], \mathbf{y}_{0,\mathbf{b},0}^{(1)} \cdot \mathbf{y}_{0,\mathbf{b},1}^{(0)} \cdot \omega[\mathbf{b}], \mathbf{y}_{0,\mathbf{b},0}^{(0)} \cdot \mathbf{y}_{0,\mathbf{b},1}^{(1)} \cdot \omega[\mathbf{b}], \mathbf{y}_{0,\mathbf{b},0}^{(0)} \cdot \mathbf{y}_{0,\mathbf{b},1}^{(0)} \cdot \omega[\mathbf{b}])$$

and use it to obtain any value $A_0^{(1)}[\mathbf{b}, \mathbf{c}]$ for $(\mathbf{b}, \mathbf{c}) \in \{0, 1\}^{r_1-2} \times \{0, 1\}^{r_2}$ via a single read into T_{A_0} . The size of T_{A_0} , again, does not depend on $\ell_2 = 2^{r_2}$. Building T_{A_0} requires at most $8 \cdot 2^{r_1-2} = 2^{r_1+1}$ multiplications and no additions.

We now explain how to obtain the values $A_1^{(1)}[\mathbf{b}, \mathbf{c}]$ and $A_2^{(1)}[\mathbf{b}, \mathbf{c}]$, again via the construction of a lookup table. Note that these depend on the *four* bits $\mathbf{f}_{(\varepsilon_1, \mathbf{b}, \varepsilon_2), \mathbf{c}}$, $\varepsilon_1, \varepsilon_2 \in \{0, 1\}$, so our tables will need $2^4 = 16$ entries (as opposed to 4 as in T_{A_0}) per index $\mathbf{b}$. The prover proceeds as follows:

- It computes a table $T_{S_1} \in (\mathbb{K}^4)^{\{0,1\}^{r_1-2}}$ analogous to T_{A_0} , which allows to obtain the values

$$S_1^{(1)}[\mathbf{b}, \mathbf{c}] := E_{10}^{(1)}[\mathbf{b}, \mathbf{c}] \cdot E_{11}^{(1)}[\mathbf{b}, \mathbf{c}] \cdot \omega[\mathbf{b}],$$

(observe that T_{A_0} allows to obtain the values $E_{00}^{(1)}[\mathbf{b}, \mathbf{c}] \cdot E_{01}^{(1)}[\mathbf{b}, \mathbf{c}] \cdot \omega[\mathbf{b}]$).

- It computes the table $T_{A_2} \in (\mathbb{K}^{16})^{\{0,1\}^{r_1-2}}$ given by

$$T_{A_2}[\mathbf{b}] = \left((\mathbf{y}_{1,\mathbf{b},0}^{(x_1)} - \mathbf{y}_{0,\mathbf{b},0}^{(x_0)}) \cdot (\mathbf{y}_{1,\mathbf{b},1}^{(x'_1)} - \mathbf{y}_{0,\mathbf{b},1}^{(x'_0)}) \cdot \omega[\mathbf{b}] \right)_{(x_0, x_1, x'_0, x'_1) \in \{0,1\}^4}, \quad (41)$$

so that $A_2^{(1)}[\mathbf{b}, \mathbf{c}]$ is the element

$$A_2^{(1)}[\mathbf{b}, \mathbf{c}] = T_{A_2}[\mathbf{b}][\mathbf{f}_{(0,\mathbf{b},0),\mathbf{c}}, \mathbf{f}_{(1,\mathbf{b},0),\mathbf{c}}, \mathbf{f}_{(0,\mathbf{b},1),\mathbf{c}}, \mathbf{f}_{(1,\mathbf{b},1),\mathbf{c}}].$$

Building T_{A_2} takes at most 20 multiplications and 8 additions per index $\mathbf{b}$ ⁵. I.e. at most $5 \cdot 2^{r_1}$ multiplications and 2^{r_1+1} additions in total. [Albert: Discuss cost of other tables?](#)

- It obtains the values $A_1^{(1)}[\mathbf{b}, \mathbf{c}]$ via the identity

$$S_1^{(1)}[\mathbf{b}, \mathbf{c}] = A_0^{(1)}[\mathbf{b}, \mathbf{c}] + A_1^{(1)}[\mathbf{b}, \mathbf{c}] + A_2^{(1)}[\mathbf{b}, \mathbf{c}] \quad \text{for all } (\mathbf{b}, \mathbf{c}) \in \{0, 1\}^{r_1-2} \times \{0, 1\}^{r_2}.$$

Once the tables have been computed, obtaining $h_{1,\mathbf{c}}(X)$ for one $\mathbf{c} \in \{0, 1\}^{r_2}$ costs $3 \cdot 2^{r_1-2}$ table reads (one read per table $T_{A_0}, T_{S_1}, T_{A_2}$ per index $\mathbf{b} \in \{0, 1\}^{r_1-2}$) and $3 \cdot 2^{r_1-2} + 2$ additions: this is the cost of adding together the values read at each table, providing $\sum_{\mathbf{b}} A_0[\mathbf{b}, \mathbf{c}]$, $\sum_{\mathbf{b}} S_1[\mathbf{b}, \mathbf{c}]$, $\sum_{\mathbf{b}} A_2[\mathbf{b}, \mathbf{c}]$, and then doing two additions to obtain $\sum_{\mathbf{b}} A_1[\mathbf{b}, \mathbf{c}] = \sum_{\mathbf{b}} S_1[\mathbf{b}, \mathbf{c}] - \sum_{\mathbf{b}} A_0[\mathbf{b}, \mathbf{c}] - \sum_{\mathbf{b}} A_2[\mathbf{b}, \mathbf{c}]$. In total, this amounts to at most $3\ell/4$ table reads and $\approx 3\ell/4 + 2 \cdot \ell_2$ additions to compute $h_{1,\mathbf{c}}$ for all $\mathbf{c}$, and no multiplications (in our instantiation, additions in $\mathbb{K}$ are 128-bit XORs).

Finally, the prover computes $h_1(X)$ from the polynomials $h_{1,\mathbf{c}}(X)$ with at most $5 \cdot 2^{r_2}$ additions and $3 \cdot 2^{r_2}$ multiplications (plus 2^{r_2} multiplications to precompute the values $\text{eq}(\mathbf{c}; \boldsymbol{\alpha}^{(i,2)})$, shared by all rounds of this sumcheck, and 2^{r_1-2} multiplications for the weights $\omega[\mathbf{b}]$).

In [Table 5](#) we record the total upper bound costs for computing $h_1(X)$. In our application we obtain lower numbers since $\mathbf{y}_{\mathbf{b}}^{(0)} = 1$ and we work in characteristic two, and so in [Eq. \(41\)](#) some elements collapse to 0.

On the other hand, a standard generic approach to compute $h_1(X)$ would require $O(\ell) = O(\ell_1 \cdot \ell_2)$ multiplications and additions, even if using the few sumcheck optimizations that apply in our settings (to our knowledge) [Albert: which ones, exactly?](#). Crucially, we note that our sumcheck features

⁵For the multiplications, there are 4 multiplications to compute the 4 elements $t_{x'_0, x'_1, \mathbf{b}} := (\mathbf{y}_{1,\mathbf{b},1}^{(x'_1)} - \mathbf{y}_{0,\mathbf{b},1}^{(x'_0)}) \cdot \omega[\mathbf{b}]$, and then 16 more multiplications to compute $(\mathbf{y}_{1,\mathbf{b},0}^{(x_1)} - \mathbf{y}_{0,\mathbf{b},0}^{(x_0)}) \cdot t_{x'_0, x'_1, \mathbf{b}}$

summands of full field bit size $\log |\mathbb{K}|$, which voids some known powerful optimizations when the summands are small, e.g. the univariate skip [?] and some optimizations from [?] **Albert: This is a crucial point. It should be highlighted more. And it should be an important point of the introduction and technical overview**

On the other hand, a standard linear-time sumcheck prover [?] computes $h_1(X)$ with $\Theta(\ell) = \Theta(\ell_1 \cdot \ell_2)$ multiplications in $\mathbb{K}$, several per pair of summands, for the products $L_{r_1}(\cdot, 0, \cdot) \cdot L_{r_1}(\cdot, 1, \cdot)$ and their **eq** weights. The known techniques for lowering this cost do not apply here, because the polynomials being multiplied take arbitrary values in $\mathbb{K}$: the values $L_{r_1}(\mathbf{b}, \mathbf{c}) \in \{y_{\mathbf{b}}^{(0)}, y_{\mathbf{b}}^{(1)}\}$ are full $\log |\mathbb{K}|$ -bit field elements (the elements $g^{\gamma_{\mathbf{b}}}$ in our instantiation), and the entries of every higher level of the tree are products of them. The univariate skip of [?] gains by running the first rounds over a small base field, of which there is none here, and the small-value algorithms of [?, ?] require the multilinear polynomials to take small values, which they do not. (The one technique that does carry over is the degree reduction of [?] for the public factor $\mathbf{eq}(X; \alpha^{(i)})$, which we use to shorten the round messages.) What is small in our setting is not the summands but their *selectors*: each summand is one of a few position-dependent constants, chosen by the witness bits at that position, and the tables $T_{A_0}, T_{S_1}, T_{A_2}$ are a Four-Russians-style precomputation [?] of every such choice. This replaces the $\Theta(\ell)$ multiplications by $\Theta(\ell)$ table reads and additions (XORs in our instantiation) plus $O(\ell_1 + \ell_2)$ multiplications, cf. Table 5.

We remark that the same multiplication count for $h_1(X)$ can be obtained by grouping the summands according to the pattern of witness bits they depend on and accumulating the **eq** weights of each group before multiplying, which is the transpose of our tables. We present the table form because it is the one that extends to the later rounds and to the higher levels of the tree, as we show next: the tables absorb the verifier's challenges, and the bottom levels of the tree never have to be written down.

	Our approach	Standard/generic
Multiplications	$\leq 10 \cdot \ell_1 + 4 \cdot \ell_2$	$\Theta(\ell_1 \cdot \ell_2)$
Additions	$\leq 3 \cdot \ell_1 \cdot \ell_2 / 4 + 2 \cdot \ell_1 + 5 \cdot \ell_2$	$\Theta(\ell_1 \cdot \ell_2)$
Table reads	$3 \cdot \ell_1 \cdot \ell_2 / 4$, into tables of $6 \cdot \ell_1$ field elements	–

Table 5: The cost in field operations of the first sumcheck round of the last sumcheck in Construction 6.1 (Step 2 with $i = r_1 - 1$), where $\ell = \ell_1 \cdot \ell_2 = 2^{r_1} \cdot 2^{r_2}$ is the witness size. In our instantiation $\ell_1 \approx \ell^{0.6} = 2^{r_1}$ and $\ell_2 \approx \ell^{0.4} = 2^{r_2}$ (Section 5.7).

Subsequent prover messages in the last sumcheck in Step 2 of Construction 6.1 Next we focus on computing $h_2(X)$. Substituting X by the verifier's first sumcheck challenge ρ_1 in Eqs. (39) and (40), we have, for every $\mathbf{b} \in \{0, 1\}^{r_1-1}, \mathbf{c} \in \{0, 1\}^{r_2}$,

$$\begin{aligned}
L_{r_1}(\rho_1, \mathbf{b}, \mathbf{c}) &= (L_{r_1}(1, \mathbf{b}, \mathbf{c}) - L_{r_1}(0, \mathbf{b}, \mathbf{c})) \cdot \rho_1 + L_{r_1}(0, \mathbf{b}, \mathbf{c}) \\
&= \begin{cases} \mathbf{y}_{\mathbf{b}}^{(11)} := \left(\mathbf{y}_{1,\mathbf{b}}^{(1)} - \mathbf{y}_{0,\mathbf{b}}^{(1)} \right) \cdot \rho_1 + \mathbf{y}_{0,\mathbf{b}}^{(1)} & \text{if } \mathbf{f}_{(0,\mathbf{b}),\mathbf{c}} = 1, \mathbf{f}_{(1,\mathbf{b}),\mathbf{c}} = 1 \\ \mathbf{y}_{\mathbf{b}}^{(10)} := \left(\mathbf{y}_{1,\mathbf{b}}^{(0)} - \mathbf{y}_{0,\mathbf{b}}^{(1)} \right) \cdot \rho_1 + \mathbf{y}_{0,\mathbf{b}}^{(1)} & \text{if } \mathbf{f}_{(0,\mathbf{b}),\mathbf{c}} = 1, \mathbf{f}_{(1,\mathbf{b}),\mathbf{c}} = 0 \\ \mathbf{y}_{\mathbf{b}}^{(01)} := \left(\mathbf{y}_{1,\mathbf{b}}^{(1)} - \mathbf{y}_{0,\mathbf{b}}^{(0)} \right) \cdot \rho_1 + \mathbf{y}_{0,\mathbf{b}}^{(0)} & \text{if } \mathbf{f}_{(0,\mathbf{b}),\mathbf{c}} = 0, \mathbf{f}_{(1,\mathbf{b}),\mathbf{c}} = 1 \\ \mathbf{y}_{\mathbf{b}}^{(00)} := \left(\mathbf{y}_{1,\mathbf{b}}^{(0)} - \mathbf{y}_{0,\mathbf{b}}^{(0)} \right) \cdot \rho_1 + \mathbf{y}_{0,\mathbf{b}}^{(0)} & \text{if } \mathbf{f}_{(0,\mathbf{b}),\mathbf{c}} = 0, \mathbf{f}_{(1,\mathbf{b}),\mathbf{c}} = 0 \end{cases}
\end{aligned}$$

The idea now is that the prover can compute $h_2(X)$ in essentially the same way as $h_1(X)$, but this time, instead of starting with the table

$$T_{r_1} \in (\mathbb{K}^2)^{\{0,1\}^{r_1}} \quad \text{such that } T_{r_1}[\mathbf{b}] = (\mathbf{y}_{\mathbf{b}}^{(0)}, \mathbf{y}_{\mathbf{b}}^{(1)}) \text{ and } L_{r_1}(\mathbf{b}, \mathbf{c}) = T_{r_1}[\mathbf{b}][\mathbf{f}_{\mathbf{b},\mathbf{c}}], \quad (\mathbf{b}, \mathbf{c}) \in \{0,1\}^{r_1} \times \{0,1\}^{r_2},$$

it starts with the table

$$T_{r_1}^{(2)} \in (\mathbb{K}^4)^{\{0,1\}^{r_1-1}} \quad \text{such that } T_{r_1}^{(2)}[\mathbf{b}] = (\mathbf{y}_{\mathbf{b}}^{(11)}, \mathbf{y}_{\mathbf{b}}^{(10)}, \mathbf{y}_{\mathbf{b}}^{(01)}, \mathbf{y}_{\mathbf{b}}^{(00)}) \quad \text{for } \mathbf{b} \in \{0,1\}^{r_1-1}$$

for which we have, indexing the entries of $T_{r_1}^{(2)}[\mathbf{b}]$ with the 4 values $(1,1), (1,0), (0,1), (0,0)$,

$$L_{r_1}(\rho_1, \mathbf{b}, \mathbf{c}) = T_{r_1}^{(2)}[\mathbf{b}][(\mathbf{f}_{0,\mathbf{b}}, \mathbf{f}_{1,\mathbf{b}})]$$

This allows the prover, when needed, to obtain $L_{r_1}(\rho_1, \mathbf{b}, \mathbf{c})$ for any $(\mathbf{b}, \mathbf{c}) \in \{0,1\}^{r_1-1} \times \{0,1\}^{r_2}$ with a simple read in $T_{r_1}^{(2)}$.

We now have, for $(\mathbf{b}, \mathbf{c}) \in \{0,1\}^{r_1-3} \times \{0,1\}^{r_2}$,

$$\begin{aligned} A_0^{(2)}[\mathbf{b}, \mathbf{c}] &= L_{r_1}(\rho_1, 0, \mathbf{b}, 0, \mathbf{c}) \cdot L_{r_1}(\rho_1, 0, \mathbf{b}, 1, \mathbf{c}) \cdot \omega^{(2)}[\mathbf{b}] \\ &= T_{r_1}^{(2)}[0, \mathbf{b}, 0][(\mathbf{f}_{(0,0,\mathbf{b},0),\mathbf{c}}, \mathbf{f}_{(1,0,\mathbf{b},0),\mathbf{c}})] \cdot T_{r_1}^{(2)}[0, \mathbf{b}, 1][(\mathbf{f}_{(0,0,\mathbf{b},1),\mathbf{c}}, \mathbf{f}_{(1,0,\mathbf{b},1),\mathbf{c}})] \cdot \omega[\mathbf{b}] \end{aligned}$$

for some weights $\omega^{(2)}[\mathbf{b}]$. Hence the elements $A_0^{(2)}[\mathbf{b}, \mathbf{c}]$ can be obtained by querying a table

$$T_{A_0}^{(2)} \in (\mathbb{K}^{16})^{\{0,1\}^{r_1-3}}$$

with

$$T_{A_0}^{(2)}[\mathbf{b}] = (T_{r_1}^{(2)}[0, \mathbf{b}, 0][\mathbf{x}] \cdot T_{r_1}^{(2)}[0, \mathbf{b}, 1][\mathbf{x}'] \cdot \omega[\mathbf{b}])_{\mathbf{x}, \mathbf{x}' \in \{0,1\}^2}$$

A table $T_{S_1}^{(2)}$ is computed analogously to T_{S_1} . The value $A_2^{(2)}[\mathbf{b}, \mathbf{c}]$, however, now depends on the value taken by *eight* witness bits, this being because for $\varepsilon \in \{0,1\}$, the value $E_{1\varepsilon}^{(2)} - E_{0\varepsilon}^{(2)}$ depends on four bits. Hence computing a table for the values $A_2^{(2)}[\mathbf{b}, \mathbf{c}]$ analogous to T_{A_2} would have $2^8 = 256$ -sized vectors as entries. Instead of computing $T_{A_2}^{(2)}$, we compute two tables for the values $E_{1\varepsilon}^{(2)} - E_{0\varepsilon}^{(2)}$ (one table for each $\varepsilon \in \{0,1\}$, with one of the tables also including multiplication by $\omega[\mathbf{b}]$). Then the value $A_2^{(2)}[\mathbf{b}, \mathbf{c}]$ can be obtained by performing a lookup into each table, followed by a single multiplication.

This can be iterated in further sumcheck rounds, for computing $h_3(X), h_4(X)$, etc. The limitation is that the entries in the tables are vectors whose dimension grows quadratically. More precisely, each further round squares the number of cases per index of every table, but it only halves the number of indices. Hence, in practice we stop using lookup tables after a few rounds of each sumcheck and GKR, and after that we execute the sumcheck protocol normally. Concretely, in our implementation we never compute a table with more than 16 elements per index. This is done so that the tables can be kept in cache. [Table 6](#) displays the different tables built or not built in our implementation, the arithmetic cost of doing so, and their size at $\ell_1 = 2^{r_1} = 2^{16}$ (corresponding to a total witness size of 2^{27} bits in our implementation).

The rest of sumchecks in Step 2 of Construction 6.1 So far we have discussed the last sumcheck from Step 2 of Construction 6.1, corresponding to the lower layer of the binary tree of products of our GKR circuit for the grand products.

The sumchecks run for $i = r_1 - 2, r_1 - 3, \dots$ can be optimized in the exact same way as we did for the last sumcheck ($i = r_1 - 1$). In this case, instead of starting with the table $T_{r_1}^{(1)}$, which at each index contains the two values $(\mathbf{y}_{\mathbf{b}}^{(1)}, \mathbf{y}_{\mathbf{b}}^{(0)})$, one starts with the table

$$T_{r_1-j} \in (\mathbb{K}^{2^{2^j}})^{\{0,1\}^{r_1-j}}$$

described in Eq. (38). Since T_{r_1-j} has quadratically many more entries per index than T_{r_1-j+1} , all subsequent tables also have quadratically many more entries per index compared to the analogous tables from the $(i+1)$ -th sumcheck. And, further, they require quadratically more operations to compute.

In our implementation we want to keep all tables in cache. Because of this, the smaller i is (i.e. the further we get from the last sumcheck in the GKR protocol), the fewer number of sumcheck rounds we use tables in. Concretely, for $i = r_1 - 2$ ($j = 1$) we build and use tables $T_{r_1-1}^{(1)}, T_{A_0}^{(1)}, T_{S_1}^{(1)}, T_{r_1-1}^{(2)}$ (note the lack of $T_{A_2}^{(1)}$). For $i = r_1 - 3$ ($j = 2$) we only use the table $T_{r_1-2}^{(1)}$.

Table	Dimension	Adds/mults. in $\mathbb{K}$	Size (MB), $\ell_1 = 2^{16}$	Built?
$T_{r_1}^{(1)}$	$(\mathbb{K}^2)^{2^{r_1}}$	0/0	2.1	yes
$T_{A_0}^{(1)}$	$(\mathbb{K}^{2^2})^{2^{r_1-2}}$	0 / 2^{r_1+1}	1.0	yes
$T_{S_1}^{(1)}$	$(\mathbb{K}^{2^2})^{2^{r_1-2}}$	0 / 2^{r_1+1}	1.0	yes
$T_{A_2}^{(1)}$	$(\mathbb{K}^{2^4})^{2^{r_1-2}}$	2^{r_1+1} / $5 \cdot 2^{r_1}$	4.2	yes
$T_{r_1}^{(2)}$	$(\mathbb{K}^{2^2})^{2^{r_1-1}}$	2^{r_1+2} / 2^{r_1+1}	2.1	yes
$T_{A_0}^{(2)}$	$(\mathbb{K}^{2^4})^{2^{r_1-3}}$	0 / 2^{r_1+2}	2.1	yes
$T_{S_1}^{(2)}$	$(\mathbb{K}^{2^4})^{2^{r_1-3}}$	0 / 2^{r_1+2}	2.1	yes
$T_{A_2}^{(2)}$	$(\mathbb{K}^{2^8})^{2^{r_1-3}}$	—	33.6	no
$T_{r_1}^{(3)}$	$(\mathbb{K}^{2^4})^{2^{r_1-2}}$	2^{r_1+3} / 2^{r_1+2}	4.2	yes
$T_{A_0}^{(3)}$	$(\mathbb{K}^{2^8})^{2^{r_1-4}}$	—	16.8	no
$T_{S_1}^{(3)}$	$(\mathbb{K}^{2^8})^{2^{r_1-4}}$	—	16.8	no
$T_{A_2}^{(3)}$	$(\mathbb{K}^{2^{16}})^{2^{r_1-4}}$	—	4295	no

Table 6: Dimensions, build costs, and sizes of the tables for computing $h_1(X), h_2(X), h_3(X)$, with $\log |\mathbb{K}| \approx 128$. We use the example size $\ell_1 = 2^{r_1} = 2^{16}$, which following our instantiation ($\ell_1 \approx \ell^{0.6}$, Section 5.7) this corresponds to a total witness size $\ell \approx 2^{27}$. **Albert:** It seems we do not always build $T_{A_0}^{(2)}$ and $T_{S_1}^{(2)}$?

6.2 Other optimizations for Construction 5.4

Step 1 of Construction 5.4 via lookup tables In Step 1 of Construction 5.4, the prover computes the elements $\mu_1, \dots, \mu_{\ell_2}$ again by means of a lookup table T , which is indexed over $i \in [\ell_1/4]$ and contains the 16-length vectors

$$T[i] = \left(\sum_{k \in [4]} \varepsilon_k \cdot \pi_{\text{canon}}^{-1}(\mathbf{v}^{(1)})_{4*(i-1)+k} \right)_{\varepsilon_1, \dots, \varepsilon_4 \in \{0,1\}} \in \mathbb{Z}[X]^{16}.$$

7 Experiments

We implement $\mathbb{BitZ}$ (Constructions 5.4 and 5.13) following the guideline in Section 5.7. We then implement an end-to-end roof system, called $\mathbb{BitZ}$ -SNARK, for integer R1CS constraints as described in ?? Albert: technical overview (note that, however, $\mathbb{BitZ}$ can be used to prove much more general constraints, such as the ones in Zinc+ [AGK+26], cf. ?? Albert: some remark in TO). Following the lead of ?? and other usages of AI ?? to optimize against verified specs, we used AI to optimize the prover while keeping the verifier implementation untouched. As such, we consider our prover fairly optimal, while the our verifier’s implementation is pending significant optimization.

7.1 Raw performance of the $\mathbb{BitZ}$ PCS for the core LinBitsRings relation

We measure the performance of Construction 5.4 for proving the Core LinBitsRing relation $\mathcal{R}_{\text{Lin-BitsDual}}(\{0, 1\}, \pi, \mathbb{F}_q)$ following the instantiation in Section 5.7. Concretely, $\mathbb{F}_{2^\nu} = \mathbb{F}_{2^{128}}$ and $\mathcal{C}$ is a Reed-Solomon code. We set $|\mathbb{F}_q| < (2^{128} - 1)/\ell_1$ and we use the unique decoding radius as proximity parameter, so that Round 0 (OOD sampling) and Round 1 of Construction 5.4 do not trigger. Our goal for now is to measure the performance of the grand product and the WHIR/Ligerito commit and opening. In later experiments, Round 0 and Round 1 do trigger.

$\log_2 \ell$	Commit	Prover time (ms)				Verifier	Proof size (KB)		
	(ms)	Grand prod.	Ring switch	Ligerito	Total	(ms)	Non-Lig.	Ligerito	Total
20	0.75	4.97	1.09	4.67	11.0	1.27	15.1	44.5	59.6
21	0.85	6.76	1.25	4.62	12.9	1.52	16.0	49.2	65.2
22	1.52	10.2	1.43	4.83	16.6	1.87	16.9	53.1	69.9
23	2.28	17.1	1.72	7.82	27.1	1.73	21.7	65.1	86.8
24	4.01	28.6	2.60	9.43	41.4	2.17	22.7	70.4	93.1
25	8.32	52.0	3.91	12.7	69.0	2.64	31.4	77.0	108
26	15.2	92.8	6.77	19.5	120	2.76	32.7	89.9	123
27	32.6	173	11.3	24.6	211	3.37	33.8	96.4	130
28	64.8	337	20.4	35.0	396	3.89	50.8	102	153
29	136	721	52.0	67.6	843	4.61	52.2	116	168
30	270	2609	188	195	2999	6.05	85.6	123	209

Table 7: Albert: Use the previous shorter captions Albert: Commit inside prover time Cost of $\mathbb{BitZ}$ (Construction 5.4) for committing to $\ell = 2^n$ bits, $n = 20, \dots, 30$, and proving one claim $\langle \mathbf{v}, \mathbf{f} \rangle = \mu$ over $\mathbb{F}_q$ with $q = 2^{100} - 15$ and the tensor split $\ell_1 = 2^{\lceil 0.6n \rceil}$, $\ell_1 \cdot \ell_2 = \ell$ (Section 5.7). The commitment is opened with ring switching and Ligerito [?] over a Reed-Solomon code of rate 1/8 over $\mathbb{F}_{2^{128}}$ (initial folding of 2^4 rows, Johnson regime, BLAKE3 Merkle trees), configured for 100 bits of round-by-round security (the minimum over its levels and error terms is 100.0 bits); every other round (the GKR, the sumcheck reducing to MLE evaluation claims, the ring switch) has error at most $7 \cdot 2^{-128}$. Prover columns: *grand products* is computing the integers μ_j and the batched GKR for the ℓ_2 grand products in the exponent (Section 6); *ring switch* is the sumcheck reducing the GKR output claims to MLE evaluation claims together with the ring-switching step; *Ligerito* is the Ligerito opening. These exclude the commitment, listed separately. *Non-Ligerito* proof bytes are the μ_j , the GKR and sumcheck messages, and the ring-switch message; KB = 1000 bytes. Apple M4 (10 cores: 4 performance + 6 efficiency), 16 GB, 8 threads; medians of 5 runs after one warm-up.

Albert: 1 threaded one?

Albert: Measure Binius64 proving the same inner product directly in Bin fields?

7.2 Integer multiplication

We measure the cost of proving 2^i integer multiplications of the form $x \cdot y = z$, for x, y random 32-bit integers, and z a 64-bit integer. As we mentioned, this is proved via our implemented end-to-end proof system, called BitZ-SNARK for R1CS over the integers, with BitZ as its PCS, as described in ??.

We compare the performance of BitZ-SNARK with Binius64 [Bin26], Plonky3 [?], Zinc+ [AGK⁺26], and Limber [CXNB26] on the same task. Both Binius64 and Plonky3 implement gadgets for proving integer multiplication constraints. We use those in our benchmarks **Albert: Confirm**.

$\log_2 N$	Prover time (ms)						Verifier (ms)	Proof size (KB)			
	Commit	PIOP	Grand prod.	Ring switch	Ligerito	Total		PIOP	Non-Lig.	Ligerito	Total
15	0.79	1.50	10.2	1.76	0.86	15.5	4.88	1.7	18.2	137	157
16	1.20	2.26	16.8	2.03	1.45	24.2	7.61	1.9	21.7	162	186
17	2.12	3.47	28.6	3.03	2.10	42.2	15.0	2.0	22.7	181	206
18	3.73	5.93	50.9	4.05	2.73	70.0	24.5	2.1	27.3	202	232
19	6.99	10.3	96.3	6.57	4.09	129	45.9	2.2	29.2	234	265
20	15.1	18.6	185	10.3	6.49	240	82.7	2.3	38.0	255	296
21	30.1	37.8	393	18.2	10.6	494	153	2.4	39.1	281	323
22	54.2	80.0	805	31.9	19.1	995	284	2.5	57.0	321	381

Table 8: Cost of proving $N = 2^n$ integer multiplications $x \cdot y = z$, $n = 15, \dots, 22$, for random 32-bit integers x, y (so that z is a 64-bit integer) with ?? : one R1CS constraint per multiplication over $\mathbb{Z}$, projected to a prime q sampled after the commitment from an interval $[2^{b-1}, 2^b)$ with b between 108 and 111 depending on the shape, a Spartan PIOP over $\mathbb{F}_q$ (with a 3-variable univariate skip), bitification, and the BitZ opening of the 128 bits committed per multiplication ($\ell = 2^{n+7}$ bits in cells of $W = 1$ bit, i.e. 128 cells per multiplication). Security profile $\lambda = 100$: every round-by-round error is at most $2^{-100.0}$, the binding term being `step5_3:ligerito-tracked`; the commitment is opened with ring switching and Ligerito [?] over a Reed–Solomon code of rate 1/2 over $\mathbb{F}_{2^{128}}$ (initial folding of 2^4 rows), configured for 100 bits. Prover columns: *Commit* is bit packing plus the commitment; *PIOP* is the prime projection, the Spartan PIOP and the (microsecond-scale) bitification step; *grand products*, *ring switch* and *Ligerito* are the three parts of the BitZ opening as in Table 7; *Total* is end to end and includes the commitment. Proof columns: *PIOP* is the Spartan messages and grinding nonces; *Non-Lig.* is the opening’s integer folds, GKR, sumcheck and ring-switch messages; *Ligerito* is the Ligerito proof; KB = 1000 bytes. Apple M4 (10 cores: 4 performance + 6 efficiency), 16 GB, 8 threads; medians of 5 runs after one warm-up; witness generation and the one-time relation preparation are excluded.

Albert: 128 bits?

7.3 SHA-256 hashing followed by ECDSA signature verification

Albert: Use $\text{Id}_{2^r} \otimes M$ as virtualization matrix, leading to 2^r compressions per grand product

7.4 RSA accumulation and Poseidon hashing

We compare Limber [CXNB26], Zinc+[AGK⁺26] and our BitZon proving two RSA accumulation updates plus a Poseidon hash evaluation (see Section 7.1 of ??). An RSA accumulation consists in performing a modular exponentiation on a 2048-bit RSA modulus. This is exactly the computation with which Limber [CXNB26] experiments with, and indeed we take the corresponding R1CS constraint from their repository. Recall that Limber is a proof system for polynomial relations over the integers, (see ?? for further details on Limber, Zinc+, etc.).

We set BitZ at 114 bits of security, matching Limber⁶
 Albert: need to check the soundness achieved by Zinc+

Scheme	Commitment	Prove (ms)	Verify (ms)	Proof (KB)
BitZ (this work)	binary field, hash	257	10.1	338
Zinc+ [AGK ⁺ 26]*	integer code, hash	850	57.5	1418
Limber [CXNB26] (Brakedown)	hash	1143	42.8	5885
Limber [CXNB26] (Hyrax)	Tom-256, DL	1180	35.4	175
Arkworks/Garuda [†]	BLS12-381, pairing	231000	25	7.2
MultiSwap-xJsnark [†]	BLS12-381, pairing	88500	2	0.2

Table 9: Proving one MultiSwap [?] batch update: two RSA-2048 accumulator updates verified through Wesolowski proofs, together with a Poseidon-based hash-to-prime. The four upper rows prove the same integer Mod-R1CS statement of 6209 modular-multiplication rows and were measured on the same machine (Apple M4, 16GB), single-threaded, as medians of five runs; prover time covers commitment and proving and excludes witness generation. The two lower rows are circuit-based baselines (25.2M and 6.2M R1CS constraints respectively), for which we quote [CXNB26, Table 1] verbatim. BitZ and Limber sample their fingerprinting prime from the transcript after the commitment and reach round-by-round Fiat–Shamir soundness errors of at most 2^{-114} . * Zinc+ is benchmarked on the unwired mock circuit of [CXNB26, Table 1, footnote c], extended to 8192 slots: it omits the bit-decomposition and hash-to-prime rows and the range checks on the quotient advice, and uses a fixed rather than a transcript-sampled projecting prime, with 100 column openings. Its proof compresses to 686KB with `zstd`; the other proofs are reported uncompressed. [†] Quoted from [CXNB26]; measured on an M4 Pro, 24GB, and not reproduced here.

7.5 Hashing followed by Falcon signature verification

7.6 Comparison with Hyrax

Albert: Comparing with Hyrax may be more interesting than it sounds initially. Hyrax is pay-per-bit and opens on large prime fields: a profile that F2Z can adopt. Hence (maybe) projects using Hyrax could use F2Z as a drop-in replacement which is PQ secure and more efficient

7.7 Open questions

Albert: We may want to keep these to ourselves

Albert: This section is just scratch/notes/thoughts

- When having $\mathbb{F}_2$ -virtualization in the grand product, there’s potential optimizisatiosn. So far I’ve only found a way to cheapen the following settings: say first and two vectors in the grand product are $\mathbf{g}_1, \mathbf{g}_2$, and third is $\mathbf{g}_1 \text{ XOR } \mathbf{g}_2$. This is a tricky problem. We should look more into it
- Should we allow for $W > 1$? Currently just setting $W = 1$ and thinking of all bits as unrelated leads to better performance, but that sounds suboptimal.
- Folding instances of $\mathcal{R}_{\text{Lin-BitsDual}}$

⁶Limber states in the paper that it achieves 128 bits of security, but its fingerprint step only achieves 114 bits of security of Limbery in the implementation, because the random prime sampled to perform the fingerprint is of 128 bits, and no other measures (like, say, grinding) are taken. Albert: Double check and let the authors know.

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A Deferred proofs

A.1 Completeness and soundness of [Construction 5.13](#)

Albert: Review: AI written

We prove [Theorem 5.15](#). . For readability, we temporarily denote $\pi : \mathbb{Z} \rightarrow R$ for the reduction modulo $\text{char}(R)$ and we let

$$\mathcal{R}_{\text{gen}} := \mathcal{R}_{\text{Lin-BitsDual}}(S_{\leq B}, \psi, R, \mathcal{C}, L), \quad \mathcal{R}_{\text{core}} := \mathcal{R}_{\text{Lin-BitsDual}}(\{0, 1\}, \pi, R, \mathcal{C}), \quad \mathbf{x}' := (\mu, \mathbf{u}),$$

and $\mathcal{R}_{\text{PESAT}} = \mathcal{R}_{\text{PESAT}}(\mathbb{F}_{2^\nu}, Q_{\mathbb{B}\text{it}\mathbb{Z}}, \mathcal{C}, \text{Pack}_\nu)$, $\mathcal{R}_{\text{Lin}} = \mathcal{R}_{\text{Lin}}(\mathbb{F}_{2^\nu}, \mathcal{C}, \text{Pack}_\nu)$ (as in [Section 5.4](#)). Recall from [Definition 5.9](#) that a bitification of $\mathbf{f} \in S_{\leq B}^{\ell_0}$ is any $\mathbf{g} \in \{0, 1\}^\ell$ with $\mathbf{f} = L \cdot \mathbf{g}_S$. For such $\mathbf{g}$, we say that $((\mu, \mathbf{v}), f, \mathbf{f})$ belongs to $\mathcal{R}_{\text{gen}}^\delta$ via $\mathbf{g}$ if $\Delta(f, \text{Enc}_\mathcal{C} \circ \text{Pack}_\nu \circ \pi_2(\mathbf{g})) < \delta$ and $\langle \psi(\mathbf{f}), \mathbf{v} \rangle = \mu$. Then $((\mu, \mathbf{v}), f, \mathbf{f}) \in \mathcal{R}_{\text{gen}}^\delta$ if and only if $((\mu, \mathbf{v}), f, \mathbf{f})$ belongs to $\mathcal{R}_{\text{gen}}^\delta$ via some bitification $\mathbf{g}$ of $\mathbf{f}$. The honest prover uses $\mathbf{g}$ as witness in the call to [Construction 5.4](#). The same applies to $\mathcal{R}_{\text{gen}}$, with equality of words in place of the distance condition. We assume that $L \cdot \mathbf{g}_S \in S_{\leq B}^{\ell_0}$ for all $\mathbf{g} \in \{0, 1\}^\ell$, as is the case for the standard bitification matrix of [Definition 5.9](#) (for a general L this is needed: otherwise the prover may commit to bits $\mathbf{g}$ with $L \cdot \mathbf{g}_S \notin S_{\leq B}^{\ell_0}$ and have the verifier accept, although no witness in $S_{\leq B}^{\ell_0}$ exists **Albert: process this requirement**).

For $\mathbf{g} \in \{0,1\}^\ell$ let $\mathbf{f}_{\mathbf{g}} := L \cdot \mathbf{g}_S \in S_{\leq B}^{\ell_0}$. By Lemma 5.12, $\langle \psi(\mathbf{f}_{\mathbf{g}}), \mathbf{v} \rangle = \langle \psi(\mathbf{g}_S), \psi(L)^\top \cdot \mathbf{v} \rangle = \langle \pi(\mathbf{g}), \mathbf{u} \rangle$, and hence

$$(\mathbf{x}', f, \mathbf{g}) \in \mathcal{R}_{\text{core}}^\delta \iff ((\mu, \mathbf{v}), f, \mathbf{f}_{\mathbf{g}}) \text{ belongs to } \mathcal{R}_{\text{gen}}^\delta \text{ via } \mathbf{g}, \quad (42)$$

and likewise for $\mathcal{R}_{\text{core}}$ and $\mathcal{R}_{\text{gen}}$. Completeness then follows from (42) and Theorem 5.7.

For round-by-round knowledge soundness, we decompose Construction 5.13 as in the proof of Theorem 5.7: the first IOR Π'_1 is the instance translation $(\mu, \mathbf{v}) \mapsto \mathbf{x}'$ followed by the IOR Π_1 of that proof (of Section A.2, if Round 0 is executed), and the remaining IORs are Π_{red} and Π_{iopp} (their variants from Section A.2, if Round 0 is executed). Π'_1 has the same messages and output as Π_1 , and it is perfectly complete by (42). By Remark 4.8 it suffices to show that Π'_1 has round-by-round knowledge soundness with relaxation $(\mathcal{R}_{\text{gen}}^\delta, \mathcal{R}_{\text{PESAT}}^\delta)$ (resp. $(\mathcal{R}_{\text{gen}}^\delta, \mathcal{R}_{\text{PESAT}}^{\delta, \zeta, y})$) and with the same errors as Π_1 ; here $\kappa_0 = \theta(R, \ell)$ is computed with respect to the tensor decomposition $\mathbf{u} = \mathbf{u}^{(1)} \otimes \mathbf{u}^{(2)}$ used in Construction 5.4 (cf. Remark 5.14).

Let KState_1 be the knowledge state function of Π_1 ; recall that its extractor is the identity map. We set $\text{KState}(\mathbf{x}, y, \text{tr}, \mathbf{w}) = 1$ if and only if $(\mathbf{x}, y, \mathbf{w}) \in \mathcal{R}_{\text{gen}}^\delta$ when tr contains no verifier message, and $\text{KState}(\mathbf{x}, y, \text{tr}, \mathbf{w}) := \text{KState}_1(\mathbf{x}', y, \text{tr}, \mathbf{w})$ otherwise; and we set $\mathcal{E}_{\text{rbr}}(\mathbf{x}, y, \text{tr}, \mathbf{w}) := \mathbf{f}_{\mathbf{w}}$ if tr ends with the first verifier message and $\mathbf{w} \in \{0,1\}^\ell$, and $\mathcal{E}_{\text{rbr}}(\mathbf{x}, y, \text{tr}, \mathbf{w}) := \mathbf{w}$ otherwise. The conditions of Definition 4.6 hold: at the empty transcript by construction, at prover moves because KState is constant before the first verifier message and coincides with KState_1 afterwards, and at full transcripts because Π'_1 and Π_1 have the same output. At every verifier message after the first one, KState coincides with KState_1 before and after the message and $\mathcal{E}_{\text{rbr}}$ is the identity map, so the error of Π'_1 is that of Π_1 . Finally, let ρ be the first verifier message and tr the transcript preceding it, and suppose that $\text{KState}_1(\mathbf{x}', y, \text{tr} \parallel \rho, \mathbf{w}) = 1$ and $\text{KState}(\mathbf{x}, y, \text{tr}, \mathcal{E}_{\text{rbr}}(\mathbf{x}, y, \text{tr} \parallel \rho, \mathbf{w})) = 0$. Then $(\mathbf{x}', y, \mathbf{w}) \notin \mathcal{R}_{\text{core}}^\delta$: otherwise $\mathbf{w} = \mathbf{g} \in \{0,1\}^\ell$, $\mathcal{E}_{\text{rbr}}$ outputs $\mathbf{f}_{\mathbf{g}}$, and (42) gives $\text{KState}(\mathbf{x}, y, \text{tr}, \mathbf{f}_{\mathbf{g}}) = 1$. We distinguish three cases.

- $\rho = (q, \pi_{\text{rand}})$ (Round 1 executed, Round 0 not executed). By the definition of KState_1 , $\mathbf{w} = \mathbf{g} \in \{0,1\}^\ell$, and $\text{KState}_1(\mathbf{x}', y, \text{tr}, \mathbf{g}) = 0$ by the above. This is precisely the event bounded by κ_0 in the proof of Theorem 5.7, so the error is at most κ_0 .
- ρ is the empty message (Round 1 skipped). Then $\text{tr} \parallel \rho$ is the full transcript, so $\mathbf{w} = \mathbf{g} \in \{0,1\}^\ell$, and, since appending the empty message does not change KState_1 , the prover-move condition for KState_1 gives $\text{KState}_1(\mathbf{x}', y, \emptyset, \mathbf{g}) = 1$, i.e. $(\mathbf{x}', y, \mathbf{g}) \in \mathcal{R}_{\text{core}}^\delta$, a contradiction. The error is 0.
- $\rho = \zeta$ (Round 0 executed). By the definition of KState_1 in Section A.2, $\text{Coll}(f, \zeta)$ holds, which happens with probability at most κ_{OOD} over ζ by Lemma A.1.

A.2 Soundness of Out Of Domain Sampling (OOD) in Construction 5.4

Albert: Where to put this? If Round 0 is executed, assume moreover that the IOR obtained from Π_{red} by extending its instances with a pair $(\zeta, y) \in \mathbb{F}_{2^\nu}^2$ (which the protocol ignores and copies to its output instance) is a perfectly complete IOR from $\mathcal{R}_{\text{PESAT}}^{\zeta, y}$ to $\mathcal{R}_{\text{Lin}}^{\zeta, y}$ with round-by-round knowledge soundness with relaxation $(\mathcal{R}_{\text{PESAT}}^{\delta, \zeta, y}, \mathcal{R}_{\text{Lin}}^{\delta, \zeta, y})$, with the same errors; and, similarly, that Π_{iopp} is a perfectly complete t_{iopp} -round IOP for $\mathcal{R}_{\text{Lin}}^{\zeta, y}$ with round-by-round knowledge soundness with relaxation $(\mathcal{R}_{\text{Lin}}^{\delta, \zeta, y}, \perp)$, with the same errors (cf. ??).

We will need the following well-known result regarding the soundness of the OOD sampling step.

Lemma A.1 (OOD collision probability). *Let δ, L_δ be as in [Theorem 5.7](#). Let $f \in \mathbb{F}_{2^\nu}^m$, and let $\text{List}(f, \delta)$ denote the set of vectors $\mathbf{g} \in \mathbb{F}_2^\ell$ with $\Delta(f, \text{Enc}_\mathcal{C} \circ \text{Pack}_\nu(\mathbf{g})) < \delta$. For $\zeta \in \mathbb{F}_{2^\nu}$, let $\text{Coll}(f, \zeta)$ denote the event that there exist two distinct $\mathbf{g}, \mathbf{g}' \in \text{List}(f, \delta)$ with $\widetilde{\text{Pack}_\nu(\mathbf{g})(\zeta)} = \widetilde{\text{Pack}_\nu(\mathbf{g}')(\zeta)}$. Then **Albert**: Is ζ defined?*

$$\Pr_{\zeta \leftarrow \mathbb{F}_{2^\nu}} [\text{Coll}(f, \zeta)] \leq \binom{L_\delta}{2} \cdot \frac{\ell/\nu - 1}{|\mathbb{F}_{2^\nu}|}.$$

Proof. Observe that since $\text{Enc}_\mathcal{C} \circ \text{Pack}_\nu$ is injective, and $\mathcal{C}$ is (δ, L_δ) -list-decodable, we have $|\text{List}(f, \delta)| \leq L_\delta$. Fix two distinct $\mathbf{g}, \mathbf{g}' \in \text{List}(f, \delta)$. Since Pack_ν is injective, $\text{Pack}_\nu(\mathbf{g}) \neq \text{Pack}_\nu(\mathbf{g}')$, and hence $D := \text{Pack}_\nu(\mathbf{g}) - \text{Pack}_\nu(\mathbf{g}')$ is a nonzero multilinear polynomial in $m := \log(\ell/\nu)$ variables over $\mathbb{F}_{2^\nu}$. The substitution $X_i \leftarrow \zeta^{2^i-1}$ maps distinct multilinear monomials $\prod_{i \in T} X_i$ (for $T \subseteq [m]$) to distinct powers $\zeta^{\sum_{i \in T} 2^i-1}$, with exponents ranging over $[0.. \ell/\nu - 1]$; hence $D(\zeta)$, viewed as a polynomial in ζ , is a nonzero univariate polynomial of degree at most $\ell/\nu - 1$, and it vanishes for at most $\ell/\nu - 1$ values of $\zeta \in \mathbb{F}_{2^\nu}$. The lemma follows by a union bound over the at most $\binom{L_\delta}{2}$ pairs. This completes the proof. $\square$

Albert: **Review**: **AI written** Assume that Round 0 is executed in [Construction 5.4](#), so that $\mathcal{C}$ is (δ, L_δ) -list-decodable, possibly with $L_\delta > 1$. We follow a standard approach ?? where we work with variants of relations that include the OOD evaluation claim. Concretely, we denote by $\mathcal{R}_{\text{PESAT}}^{\zeta, y}$ the relation consisting of all triples $((\mathbf{x}, \zeta, y), \mathbf{y}, \mathbf{w})$, with $\zeta, y \in \mathbb{F}_{2^\nu}$, such that $(\mathbf{x}, \mathbf{y}, \mathbf{w}) \in \mathcal{R}_{\text{PESAT}}$ and

$$\widetilde{\text{Pack}_\nu(\mathbf{w})(\zeta)} = y$$

where ζ is the evaluation point from Round 0 of [Construction 5.4](#). The relation $\mathcal{R}_{\text{Lin}}^{\zeta, y}$ is defined analogously. The relaxations $\mathcal{R}_{\text{PESAT}}^{\delta, \zeta, y}$ and $\mathcal{R}_{\text{Lin}}^{\delta, \zeta, y}$ are obtained by replacing $\mathcal{R}_{\text{PESAT}}, \mathcal{R}_{\text{Lin}}$ with $\mathcal{R}_{\text{PESAT}}^\delta, \mathcal{R}_{\text{Lin}}^\delta$ in these definitions.

We modify the decomposition from the proof of [Theorem 5.7](#) as follows. The IOR Π_1 now consists of applying Rounds 0, 1 and 2 of [Construction 5.4](#), and outputting

$$(\mathbf{x}' = ((g^{\gamma_i})_{i \in [\ell_1]}, (g^{\mu_j})_{j \in [\ell_2]}, \zeta, y), \mathbf{y}' = f, \mathbf{w}' = \pi_2(\mathbf{f})),$$

so that Π_1 is an IOR from $\mathcal{R}_{\text{Lin-BitsDual}}$ to $\mathcal{R}_{\text{PESAT}}^{\zeta, y}$. Formally, Π_1 is now a 3-round IOR whose third verifier message is empty: the prover's first message is empty, the verifier sends ζ , the prover sends (y, μ') , the verifier replies with (q, π_{rand}) , and the prover then sends μ . The IORs Π_2 and Π_3 are replaced by the IOR from $\mathcal{R}_{\text{PESAT}}^{\zeta, y}$ to $\mathcal{R}_{\text{Lin}}^{\zeta, y}$ obtained from Π_{red} , and by the IOP Π_{iopp} for $\mathcal{R}_{\text{Lin}}^{\zeta, y}$, as in the additional hypotheses of the theorem; their errors are unchanged. Below we show that Π_1 is perfectly complete and has round-by-round knowledge soundness with relaxation $(\mathcal{R}_{\text{Lin-BitsDual}}^\delta, \mathcal{R}_{\text{PESAT}}^{\delta, \zeta, y})$, with errors $(\kappa_{\text{OOD}}, \kappa_0, 0)$; as before, the empty third verifier message contributes no error, the last prover message of Π_1 can be sent together with the first prover message of Π_2 , and in this way [Construction 5.4](#) has $2 + t_{\text{red}} + t_{\text{iopp}}$ rounds. The theorem then again follows from [Remark 4.8](#).

We extend the knowledge state function KState from the case above; recall the collision event $\text{Coll}(f, \zeta)$ from [Lemma A.1](#). For the empty transcript, and for the transcript $(\perp)$ consisting of the empty first prover message, we set $\text{KState} = 1$ if and only if $(\mathbf{x}, \mathbf{y}, \mathbf{w}) \in \mathcal{R}_{\text{Lin-BitsDual}}^\delta$, as before. For $\text{tr} = (\perp, \zeta)$, we set $\text{KState}(\mathbf{x}, \mathbf{y}, \text{tr}, \mathbf{w}) = 1$ if and only if $(\mathbf{x}, \mathbf{y}, \mathbf{w}) \in \mathcal{R}_{\text{Lin-BitsDual}}^\delta$ or $\text{Coll}(f, \zeta)$ holds. For $\text{tr} = (\perp, \zeta, (y, \mu'))$, we set $\widetilde{\text{KState}}(\mathbf{x}, \mathbf{y}, \text{tr}, \mathbf{w}) = 1$ if and only if $\text{Coll}(f, \zeta)$ holds, or both $(\mathbf{x}, \mathbf{y}, \mathbf{w}) \in \mathcal{R}_{\text{Lin-BitsDual}}^\delta$ and $\widetilde{\text{Pack}_\nu \circ \pi_2(\mathbf{w})(\zeta)} = y$. For $\text{tr} = (\perp, \zeta, (y, \mu'), (q, \pi_{\text{rand}}))$, we set

$\text{KState}(\mathbf{x}, \mathbf{y}, \mathbf{tr}, \mathbf{w}) = 1$ if and only if $\text{Coll}(f, \zeta)$ holds, or the conditions required in the case above hold (i.e. the verifier's degree and range checks on μ' pass, $\mathbf{w} = \mathbf{f} \in \{0, 1\}^\ell$, and (25)) together with $\widetilde{\text{Pack}_\nu \circ \pi_2(\mathbf{f})}(\zeta) = y$. Finally, for $\mathbf{tr} = (\perp, \zeta, (y, \mu'), (q, \pi_{\text{rand}}), \mu)$, or for the full transcript obtained by appending the empty third verifier message, we set $\text{KState}(\mathbf{x}, \mathbf{y}, \mathbf{tr}, \mathbf{w}) = 1$ if and only if the verifier accepts on $\mathbf{tr}$ and $(\mathbf{x}', \mathbf{y}', \mathbf{w}) \in \mathcal{R}_{\text{PESAT}}^{\delta, \zeta, y}$, where $(\mathbf{x}', \mathbf{y}')$ is the output of the verifier described above; note that the collision escape $\text{Coll}(f, \zeta)$ is dropped in this last step. As before, the extractor is the identity map $\mathcal{E}_{\text{rbr}}(\mathbf{x}, \mathbf{y}, \mathbf{tr}, \mathbf{w}) = \mathbf{w}$.

We verify the conditions of Definition 4.6. The conditions at the empty and full transcripts hold by construction. For the prover message (y, μ') : if $\text{KState}(\mathbf{x}, \mathbf{y}, (\perp, \zeta), \mathbf{w}) = 0$, then $\text{Coll}(f, \zeta)$ fails and $(\mathbf{x}, \mathbf{y}, \mathbf{w}) \notin \mathcal{R}_{\text{Lin-BitsDual}}^\delta$, and hence $\text{KState}(\mathbf{x}, \mathbf{y}, (\perp, \zeta, (y, \mu')), \mathbf{w}) = 0$ for every prover message (y, μ') . For the prover message μ : suppose $\text{KState}(\mathbf{x}, \mathbf{y}, (\perp, \zeta, (y, \mu'), (q, \pi_{\text{rand}})), \mathbf{w}) = 0$ and, towards a contradiction, $\text{KState}(\mathbf{x}, \mathbf{y}, (\perp, \zeta, (y, \mu'), (q, \pi_{\text{rand}}), \mu), \mathbf{w}) = 1$ for some μ . The former excludes $\text{Coll}(f, \zeta)$. The latter gives that the verifier accepts and that $\mathbf{w}$ satisfies the defining conditions of $\mathcal{R}_{\text{PESAT}}^\delta$ together with $\widetilde{\text{Pack}_\nu \circ \pi_2(\mathbf{w})}(\zeta) = y$; by the second statement proved in the case above, the former two imply that $\mathbf{w} = \mathbf{f}$ satisfies the verifier's checks on μ' and (25). Hence $\text{KState}(\mathbf{x}, \mathbf{y}, (\perp, \zeta, (y, \mu'), (q, \pi_{\text{rand}})), \mathbf{w}) = 1$, a contradiction.

It remains to bound the errors at the two nonempty verifier messages. For the message ζ : since the extractor is the identity map, the event in Definition 4.7 requires some $\mathbf{w}$ with $(\mathbf{x}, \mathbf{y}, \mathbf{w}) \notin \mathcal{R}_{\text{Lin-BitsDual}}^\delta$ and $\text{KState}(\mathbf{x}, \mathbf{y}, (\perp, \zeta), \mathbf{w}) = 1$, which forces $\text{Coll}(f, \zeta)$; by Lemma A.1, this occurs with probability at most $\binom{L_\delta}{2} \cdot (\ell/\nu - 1)/|\mathbb{F}_{2^\nu}| = \kappa_{\text{OOD}}$ over ζ . For the message (q, π_{rand}) : assume $\text{KState}(\mathbf{x}, \mathbf{y}, (\perp, \zeta, (y, \mu'), (q, \pi_{\text{rand}})), \mathbf{w}) = 1$ and $\text{KState}(\mathbf{x}, \mathbf{y}, (\perp, \zeta, (y, \mu')), \mathbf{w}) = 0$. The latter excludes $\text{Coll}(f, \zeta)$, so the former gives that $\mathbf{w} = \mathbf{f} \in \{0, 1\}^\ell$ satisfies (25) and $\widetilde{\text{Pack}_\nu \circ \pi_2(\mathbf{f})}(\zeta) = y$; the latter then further gives $(\mathbf{x}, \mathbf{y}, \mathbf{f}) \notin \mathcal{R}_{\text{Lin-BitsDual}}^\delta$. Since $\text{Coll}(f, \zeta)$ fails, there is at most one vector $\mathbf{g} \in \mathbb{F}_2^\ell$ with $\Delta(f, \text{Enc}_C \circ \text{Pack}_\nu(\mathbf{g})) < \delta$ and $\text{Pack}_\nu(\mathbf{g})(\zeta) = y$; hence $\pi_2(\mathbf{f})$ is this vector, and it is determined by (f, ζ, y) . In particular, $\mathbf{f}$, and with it the polynomial h from the case above, is fixed before (q, π_{rand}) is sampled. The rest of the argument is exactly as in the case above: (26) holds, h is nonzero, has total degree at most $2 \cdot D$ and coefficients in $[-2 \cdot T \cdot \ell \cdot N_R^2 \dots 2 \cdot T \cdot \ell \cdot N_R^2]$, satisfies $\pi_{\text{canon}}(h) \neq 0$, and the event at hand implies $\pi_{\text{rand}}(h) = 0$, which occurs with probability at most $\theta(R, \ell) = \kappa_0$ over (q, π_{rand}) .

Finally, Π_1 is perfectly complete: the honest prover's message y equals $\widetilde{\text{Pack}_\nu \circ \pi_2(\mathbf{f})}(\zeta)$, so the output triple belongs to $\mathcal{R}_{\text{PESAT}}^{\zeta, y}$ whenever the input triple belongs to $\mathcal{R}_{\text{Lin-BitsDual}}$, the remaining conditions being as in the case above.

B Ring switching via Galois orbits

Albert: Context: how does it relate to other references?

Let $b_1, \dots, b_m \in \mathbb{F}_2^{2^n}$. **Albert:** Use boldface. Using an $\mathbb{F}_2$ -basis $\beta_1, \dots, \beta_m$ of $\mathbb{F}_{2^m}$ we pack m inner products

$$\langle b_i, \mathbf{v} \rangle = \sum_{j \in \{0, 1\}^n} b_{i,j} \cdot \mathbf{v}_j, \quad i = 1, \dots, m,$$

coefficient-wise into a single inner product

$$b := b_1 \cdot \beta_1 + \dots b_m \cdot \beta_m \in \mathbb{F}_{2^m}^{2^n}$$

$$\begin{aligned}
\langle b, \mathbf{v} \rangle &= \langle b_1, \mathbf{v} \rangle \cdot \beta_1 + \dots + \langle b_m, \mathbf{v} \rangle \cdot \beta_m \\
&= \sum_{j \in \{0,1\}^n} (b_{1,j} \cdot \beta_1 + \dots + b_{m,j} \cdot \beta_m) \cdot \mathbf{v}_j.
\end{aligned}$$

The Frobenius automorphism $\phi(z) = z^2$ keeps $\mathbb{F}_2 \subset \mathbb{F}_{2^m}$ fixed, and has no other fixed points. We extract the value of $\langle b_i, \mathbf{v} \rangle$ from the values of $\langle b, \mathbf{v} \rangle$ along the Galois orbit

$$\mathbf{v}, \mathbf{v}^2, \mathbf{v}^{2^2}, \dots, \mathbf{v}^{2^{m-1}},$$

as follows, where by $\mathbf{v}^t$ we mean that each component of $\mathbf{v}$ is raised to the power t . The Moore matrix

$$M = \begin{pmatrix} \beta_1 & \beta_2 & \dots & \beta_m \\ \beta_1^2 & \beta_2^2 & \dots & \beta_m^2 \\ \beta_1^{2^2} & \beta_2^{2^2} & \dots & \beta_m^{2^2} \\ \vdots & & & \\ \beta_1^{2^{m-1}} & \beta_2^{2^{m-1}} & \dots & \beta_m^{2^{m-1}} \end{pmatrix}$$

consists of the Galois orbits of the basis vectors, and is invertible. (See below.) Using that the component polynomials are over $\mathbb{F}_2$, we see that

$$\phi^k \left(\langle b, \phi^{-k}(\mathbf{v}) \rangle \right) = \langle b_1, \mathbf{v} \rangle \cdot \phi^k \beta_1 + \dots + \langle b_m, \mathbf{v} \rangle \cdot \phi^k \beta_m,$$

for any k . To this end, observe that

$$M \cdot \begin{pmatrix} \langle b_1, \mathbf{v} \rangle \\ \langle b_2, \mathbf{v} \rangle \\ \langle b_3, \mathbf{v} \rangle \\ \vdots \\ \langle b_m, \mathbf{v} \rangle \end{pmatrix} = \begin{pmatrix} \langle b, \mathbf{v} \rangle \\ \phi \left(\langle b, \phi^{-1}(\mathbf{v}) \rangle \right) \\ \phi^2 \left(\langle b, \phi^{-2}(\mathbf{v}) \rangle \right) \\ \vdots \\ \phi^{m-1} \left(\langle b, \phi^{-(m-1)}(\mathbf{v}) \rangle \right) \end{pmatrix}.$$

Multiplying both sides by M^{-1} (M is a Moore matrix and hence invertible because the $\beta_1, \dots, \beta_m$ are invertible) we obtain an expression of the values $\langle b_i, \mathbf{v} \rangle$ as a linear expression of the values $\phi^k \left(\langle b, \phi^{-k}(\mathbf{v}) \rangle \right)$.

Now one can prove the inner product claims $\langle b, \phi^{-i}(\mathbf{v}) \rangle$ for all i by first batching them all into one single claim on the value of $\langle b, \mathbf{u} \rangle$, and then proving this later claim normally, since we had committed to the vector b initially. The batching step introduces the error term

$$1/|\mathbb{F}_{2^\nu}|.$$

Ulrich: Invertibility of M follows from that of M^t : any linear functional $(c_1, \dots, c_m)$ in the kernel of M^t gives rise to the linearizable polynomial

$$p(X) = \sum_{i=0}^{m-1} c_i \cdot X^{2^i}$$

which has $\beta_1, \dots, \beta_m$, and hence the whole space F as roots. By dimension, $p(X)$ must be trivial, meaning that all its coefficients are zero.